# Venture Capital and Engineering Quality: Evidence from Software Startups*

João Dimas† Lampros Pnevmatikos‡

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**Abstract**

We investigate a novel channel of venture capital value creation: the practices through which startups build and maintain their technology. We construct a new panel of 569 firms using more than 60,000 full codebase snapshots. We compare changes in engineering quality following first VC investment with changes at similar firms that remain without VC backing. We also construct a dollar-denominated measure of technical debt, the estimated maintenance liability embedded in software. Relative to codebase size, serious deviations from coding best practices decline by 15% and technical debt by 12%. These improvements accompany substantial codebase growth. The decline in the frequency of serious coding problems persists after matching on pre-financing growth, and quality gains extend to firms already generating revenue. In contrast, a placebo test around later financing rounds shows no evidence of quality improvement. We also analyze 5.6 million code contributions (commits). After first VC investment, teams more often document collaboration and copyright compliance, the share of weekend contributions declines, and existing engineers become less likely to introduce bugs. The findings suggest that VC professionalization extends to the engineering practices through which teams build and maintain their products.

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†Olin Business School at Washington University in St. Louis.

‡Olin Business School at Washington University in St. Louis. Send correspondence to Lampros Pnevmatikos; e-mail: l.pnevmatikos@wustl.edu.

# 1 Introduction

An important question in corporate finance concerns how venture capitalists contribute to the success of the startups they finance (Gompers et al., 2020; Amornsiripanitch, Gompers, and Xuan, 2019; Hsu, 2004). While venture capitalists select promising firms, their involvement after investment can also shape firm performance. For example, Bernstein, Giroud, and Townsend (2016) show that reductions in the cost of monitoring portfolio companies improve performance, especially innovation. Prior work documents that VCs professionalize startups by reshaping governance structures, leadership, and organizational design: Hellmann and Puri (2002) find that VC-backed firms are more likely to hire professional CEOs, implement stock option plans, and formalize HR policies, and that these changes occur earlier than in comparable non-VC-backed firms. Yet a fundamental operational margin remains unexplored: do venture capitalists also change the *technical practices* through which startups build their products?

For technology startups—a central part of the venture capital ecosystem—long-run value depends not only on strategy and governance, but also on the quality of the underlying software. Design choices and engineering practices drive the accumulation of technical debt (TD)—the future cost of fixing coding problems—and can shape a startup's ability to scale reliably, innovate, and manage operational risk as the organization grows (Sculley et al., 2014, 2015).

Technical debt is not recorded in financial statements, yet it behaves like an economically meaningful liability. Deviations from engineering best practices can increase the future resources required to maintain and modify a product, and, if left unaddressed, can compound into higher defect risk and slower feature development, similar to postponing CapEx (Banker, Liang, and Ramasubbu, 2021; Krasner, 2022; McKinsey & Company, 2020, 2023). In that sense, TD is closer to an off-balance-sheet operating liability than something that should only concern engineers. These liabilities are difficult to compare across firms because the effort needed to remedy coding problems depends on code complexity and connections between components, while measurement tools use different rules and definitions (Krasner, 2022). Source code is also usually proprietary, limiting researchers' access to the information needed to measure these liabilities. Our setting allows us to bring the industry-standard concept of TD into empirical finance by constructing a consistent, dollar-denominated measure from code-level data.

To accomplish this, we assemble longitudinal codebase data for 569 open-core technology firms. These firms commercialize software whose core source code is publicly available, allowing us to observe development histories for an economically relevant part of their products. We combine 61,624 full codebase snapshots with financing records from PitchBook and employment histories from LinkedIn. Each snapshot captures the complete codebase at a particular point in time. We evaluate each snapshot using SonarQube, an industry-standard tool that identifies deviations from coding best practices. Our focal event is a firm's first completed Seed round, or its first completed Series A round

when no Seed is recorded.

VC contracting and monitoring provide a rationale for changes in technical practice after financing. Staged capital provision gives investors opportunities to review progress and condition future funding on performance (Gompers, 1995), while control and board rights facilitate intervention (Kaplan and Strömberg, 2003). Investor advice can also affect entrepreneurial decisions when both parties contribute to the venture (Casamatta, 2003). In software firms, these activities may encourage more systematic review, testing, and coordination of development work. Financing can also fund engineers and tools directly. The net effect on quality is uncertain, however, because faster development and larger teams can increase complexity and coordination costs.

To examine whether first institutional VC entry is associated with changes in engineering quality, we estimate a stacked difference-in-differences model comparing changes in newly funded firms (treated) with contemporaneous changes in firms that remain without VC backing throughout the event window (control). We measure coding problems relative to codebase size so that firms with different amounts of software can be compared. The frequency of serious deviations from coding best practices falls by 15% of the treated pre-investment mean. These are problems that can make software less stable, less secure, or more costly to maintain. We also find fewer problems that make software harder to modify or extend. Changes in minor code issues, in code patterns associated with software failures, and in security-sensitive code flagged for human review are not statistically significant.

Our technical-debt measure translates these coding problems into estimated future maintenance costs. Relative to codebase size, this liability falls by 12% of the treated pre-investment mean. The decline therefore represents a lower estimated maintenance burden for a given amount of software.

Event-study plots show that serious coding problems become less frequent around the first financing round, with little divergence between treated and control firms before funding. This timing pattern is consistent with a change in development practices accompanying investor entry. The decline occurs as codebases expand substantially. A growing codebase can have fewer problems per unit of code even if existing problems are not removed, provided that problem counts grow more slowly than code volume, which represents an increase in average code quality. Because we do not track the resolution of individual problems, our comparisons cannot distinguish between correcting existing code and adding cleaner code, but both are important margins of improvement. The decline in serious coding problems also persists among firms with slow pre-financing codebase growth.

We also acknowledge that VC funding is inherently selective and non-random: investors finance firms they expect to grow. A natural concern is that the quality improvements reflect growth trajectories that were already different before financing. Although engineering-quality measures are well balanced before funding, treated firms are smaller and grow faster than controls. To address this concern, we match treated and control firms on pre-financing headcount growth within each financing cohort. The declines in serious coding problems and problems that make software harder to maintain persist across a range of matching choices. The estimated reduction in maintenance liabilities also remains similar

in magnitude. These comparisons show that the coding-quality findings persist when newly funded firms are compared with startups on similar prior growth trajectories.

To examine whether quality improvements recur with additional financing, we implement a placebo test centered on each firm's first Series B-or-later round. Around these later rounds, total headcount and software-engineering headcount rise substantially, as they do around first VC entry. Yet we find no statistically significant improvement in coding quality or estimated maintenance liabilities. This combination of substantial hiring and no detectable quality gains is consistent with a change in technical practice associated with first institutional entry.

We also test whether the quality improvements reflect a transition to commercialization around the first financing round. Most treated firms were already generating revenue before financing, and quality gains extend to these firms. We find no evidence of larger improvements among firms that had not yet begun selling their products. Development activity expands, and employment grows in both engineering and commercial functions, with a larger shift toward engineering. However, developers' descriptions of code changes do not show evidence of an increase in commercialization features, such as pricing, subscriptions, or billing. The quality improvement is therefore not confined to firms making the initial transition to revenue generation.

We then examine engineering processes by distilling information from 5.6 million individual code contributions (commits). Following first VC entry, firms document collaboration and copyright and licensing assurances more frequently, a larger share of commit messages references numbered records of development work, and a smaller share of commits is made on weekends. We also trace the lines changed in later bug fixes to the earlier commits that last modified them. Among engineers active before financing, the share of their commits linked to subsequent fixes declines after VC entry. These links identify earlier work associated with later repairs, with attribution depending on which bugs are discovered and documented. Together, these findings reveal more formal engineering procedures and improvements in the work of existing teams.

Our results contribute to several strands of the literature. First, we contribute to the venture capital literature on post-investment value creation. While prior work has emphasized governance and leadership professionalization, we provide evidence on the technical practices through which startups build their core products. Our findings suggest that engineering discipline improves around VC entry, with reductions in serious problems and in problems that make software difficult to maintain.

Second, the results complement evidence on how private equity investors change the day-to-day operations of mature firms. Bernstein and Sheen (2016), for example, document improvements in safety and maintenance at restaurants following buyouts. We document changes in engineering practices at an earlier stage of firm development, when the first institutional investor enters. Later rounds accompany substantial hiring without a statistically significant quality gain. Read alongside the commit histories, this difference supports a governance interpretation in which first institutional entry strengthens engineering discipline.

Third, our measurement of software maintenance liabilities informs research on intangible assets and software-intensive firms (Eisfeldt and Papanikolaou, 2013; Peters and Taylor, 2017). We construct, to our knowledge, the first comparable, dollar-denominated measure of technical debt in the finance literature. Combining this measure with development histories allows researchers to study how firms build and maintain software assets, with applications to technology-firm valuation and M&A due diligence.

The remainder of the paper proceeds as follows. Section 2 describes the data, sample construction, and measures of engineering quality and technical debt. Section 3 presents the empirical design, results, and mechanism evidence. Section 4 examines matching, later financing rounds, commercialization, and specification sensitivity. Section 5 concludes.

# 2 Data & Background

We combine three data sources to follow software development, employment, and financing at open-core technology firms: **(i)** our comprehensive evaluations of source code obtained from GitHub and GitLab; **(ii)** employee–firm matched data from LinkedIn; and **(iii)** funding and deal records from PitchBook.

Firms typically restrict access to proprietary source code, making their software assets difficult to study. We therefore focus on open-core firms: they make the core of their product publicly available, allowing us to observe its development history and technical choices.

## 2.1 The Open-Core Business Model

The open-core model combines the collaborative advantages of open-source software with the revenue potential of proprietary licensing. Under this model, a firm releases its core product under an open-source license, making the source code freely accessible to developers and researchers. The firm retains exclusive rights to enterprise-grade features, proprietary modules, or support services, which it monetizes through subscription or licensing agreements.

A prominent example is Posit (formerly RStudio, Inc.), the company behind RStudio, a programming environment for R. While the core software is free and open source, the company generates revenue by licensing enhanced versions for commercial applications. More broadly, research on open-source software documents knowledge spillovers among projects and productivity gains for the firms that use it (Fershtman and Gandal, 2011; Nagle, 2019).

The open-core setting also has implications for external validity. Relative to closed-source startups, open-core firms expose substantial portions of their codebase (and often issue trackers) to public scrutiny, which can create baseline quality discipline through reputational concerns, external bug reporting, and community review. If this community discipline substitutes for some of the monitoring

and process improvements that VCs bring, the marginal effect of VC entry on engineering quality could be attenuated in our setting and potentially larger in proprietary environments. Conversely, transparency may complement VC involvement by making engineering performance more observable and by facilitating rapid community-driven fixes, so the direction of any external-validity bias is not a priori clear. We therefore interpret our estimates as evidence that VC entry is associated with changes in engineering outcomes in an observable, open-core environment.

## 2.2 Codebase Data

We define our sample universe using OSSRank, a leading analytics platform for the open-source ecosystem, to identify 649 unique companies that commercialize open-source repositories. These firms primarily operate in the technology sector.

We match each firm to the repositories on GitHub and GitLab that hold the code underlying its core product. A repository is the central storage location for a software project's source code. GitHub and GitLab are industry-standard platforms for software development and version control, widely adopted across both startups and large enterprises. These platforms record changes to the code as timestamped "commits" that identify the author, describe the change, and include an accompanying message.

We construct a monthly panel of codebases by retrieving the final version of each repository for every month in our sample period (2010–2024). We evaluate codebases with SonarQube using a consistent measurement procedure across firms and over time. While most repositories are hosted on GitHub, some firms use GitLab. Since both platforms offer nearly identical version control features, we pool these data sources into a combined dataset.

Because a firm can maintain several repositories, we measure engineering quality at the firm–month level by aggregating across its available repositories. We sum lines of code, estimated repair time, and issue counts, and recompute ratio measures from these totals. This construction captures development across the firm's covered projects while retaining the firm as the unit of analysis. Section 4.4 shows that the main estimates change little when we exclude firms with more than five scanned repositories.

Our company–repository mapping links 703 repositories to 609 firms, 569 of which contribute at least one covered codebase month. We generate 61,624 repository–month observations and fill gaps between observed months by linearly interpolating metrics within firms. The monthly panel contains 61,768 firm–months with a positive line count. For estimation, we retain firm–months with at least one recorded repository scan and a positive line count. Appendix N reports the sample flow and comparisons using complete observation windows and fixed pre-financing repository sets.

## 2.3 Funding & Firm Data

We date VC entry using PitchBook financing records. For each firm, we extract its investment history, including deal dates, transaction types, investor composition, and funding stages. We first remove rounds recorded as rumored, failed or cancelled, announced or in progress, upcoming, or postponed. We then identify the first completed Seed round, or the first completed Series A round when no completed Seed is recorded. To focus on early-stage venture investment, we exclude firms that enter the data only through a public listing, debt issuance, management buyout, acquisition, or Series B or later round with no prior Seed or Series A financing. We separately examine later financing rounds in our Series B+ comparisons.

Since the firms in our sample are predominantly private and early stage, standard financial disclosures and coverage in databases such as Compustat are limited. To measure firm size and growth, we use employee–firm matched LinkedIn records to construct monthly observations of total headcount and software-engineering headcount. These histories track organizational expansion alongside changes in the codebase.

## 2.4 Measuring Engineering Quality

We measure engineering quality using SonarQube, a widely used analysis tool that scans source code and flags deviations from common coding and security practices. For each historical snapshot, we apply language-specific checks directly to its source code. Using the same measurement procedure across firms and months supports comparisons of firms using different programming languages. Appendix B details the configurations for selected languages.

### 2.4.1 Taxonomy of Coding Problems

Each SonarQube rule checks for a specific pattern in the code. When the scanner finds that pattern, it records an "issue"; these issues are the basic unit behind our engineering-quality measures.

SonarQube identifies three types of coding problems: bugs, security vulnerabilities, and maintainability violations. Bugs can cause failures or incorrect behavior; security vulnerabilities can expose software to attack. Maintainability violations make software harder to understand, modify, or extend. We measure the frequency of each type per 1,000 lines of code.

SonarQube also classifies violations by severity. Our principal quality measure, Critical Violations, counts serious deviations from coding best practices that the tool classifies as critical. Most critical violations in our data are maintainability violations.[1] We also measure total violations across all severity levels.

[1] SonarQube's traditional measures do not split critical violations by type. We therefore infer the issue-type split from its alternative classification framework, which reports high-severity issues by the quality they affect: reliability, security or maintainability. See Appendix M.4.

## 2.5 Technical Debt

Technical Debt measures the stock of maintenance liabilities embedded in a codebase. In the software engineering literature, technical debt captures development choices that save time today but raise the cost of future changes (Sculley et al., 2014). Firms effectively pay interest through additional maintenance work and face costly paydowns when refactoring—reorganizing existing code without changing its intended functionality—becomes unavoidable. For example, duplicating code can accelerate a feature's introduction but requires developers to update several copies whenever that feature changes. Although this tradeoff is familiar in software engineering, it has received little direct measurement in finance; our approach expresses these software liabilities in dollars so they can be compared across firms and over time.

We measure technical debt using SonarQube's estimate of the time required to resolve maintainability violations. We sum these times across a firm's covered repositories each month, yielding estimated maintenance effort in hours. Violations that require more work to resolve receive greater weight.

The estimated maintenance effort depends on the coding rules used in the scans and the time assigned to resolve each violation. We therefore apply a common scaling factor, $\kappa$, to calibrate this measure to an external economic benchmark. This produces a dollar-denominated proxy for maintenance liabilities, with a consistent unit of measurement across firms and monthly codebase evaluations. The calibration concerns the level of the liability; it does not assume that firms incur the modeled remediation costs as current cash expenditures.

The calibration proceeds in three steps. First, we construct a proxy for the replacement cost of the observed software assets. We set development time at $\ell = 0.5$ hours (30 minutes) per non-comment line of code. For a codebase with $\text{LoC}_{i,t}$ lines of code, our proxy for replacement effort is:

$$\text{Re-development Effort}_{i,t} = \ell \times \text{LoC}_{i,t}. \tag{1}$$

Multiplying by an assumed common developer wage of $w = \$55$ per hour yields Software Assets in dollars:

$$\text{Software Assets}_{i,t} = w \times \text{Re-development Effort}_{i,t}. \tag{2}$$

Second, we choose the scaling factor $\kappa$ to place estimated maintenance effort on an economic scale. Surveyed technology executives estimate technical debt at 20–40% of the value of their technology assets before depreciation (McKinsey & Company, 2023). We use the lower endpoint as a calibration target, setting median debt equal to 20% of the median replacement-cost proxy.[2] Let $H_{i,t}$ denote estimated maintenance effort in hours for firm $i$ in month $t$. Letting $\varphi = 0.20$, we choose $\kappa$

[2]The survey covers broader technology assets than the source code we observe. We use it to calibrate the level of the measure.

such that:

$$\text{Median}\left(w \times \kappa \times H_{i,t}\right) = \varphi \times \text{Median}\left(w \times \ell \times \text{LoC}_{i,t}\right), \tag{3}$$

where we compute medians over nonmissing firm–month observations in the source panel. Simplifying yields:

$$\kappa = \varphi \times \frac{\text{Median}\left(\ell \times \text{LoC}_{i,t}\right)}{\text{Median}\left(H_{i,t}\right)}. \tag{4}$$

In the main sample, this produces $\kappa = 68.60$, which is constant across firms and time.

Third, we apply this calibration to construct Technical Debt for each firm–month:

$$\text{Technical Debt}_{i,t} = w \times \kappa \times H_{i,t}. \tag{5}$$

Because $\kappa$ is constant across firms and time, it rescales estimated maintenance effort proportionally, leaving statistical significance and changes relative to the sample mean unchanged. The calibration makes the estimated maintenance burden comparable in dollar terms, with its level determined by the common wage and calibration target.

We also report the Technical Debt Ratio, which expresses calibrated debt as a percentage of the replacement-cost proxy:

$$\text{Technical Debt Ratio}_{i,t} = 100 \times \frac{\text{Technical Debt}_{i,t}}{\text{Software Assets}_{i,t}} = 100 \times \frac{\kappa \times H_{i,t}}{\ell \times \text{LoC}_{i,t}}. \tag{6}$$

Both measures use the same observations with positive lines of code. The wage cancels from the replacement-cost ratio, while its level reflects the calibration target.[3] Changes in the ratio, measured in percentage points, indicate how the maintenance burden evolves relative to the size of the software asset.

## 2.6 Key Variables

We express all issue counts per 1,000 non-comment lines of code (kLoC). Our primary measures use SonarQube's traditional classification. The high- and low-severity maintainability and reliability measures use an alternative classification, described in Appendix M.4. *Reliability Issues* and *Bugs* therefore assess related risks of incorrect software behavior using different classifications. The severity-specific measures are not a breakdown of the traditional totals.

*Security Hotspots* identify security-sensitive code segments that require human review, such as

[3] The calibration equates median Technical Debt with 20% of median Software Assets. The median firm–month ratio is 30.2%, since the median of a ratio need not equal the ratio of medians.

code handling authentication or sensitive data. A hotspot marks a location that warrants inspection; it becomes evidence of a vulnerability only after the security context has been assessed. This distinction separates our measure of security-review needs from confirmed security failures.

In addition to these engineering measures, we use the economic measures of Technical Debt described in Section 2.5: debt in dollars per line of code and the Technical Debt Ratio, which scales debt by software assets. We measure codebase size using non-comment lines of code and organizational scale using total and software-engineering headcount. Regressions use code volume as log(LoC), total headcount as log(Headcount), and software-engineering headcount as log(SWE).

The VC indicator, $VC_{i,c,t}$, equals one for a firm treated in cohort $c$ from its qualifying financing month onward. To examine heterogeneity, *High Violations* identifies firms whose critical violation density in the month before the cohort's event exceeds the median across baseline firm–cohort observations of firms treated somewhere in the sample; *Large Round* identifies financing rounds above the treated-firm median deal size. Table 19 in Appendix A provides the full variable definitions.

## 2.7 Summary Statistics

Table 1 reports summary statistics for the firm–month panel before stacking. The median covered firm–month has 5.16 critical violations per 1,000 lines of code and a Technical Debt Ratio of 30.2%. The distributions are right-skewed: the corresponding means are 10.45 and 43.4%.

Table 1: Summary Statistics

| | | Distribution | | | | |
|---|---|---|---|---|---|---|
| **Variable** | **N** | **Mean** | **SD** | **P25** | **Median** | **P75** |
| **Panel A: Level Variables** | | | | | | |
| Technical Debt (in $) | 61,768 | 5,263,622.30 | 11,987,074.44 | 164,190.37 | 938,639.44 | 4,201,349.26 |
| Comment Lines | 61,768 | 117,976.35 | 329,153.82 | 4,707.75 | 21,377.50 | 98,217.00 |
| Total Violations | 61,768 | 11,732.45 | 26,172.37 | 399.00 | 2,131.00 | 10,016.25 |
| Critical Violations | 61,768 | 4,133.78 | 9,799.54 | 83.00 | 619.00 | 3,210.00 |
| Bugs | 61,768 | 593.95 | 1,920.59 | 5.00 | 49.00 | 279.00 |
| Security Hotspots | 61,768 | 275.21 | 632.14 | 9.00 | 56.00 | 223.00 |
| Maintainability Violations (All) | 61,768 | 10,967.73 | 24,626.64 | 363.00 | 2,042.00 | 9,368.92 |
| Maintainability Issues (High) | 61,768 | 4,106.40 | 9,747.16 | 79.00 | 608.00 | 3,196.00 |
| Maintainability Issues (Low) | 61,768 | 3,445.93 | 8,202.71 | 63.00 | 472.00 | 2,522.00 |
| Reliability Issues (Total) | 61,768 | 1,135.75 | 4,286.75 | 13.00 | 114.00 | 579.00 |
| Reliability Issues (High) | 61,768 | 28.33 | 75.54 | 0.00 | 3.00 | 20.67 |
| Reliability Issues (Low) | 61,768 | 463.55 | 3,109.39 | 0.00 | 11.00 | 134.00 |
| **Panel B: Per-LoC Variables** | | | | | | |
| Technical Debt per LoC ($) | 61,768 | 11.92 | 15.99 | 3.55 | 8.29 | 14.26 |
| Technical Debt Ratio (%) | 61,768 | 43.36 | 58.15 | 12.90 | 30.16 | 51.84 |
| Violations per 1,000 LoC | 61,768 | 29.44 | 40.29 | 8.14 | 16.84 | 34.90 |
| Critical Violations per 1,000 LoC | 61,768 | 10.45 | 18.74 | 1.44 | 5.16 | 9.75 |
| Bugs per 1,000 LoC | 61,768 | 1.77 | 5.47 | 0.04 | 0.45 | 1.39 |
| Security Hotspots per 1,000 LoC | 61,768 | 1.37 | 10.60 | 0.10 | 0.43 | 1.05 |
| Maintainability Violations per 1,000 LoC | 61,768 | 27.38 | 37.68 | 7.72 | 16.04 | 32.81 |
| Maintainability Issues (High) per 1,000 LoC | 61,768 | 10.35 | 18.68 | 1.37 | 5.09 | 9.67 |
| Maintainability Issues (Low) per 1,000 LoC | 61,768 | 7.80 | 15.76 | 1.24 | 3.86 | 7.92 |
| Reliability Issues (Total) per 1,000 LoC | 61,768 | 3.28 | 9.91 | 0.15 | 0.83 | 2.53 |
| Reliability Issues (High) per 1,000 LoC | 61,768 | 0.10 | 0.40 | 0.00 | 0.02 | 0.09 |
| Reliability Issues (Low) per 1,000 LoC | 61,768 | 1.36 | 5.73 | 0.00 | 0.09 | 0.51 |
| **Panel C: Scale Variables (Log)** | | | | | | |
| Log(Lines of Code) | 61,768 | 11.78 | 2.07 | 10.66 | 12.05 | 13.22 |
| Log(Headcount) | 32,313 | 3.58 | 2.59 | 1.79 | 3.14 | 4.88 |
| **Panel D: Treatment Indicators (Post-Funding)** | | | | | | |
| Post Seed Round (=1) | 111,780 | 0.16 | 0.37 | – | – | – |
| Post Series A (=1) | 111,780 | 0.10 | 0.30 | – | – | – |
| Post Top-Tier VC (=1) | 111,780 | 0.05 | 0.22 | – | – | – |
| Post Mature Round (=1) | 111,780 | 0.12 | 0.32 | – | – | – |

*Note:* The table reports summary statistics for the firm–month panel before stacking. SonarQube counts and remediation effort are summed across each firm's covered repositories, with gaps interpolated between observed months. Panels A–C use firm–months with a positive non-comment line count: the code-quality and both debt measures cover 61,768 firm–months, while headcount covers 32,313 firm–months. Panel A reports levels. Panel B divides issue counts by thousands of non-comment lines of code and Technical Debt by lines of code; the Technical Debt Ratio is calibrated debt divided by the replacement-cost proxy, multiplied by 100. Both debt measures use $\kappa = 68.60$ (Section 2.5). Panel C reports the natural-log scale transformations defined in Table 19. Panel D reports indicators equal to one from the relevant financing month onward, over the full firm–month grid. Post Top-Tier VC uses a fixed list: Andreessen Horowitz, Lightspeed Venture Partners, Accel, Kleiner Perkins, Benchmark Capital Holdings, Greylock, Bessemer Venture Partners, and First Round Capital. Post Mature Round includes Series B and later venture rounds, acquisition financing, recapitalizations, corporate transactions, divestitures, buyouts, spin-offs, bonds, and private investments in public equity. These two descriptive indicators do not define the treatment used in our regressions. Maintainability Violations use the traditional classification; the high- and low-severity maintainability issues use the alternative classification described in Appendix M.4 and are not a breakdown of the total. Bugs also use the traditional classification; the reliability issues use the alternative classification and are not a breakdown of Bugs.

# 3 Estimation & Empirical Results

To estimate how first institutional VC entry changes engineering quality, we exploit differences in financing dates across firms using a stacked difference-in-differences (DiD) design. With staggered treatment, a conventional two-way fixed-effects regression can compare newly funded firms with firms whose outcomes already reflect earlier financing. Under heterogeneous treatment effects, these comparisons can contaminate the estimated response and even reverse its sign (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). Following Gormley and Matsa (2011), we construct a separate comparison group for each financing cohort using firms that remain without a first round throughout its event window.

We group firms by the calendar month of their first completed Seed round, or their first completed Series A round when no Seed is recorded. For each cohort, controls are firms with no qualifying first round through the end of the window. The cohort panel covers 25 months, from event month $e = -12$ through $e = +12$ relative to financing. We then stack these panels.[4] Every estimation sample excludes fixed-effect singletons and cohorts without both treated and control firms. The stacked cohort panels contain 99 cohorts, 274 treated firms, 412 firms appearing as controls, and 484 unique firms. The baseline regression uses 487,577 observations from 483 unique firms, including 273 treated firms, after excluding 18 fixed-effect singletons. Appendix C describes the estimand and its relation to the stacked DiD regression.

We estimate

$$M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}, \tag{7}$$

where $i$, $c$, and $t$ index firm, cohort, and calendar month. $M_{i,c,t}$ is an engineering outcome, and $VC_{i,c,t}$ is $\text{Treated}_{i,c} \times \text{Post}_{c,t}$, which equals one for treated firms from the financing month onward. Cohort × Firm fixed effects, $\chi_{c,i}$, absorb persistent firm differences within each cohort; Month × Cohort fixed effects, $\delta_{c,t}$, absorb common monthly shocks within that cohort. $X_{i,c,t}$ denotes additional regressors in the heterogeneity specifications, and $\varepsilon_{i,c,t}$ is the error term. The baseline omits $X_{i,c,t}$.

The coefficient $\beta$ captures the post-financing change in treated firms relative to their cohort controls. The DiD coefficient weights these comparisons according to the observations and residual treatment variation in the stacked sample; Appendix C provides the formal interpretation. We cluster standard errors by cohort and firm to account for dependence within cohorts and across repeated appearances of the same firm.

We also examine two hypotheses about the scope of the improvement. Firms with more critical violations before financing may have greater room to improve, while firms raising larger rounds can devote more resources to engineers, tools, and remediation. The heterogeneity columns therefore

[4] A control firm–month can appear in several cohort panels, and a not-yet-treated firm can later enter as treated in its own cohort. The stacked observation count therefore exceeds the number of underlying firm–months.

interact treatment with *High Violations*, defined using critical-violation density at relative month $-1$, and *Large Round*, defined using the treated-firm median round size. These comparisons distinguish where the gains are concentrated and motivate the more direct mechanism tests below. Appendix F extends the analysis to founding-team characteristics using LinkedIn career histories, and Table 19 provides the complete variable definitions.

To examine the timing of the response, we interact treated status with event-quarter indicators while retaining monthly fixed effects. We index the event-quarter bins from $q = -3$ to $q = 5$, omitting $q = 0$, and plot them at their month-bin midpoints. The omitted quarter contains relative months $\{-4, -3, -2\}$, and the quarter around financing, $q = 1$, contains $\{-1, 0, 1\}$. The latter bin includes one pre-financing month, so a change that begins only after the round closes is averaged with an untreated month. Quarters $q = 2$ through $q = 5$ contain only post-financing observations, with the final bin ending at month +12. These estimates compare pre-financing trajectories and trace how the improvement develops during the first year of institutional backing.

## 3.1 Identification Strategy

Venture capitalists select firms whose teams, products, and growth prospects they find promising. A central identification concern is therefore that VCs select firms whose engineering quality is already improving, or whose founders have planned improvements that have yet to appear in the code. Our identification does not require financing to be randomly assigned. It requires that, absent first VC entry, treated firms and their cohort controls would have followed parallel trends in engineering quality, together with no anticipatory response and no spillovers to controls.

Several features of our data and design address selection on an improving engineering trajectory. First, the event studies show little divergence in critical violations per 1,000 lines and technical debt per line before financing, followed by reductions around the first round. To explain this pattern, an omitted change in product demand, management, or engineering priorities would need to coincide with financing and produce the observed change in quality. A gradual improvement already under way before investor entry would instead predict a divergence in the pre-treatment coefficients.

Second, the timing of engineering quality differs from that of firm growth. Treated firms expand their headcount and codebases more rapidly than controls before financing, but their critical-violation density does not show a corresponding improvement. This distinction matters because VC selection on fast-growing firms need not imply selection on improving code quality. We develop this argument in the firm-scale results and test it directly by matching on pre-financing headcount growth in Section 4.1.

Third, we examine specific alternatives to a change in engineering practices. Commercial-stage tests ask whether the quality improvement is concentrated in firms making the initial transition to revenue generation. A Series B+ placebo asks whether the same quality gains recur when firms that

already have institutional backing receive additional financing. Section 4 presents these tests and their implications for the interpretation of first VC entry.

### Baseline Balance

Table 2 reports covariate means at the last pre-treatment month ($t = -1$). Treated and control firms enter the event with similar engineering quality: the standardized difference is 0.05 for critical violations per thousand lines of code, −0.03 for total violations, and −0.01 for both technical debt per line and the Technical Debt Ratio. The absence of a substantial baseline quality gap is informative for two competing selection stories. Treated firms neither start with markedly better code nor have systematically more violations to remove, as a simple room-for-improvement explanation would predict.

This balance also has an institutional rationale. Survey evidence indicates that VCs place particular weight on the management team and assess the product, technology, and market when selecting investments (Gompers et al., 2020). These assessments concern the commercial promise and capabilities of the venture; the detailed rule violations we measure capture a different margin, namely the implementation of coding practices within the existing product. An investor can therefore select a promising technology firm without selecting a codebase with low violation density. The observed balance is consistent with this distinction between commercial due diligence and the specific engineering outcomes in our analysis.

We next test whether baseline engineering quality or codebase size predicts which firms within a cohort receive financing. Using each firm's measures in the month before financing ($t = -1$), as in Table 2, we regress an indicator equal to one for the cohort's newly financed firms and zero for its controls on three quality measures (critical and total violations per 1,000 lines and technical debt per line), and then on those measures plus log lines of code. Both linear probability models include cohort fixed effects and cluster standard errors by firm. The quality measures are not jointly statistically significant ($p = 0.24$) and explain 0.02% of the within-cohort variation in financing. Doubling a firm's lines of code is associated with a 0.1 percentage-point lower probability of financing ($p < 0.01$), but adding size raises the explained share only to 0.1%. Selection on these observables therefore concerns scale, while baseline engineering quality has little predictive power for financing.

The balance table shows these scale differences directly. Treated firms have lower total headcount (standardized difference −0.65), lower software-engineering headcount (−0.69), and smaller codebases (−0.31 in log lines of code). Cohort × Firm fixed effects absorb persistent differences in levels, and the matched specification addresses the different pre-financing growth trajectories. These comparisons support the design's focus on changes in quality around financing, while the causal interpretation ultimately requires that investors are not simply forecasting an unobserved engineering improvement that would have occurred at the same time without their entry.

Table 2: Baseline Balance: Treated vs. Control Firms at $t = -1$

| | **Treated** | **Control** | **Std. Diff.** |
|---|---|---|---|
| Log(Lines of Code) | 11.07 | 11.70 | −0.31 |
| Log(Headcount) | 2.05 | 3.58 | −0.65 |
| Log(SWE Headcount) | 1.07 | 2.58 | −0.69 |
| Critical Violations / kLoC | 12.37 | 11.16 | 0.05 |
| Technical Debt per LoC | 12.45 | 12.68 | −0.01 |
| Violations / kLoC | 30.77 | 31.86 | −0.03 |
| Technical Debt Ratio | 45.28 | 46.09 | −0.01 |
| **# Firm–cohort observations** | 247 | 19,686 | |

*Note:* This table reports covariate means at the last pre-treatment period ($t = -1$) for treated and control firms, using all control firms available in the stacked DiD panel. Firms can appear as controls in more than one cohort, so the observation counts refer to firm–cohort pairs. kLoC denotes 1,000 non-comment lines of code; SWE denotes software engineering. Log transformations follow Table 19. Technical Debt per LoC is measured in dollars per line, and the Technical Debt Ratio is calibrated debt as a percentage of the codebase's replacement cost. The observation-count row applies to code volume and quality; total and SWE headcount each have 201 treated and 10,080 control observations. Std. Diff. = (Treated Mean − Control Mean) / pooled standard deviation. See Appendix A, Table 19, for full variable definitions. Section 4.1 reports the matched-growth comparison.

## 3.2 Critical Violations

Critical and maintainability violations decline, while the bug and reliability estimates remain close to zero (Figure 1).

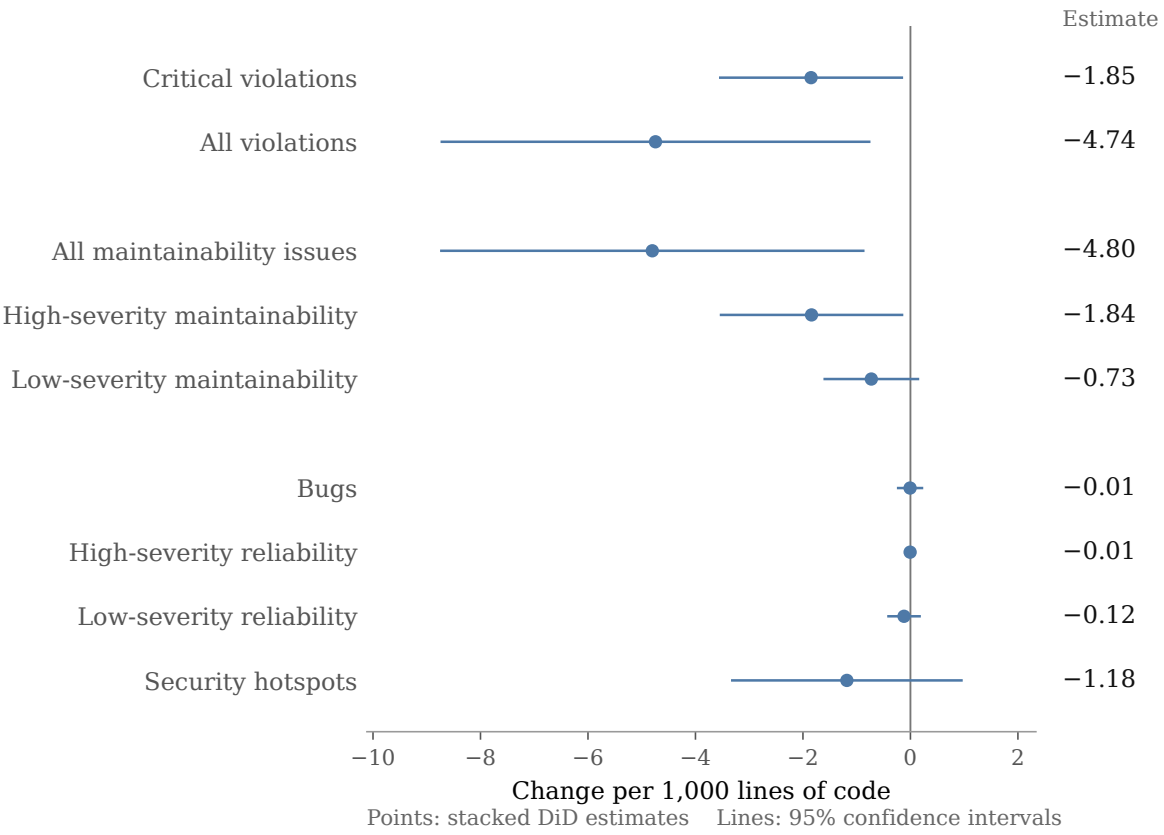



Figure 1: The Scope of the Engineering-Quality Improvement. Baseline estimates from Equation (7), in issues per 1,000 non-comment lines of code. Issue categories overlap. High- and low-severity maintainability and reliability use the alternative classification described in Appendix M.4 and are not breakdowns of the overall maintainability count or of Bugs. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

We first examine *Critical Violations*, severe deviations from coding best practices that can compromise the stability, security, or maintainability of a product. Unlike a deliberate decision to postpone a minor improvement, allowing serious code problems to accumulate can make subsequent development both riskier and more costly. A decline in this category therefore speaks to the discipline with which firms manage consequential engineering problems.

Critical violations fall by 1.85 per 1,000 lines of code after financing relative to controls (14.9% of the pre-treatment mean; $p = 0.03$; Table 3, Column (1)). More broadly, total rule violations across all severity levels decline by 4.74 per 1,000 lines (14.4% of the pre-treatment mean; $p = 0.02$; Column (4)). The critical-violation decline is concentrated among firms whose critical-violation density in the month before financing exceeds the treated-firm median: their decline is 3.9 per 1,000 lines larger than that of other treated firms ($p = 0.04$; Column (2)). Defining the groups by the six-month pre-financing average gives a difference of 3.6 ($p = 0.03$). Appendix N reports these alternatives and sensitivity to excluding firms with the highest initial violation densities. Firms raising

rounds above the treated-firm median show no statistically significant difference in the critical-violation decline ($p = 0.52$; Column (3)).

Table 3: Difference-in-Differences Estimates for Critical Violations and All Violations

| | **Critical Violations** | | | **Violations** | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Treated × Post | −1.85** | −0.28 | −2.29** | −4.74** | −4.60 | −3.81** |
| | (0.86) | (0.39) | (1.15) | (2.02) | (2.98) | (1.72) |
| Post × High Violations (=1) | | −1.31*** | | | −2.17** | |
| | | (0.32) | | | (1.00) | |
| Treated × Post × High Violations (=1) | | −3.87** | | | −0.81 | |
| | | (1.89) | | | (3.74) | |
| Post × Large Round | | | 0.02 | | | −0.58 |
| | | | (0.43) | | | (0.86) |
| Treated × Post × Large Round | | | 1.00 | | | −1.74 |
| | | | (1.57) | | | (4.23) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 487,577 | 477,107 | 487,577 | 487,577 | 477,107 | 487,577 |
| $R^2$ | 0.88 | 0.89 | 0.88 | 0.86 | 0.86 | 0.86 |

This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Treated × Post indicator. Critical Violations are severe SonarQube rule violations per 1,000 lines of code. Violations (All) aggregate rule violations across all severity levels per 1,000 lines of code. See Appendix A, Table 19, for full variable definitions.

Parenthetical percentages express each decline relative to the corresponding treated pre-treatment mean, using estimates and arithmetic means over observed treated firm–cohort–months −12 to −1 in the fitted estimation sample; each observed month receives equal weight. The means are 12.40 critical violations and 32.89 total violations per 1,000 lines.

High Violations is a dummy equal to one if the firm's pre-treatment critical violation density (at $t = -1$) exceeds the median across baseline firm–cohort observations of firms treated somewhere in the sample, and zero otherwise.

Large Round is a dummy equal to one if the VC deal size (first Seed or Series A round amount, in USD) exceeds the treated-firm median, and zero otherwise.

*Note:* Standard errors in parentheses. Stars report unadjusted pointwise tests: * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Figure 2 traces the timing of this improvement. The pre-investment estimates show no systematic divergence between treated and control firms, while negative coefficients emerge around financing and become larger over the post-investment quarters. The timing supports a shift in development practices following investor entry: it is not the pattern predicted by a gradual decline in critical-violation density

already under way before the round. The fact that the improvement develops over subsequent quarters is also consistent with engineering practices taking time to affect the accumulated codebase.

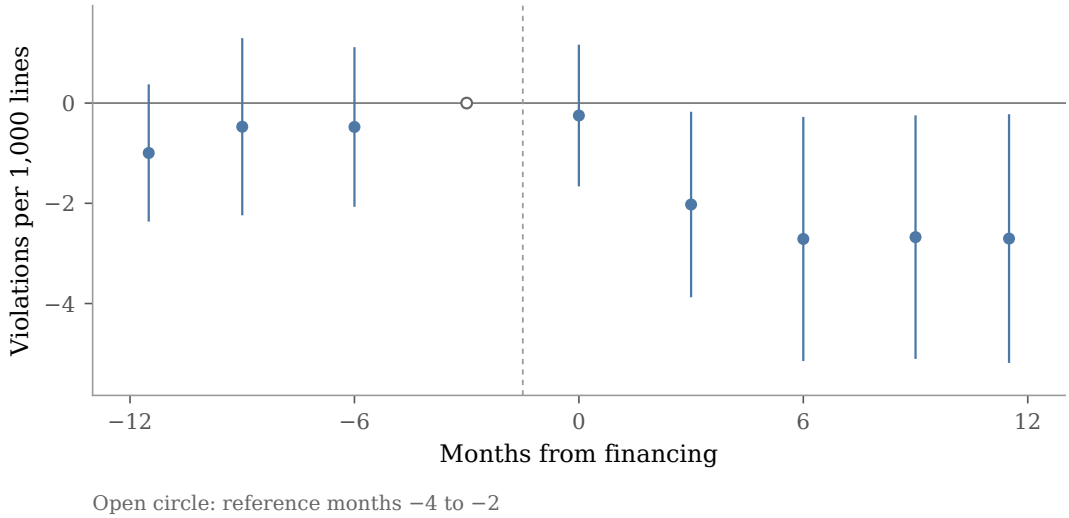



Figure 2: Critical Rule Violations per 1,000 Lines of Code. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

Figure 3 shows that the decline extends beyond critical violations and grows over the post-financing period. The pre-financing estimates also suggest some decline in violation density, although they are noisy. When we match firms on their pre-financing change in violation density, the pre-financing estimates move closer to zero while the post-financing decline remains (Figure 17).

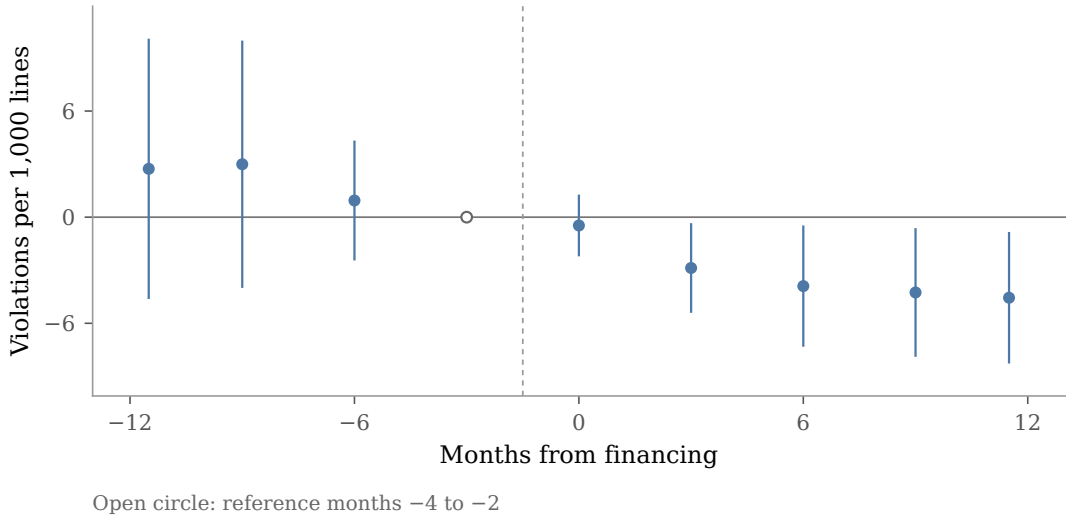



Figure 3: Rule Violations (All Severities) per 1,000 Lines of Code. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

## 3.3 Maintainability Issues and Security Hotspots

We next examine whether first VC entry improves the maintainability of software: the ease with which a codebase can be repaired, modified, or extended. Maintainability matters economically because an engineering shortcut that accelerates one release can increase the work required for every subsequent change. SonarQube flags code structures associated with this burden. Alongside the overall measure under the traditional classification, we examine high- and low-severity maintainability issues under the alternative classification described in Appendix M.4.

Maintainability violations decline by 4.8 per 1,000 lines of code after financing relative to controls (15.6% of the pre-treatment mean; $p = 0.01$; Table 4, Column (1)). When controls are matched on their pre-financing change in maintainability density, the decline is 3.3 per 1,000 lines ($p < 0.01$). High-severity issues fall by 1.84 per 1,000 lines (15.0% of the pre-treatment mean; $p = 0.03$; Column (4)). Low-severity issues decline by 0.73 per 1,000 lines (10.0% of the pre-treatment mean; $p = 0.10$; Column (7)). The improvement thus includes the code structures that make future development most costly.

Table 4: Difference-in-Differences Estimates for Maintainability Violations

| | **Maintainability Violations (All)** | | | **Maintainability Issues (High)** | | | **Maintainability Issues (Low)** | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Treated × Post | −4.80** | −4.71 | −3.92** | −1.84** | −0.27 | −2.29** | −0.73 | −0.68* | −1.03* |
| | (1.99) | (2.98) | (1.67) | (0.86) | (0.39) | (1.14) | (0.45) | (0.35) | (0.54) |
| Post × High Violations (=1) | | −1.99** | | | −1.32*** | | | −0.03 | |
| | | (0.95) | | | (0.32) | | | (0.51) | |
| Treated × Post × High Violations (=1) | | −0.67 | | | −3.86** | | | −0.14 | |
| | | (3.66) | | | (1.89) | | | (0.68) | |
| Post × Large Round | | | −0.67 | | | 0.01 | | | −0.62* |
| | | | (0.81) | | | (0.43) | | | (0.35) |
| Treated × Post × Large Round | | | −1.58 | | | 1.03 | | | 1.11** |
| | | | (4.14) | | | (1.57) | | | (0.51) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 487,577 | 477,107 | 487,577 | 487,577 | 477,107 | 487,577 | 487,577 | 477,107 | 487,577 |
| $R^2$ | 0.85 | 0.85 | 0.85 | 0.88 | 0.88 | 0.88 | 0.84 | 0.85 | 0.84 |

This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Treated × Post indicator. Maintainability Violations (All) use the traditional classification. The high- and low-severity measures use the alternative classification described in Appendix M.4 and are not a breakdown of the traditional overall count. All three measures are counts per 1,000 non-comment lines of code; high and low distinguish the assessed impact on maintainability. See Appendix A, Table 19, for full variable definitions.

See Table 3 for the definitions of the High Violations and Large Round dummies.

Parenthetical percentages use estimates and arithmetic means over observed treated firm–cohort–months −12 to −1 in the fitted estimation sample; each observed month receives equal weight. The means are 30.83 for all maintainability violations, 12.29 for high-severity issues and 7.26 for low-severity issues, per 1,000 lines.

*Note:* Standard errors in parentheses. Stars report unadjusted pointwise tests: * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Figure 4 traces the maintainability response around financing. In the baseline comparison, the early pre-financing estimates are noisy (Appendix D, Figure 18). The figure therefore shows the matched comparison, which leaves 193 of the 273 treated firms. In this comparison, the pre-financing estimates are close to zero. Figure 5 traces high-severity issues.

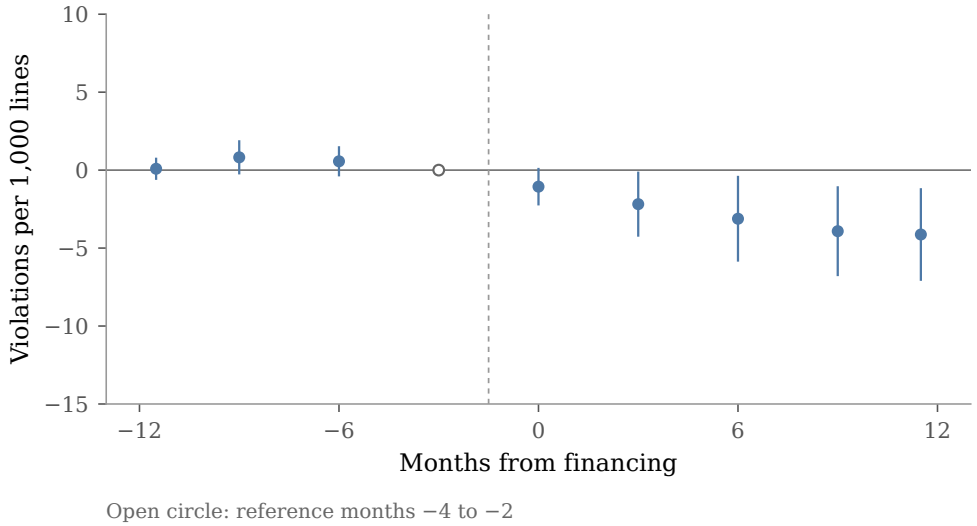



Figure 4: Maintainability Violations per 1,000 Lines of Code, Matched on Pre-Financing Density Change. Controls are matched on the change in maintainability violations per 1,000 lines between months $[-12, -11]$ and $[-4, -2]$ (193 treated firms in 90 cohorts; 335,092 stacked observations). Appendix D reports the matching procedure and the baseline comparison. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

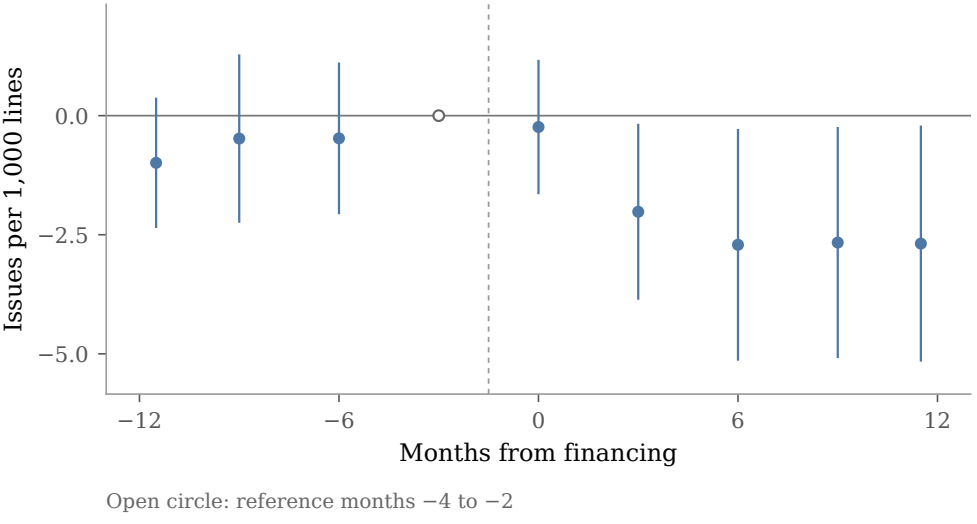



Figure 5: Maintainability Issues (High-Severity) per 1,000 Lines of Code. This measure uses the alternative classification described in Appendix M.4. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

Table 5: Difference-in-Differences Estimates for Security Hotspots

| | **Security Hotspots** | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Treated × Post | −1.18 | −2.08 | −0.02 |
| | (1.09) | (1.93) | (0.09) |
| Post × High Violations (=1) | | 0.10 | |
| | | (0.09) | |
| Treated × Post × High Violations (=1) | | 2.05 | |
| | | (1.92) | |
| Post × Large Round | | | −0.10 |
| | | | (0.11) |
| Treated × Post × Large Round | | | −2.59 |
| | | | (2.38) |
| Cohort × Firm FE | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes |
| **# Obs** | 487,577 | 477,107 | 487,577 |
| $R^2$ | 0.72 | 0.70 | 0.72 |

This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Treated × Post indicator. Security Hotspots measure security-sensitive code segments requiring manual audit per 1,000 lines of code; unlike rule violations, hotspots represent potential risk areas requiring human assessment. See Appendix A, Table 19, for full variable definitions.

See Table 3 for the definitions of the High Violations and Large Round dummies.

Parenthetical percentages use estimates and arithmetic means over observed treated firm–cohort–months −12 to −1 in the fitted estimation sample; each observed month receives equal weight. The mean is 2.16 hotspots per 1,000 lines.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

We also examine *Security Hotspots*. Security Hotspots identify security-sensitive code segments that require human review to determine whether a vulnerability exists. Table 5 reports the estimates.

The baseline estimate in Column (1) is a decline of 1.18 hotspots per 1,000 lines of code (54.7% of the pre-treatment mean), but is not statistically significant ($p = 0.27$). The large-round interaction

is also not statistically significant ($p = 0.27$). We therefore find no systematic change in security hotspots after first VC entry.

## 3.4 Bugs and Reliability

We next examine bugs and reliability issues, which concern incorrect behavior or failures during software execution. SonarQube's Bugs measure identifies code patterns that are likely to produce an incorrect result or unexpected behavior. The reliability measures capture related risks under the alternative classification, with high-severity issues carrying greater potential impact and low-severity issues carrying smaller risks. Both assess problems identifiable from source code, allowing us to examine whether the gains in maintainability extend to the software's expected behavior in use.

Table 6: Difference-in-Differences Estimates for Bugs and Reliability Issues

| | **Bugs** | | | **Reliability Issues (High)** | | | **Reliability Issues (Low)** | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Treated × Post | −0.01 | −0.01 | 0.04 | −0.01 | 0.00 | 0.00 | −0.12 | −0.31 | −0.18 |
| | (0.12) | (0.09) | (0.15) | (0.02) | (0.01) | (0.02) | (0.16) | (0.25) | (0.22) |
| Post × High Violations (=1) | | −0.23** | | | 0.01 | | | −0.08 | |
| | | (0.11) | | | (0.02) | | | (0.10) | |
| Treated × Post × High Violations (=1) | | −0.02 | | | −0.02 | | | 0.43 | |
| | | (0.27) | | | (0.03) | | | (0.28) | |
| Post × Large Round | | | 0.10 | | | 0.01 | | | −0.03 |
| | | | (0.08) | | | (0.01) | | | (0.07) |
| Treated × Post × Large Round | | | −0.18 | | | −0.02 | | | 0.17 |
| | | | (0.25) | | | (0.03) | | | (0.22) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 487,577 | 477,107 | 487,577 | 487,577 | 477,107 | 487,577 | 487,577 | 477,107 | 487,577 |
| $R^2$ | 0.90 | 0.90 | 0.90 | 0.51 | 0.51 | 0.51 | 0.90 | 0.90 | 0.90 |

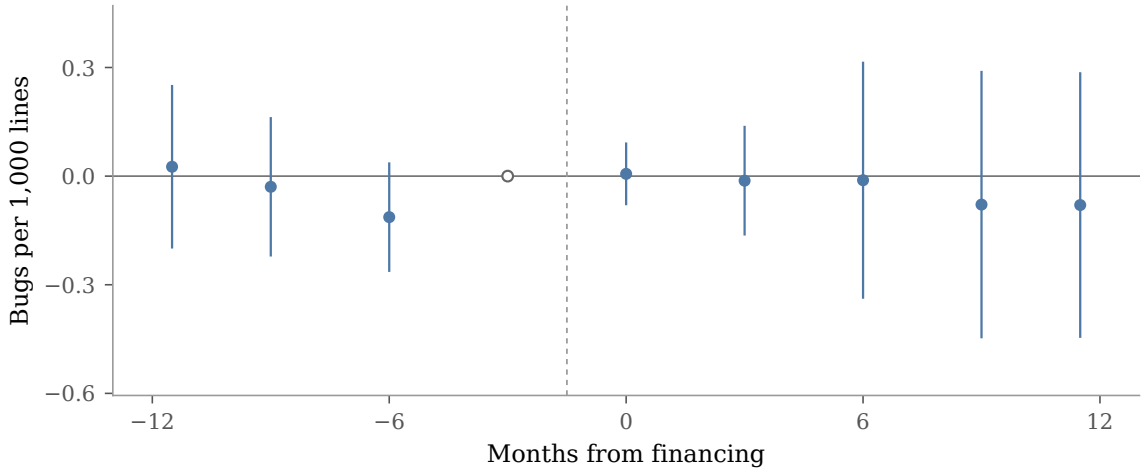
This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Treated × Post indicator. Bugs are patterns classified by SonarQube as likely to cause incorrect or unexpected behavior per 1,000 lines of code. High- and Low-Severity Reliability Issues capture stability-related rule violations at the stated severity. Bugs use the traditional classification; the reliability measures use the alternative classification described in Appendix M.4 and are not a breakdown of Bugs. See Appendix A, Table 19, for full variable definitions.

See Table 3 for the definitions of the High Violations and Large Round dummies.

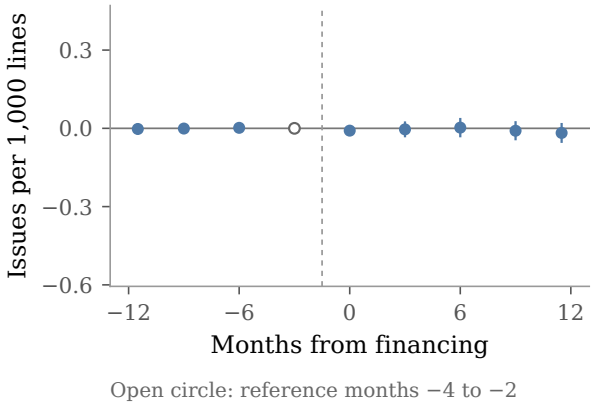
Parenthetical percentages use estimates and arithmetic means over observed treated firm–cohort–months −12 to −1 in the fitted estimation sample; each observed month receives equal weight. The Bugs mean is 1.82 per 1,000 lines.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

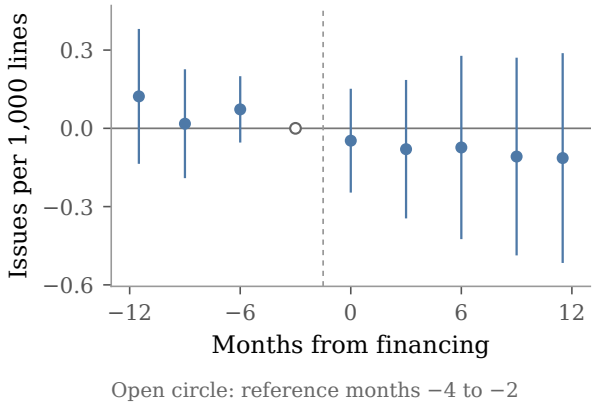
We find no statistically significant improvement in bugs or reliability (Table 6). The Bugs estimate in Column (1) shows no statistically significant change ($p = 0.95$). The high- and low-severity reliability estimates in Columns (4) and (7) are also not statistically significant ($p = 0.70$ and $p = 0.45$, respectively). Figure 6 presents the corresponding event studies. Together with the security-hotspot results, these findings indicate that the measurable gains are concentrated in rule compliance and maintainability. Firms face a lower burden of modifying and extending the product, while we detect no comparable change in the code patterns associated with software failures.




(a) Bugs per 1,000 Lines of Code




(b) Reliability Issues (High Severity) per 1,000 lines of code




(c) Reliability Issues (Low Severity) per 1,000 lines of code

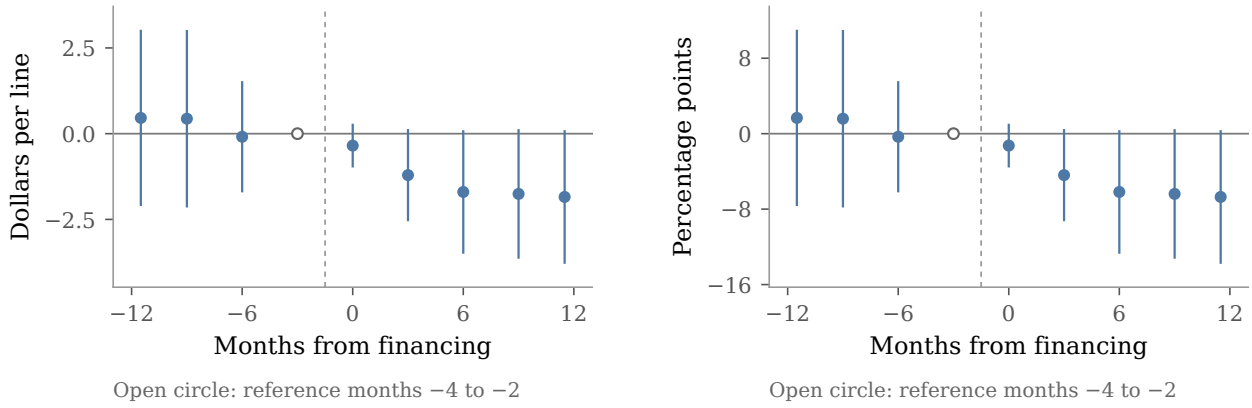
Figure 6: Bugs and Reliability Issues. Bugs use the traditional classification; the reliability measures use the alternative classification described in Appendix M.4. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

## 3.5 Technical Debt

Better coding discipline can reduce the future cost of maintaining software. We measure this liability with *Technical Debt*, which converts SonarQube's estimated time required to fix maintainability problems into dollars using the calibration in Section 2.5. The *Technical Debt Ratio* expresses the same liability as a percentage of the software asset's estimated replacement cost. Both measures translate engineering problems into an estimated maintenance burden.

Table 7, Column (1), shows that technical debt declines by \$1.57 per line of code after financing relative to controls (12.0% of the pre-treatment mean; $p = 0.04$). Correspondingly, the Technical Debt Ratio falls by 5.7 percentage points (12.0% of the pre-treatment mean; $p = 0.04$; Column (4)). Under the common calibration, this ratio decline means that the estimated maintenance liability falls by \$5.69 for every \$100 of software replacement cost. The improvement therefore represents a meaningful reduction in the future work associated with maintaining a given amount of software, in addition to the decline in unweighted violation counts. Figure 7 shows how the two measures evolve around financing.

Because we examine several quality outcomes, we also adjust their p-values for multiple testing with the Holm method, which controls the probability of any false rejection across a group of outcomes. The declines in critical violations and technical debt per line each have an adjusted p-value of 0.06, and adding the other violation measures to the group leaves the adjusted values between 0.06 and 0.07 (Appendix Table 58).




(a) Technical Debt per Line of Code (\$)

(b) Technical Debt Ratio

Figure 7: Technical Debt Measures. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

Table 7: Difference-in-Differences Estimates for Technical Debt

| | **Technical Debt per LoC ($)** | | | **Technical Debt Ratio** | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Treated × Post | −1.57** | −1.45 | −1.25 | −5.69** | −5.28 | −4.54 |
| | (0.77) | (0.98) | (0.81) | (2.78) | (3.56) | (2.96) |
| Post × High Violations (=1) | | −0.52 | | | −1.91 | |
| | | (0.52) | | | (1.88) | |
| Treated × Post × High Violations (=1) | | −0.40 | | | −1.45 | |
| | | (1.46) | | | (5.30) | |
| Post × Large Round | | | −0.19 | | | −0.71 |
| | | | (0.36) | | | (1.32) |
| Treated × Post × Large Round | | | −0.60 | | | −2.16 |
| | | | (1.51) | | | (5.49) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 487,577 | 477,107 | 487,577 | 487,577 | 477,107 | 487,577 |
| $R^2$ | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 | 0.83 |

This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Treated × Post indicator. Technical Debt per Line of Code is the calibrated remediation cost in dollars per line. The Technical Debt Ratio is calibrated technical debt as a percentage of the firm's software assets, the replacement cost of its codebase. Section 2.5 defines both outcomes.

See Table 3 for the definitions of the High Violations and Large Round dummies.

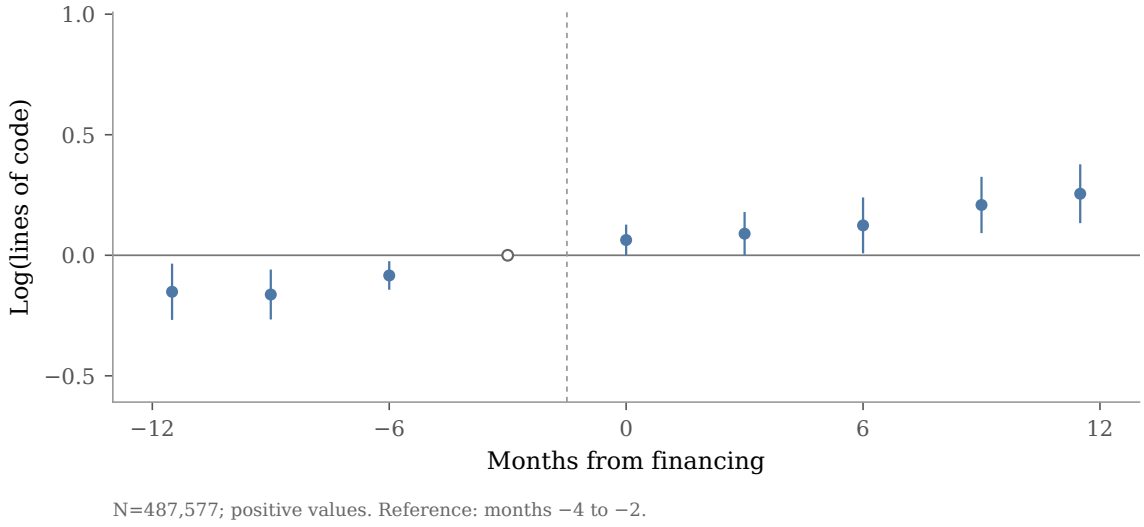
Parenthetical percentages use estimates and arithmetic means over observed treated firm–cohort–months −12 to −1 in the fitted estimation sample; each observed month receives equal weight. The means are $13.09 per line and 47.6% for the ratio. The ratio equals 40/11 times debt dollars per line on the same observations.

*Note:* Standard errors in parentheses. Stars report unadjusted pointwise tests: * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

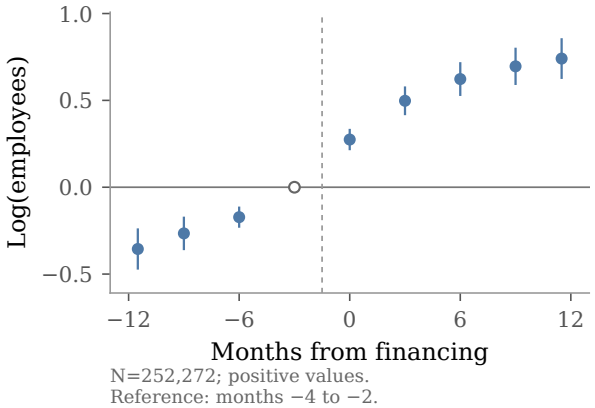
## 3.6 Firm Scale and Engineering Capacity

We next examine the expansion in firm scale and engineering capacity that accompanies first VC entry. This expansion provides the setting in which the quality improvements occur: firms are adding employees and extending their products while reducing violation density. Figure 8 plots the

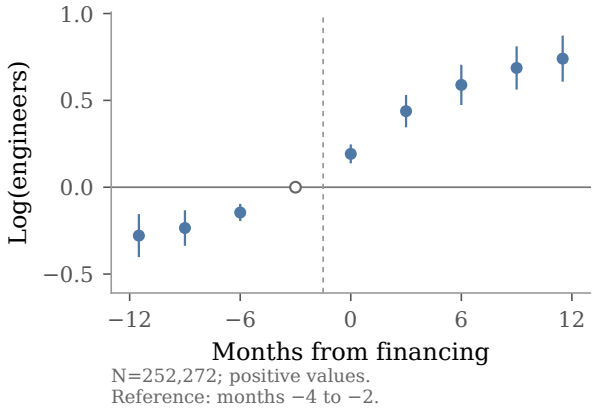
dynamic estimates for code volume in Panel (a), total headcount in Panel (b), and software-engineering headcount in Panel (c).




(a) Log(Lines of Code)




(b) Log(Total Employees)




(c) Log(Software Engineers)

Figure 8: Firm and Product Scaling. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

Organizational scale rises substantially after financing. By the end of the event window, the growth rates in total headcount and software-engineering headcount are each approximately 74 percentage points greater than in controls. Code volume also expands, although by less than headcount. This gap is consistent with the demands of building a larger engineering organization—including onboarding, review, and coordination—that need not translate immediately into proportionate additions to the codebase. It also cautions against treating the number of lines produced as a sufficient measure of engineering output.

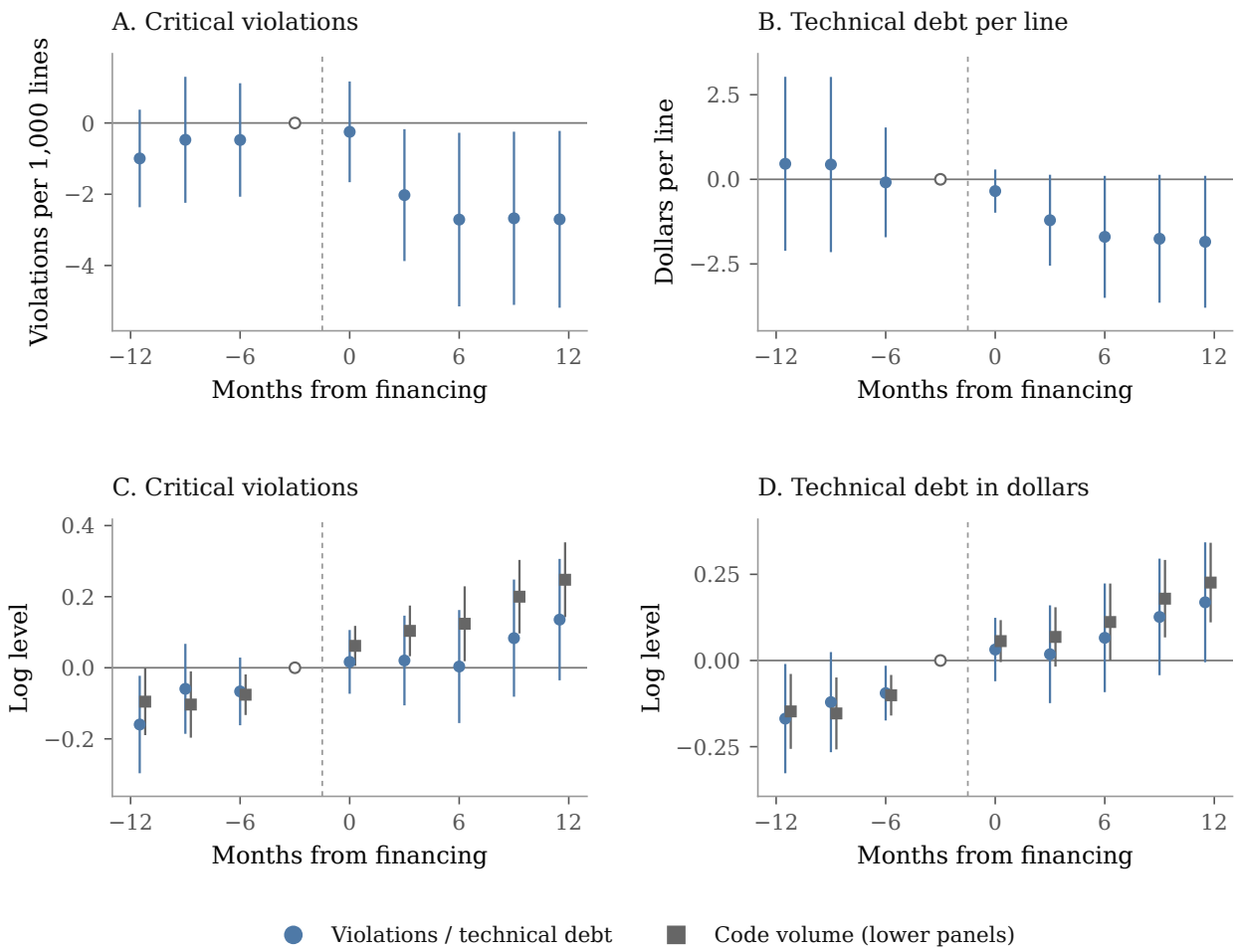



Figure 9: Engineering Quality During Codebase Expansion. Panels A–B show critical violations per 1,000 lines and technical debt in dollars per line. Panels C–D compare critical-violation counts and technical-debt levels with code volume, using the transformations defined in Table 19. Each lower-panel code-volume series uses the same fitted observations as its stock series. Log specifications exclude zero stock values. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

The pre-treatment coefficients show that treated firms were already expanding more rapidly in all three scale measures before financing. Parallel trends therefore do not hold for these outcomes, consistent with VC selection on fast-growing firms. The scaling estimates describe the resource expansion surrounding investor entry; the identification of our quality results depends on the trajectories of the engineering outcomes themselves.

Figure 9 shows how quality changes as the codebase expands. The upper panels report critical violations per 1,000 lines and technical debt per line; the lower panels compare total critical violations and total technical debt with code volume. After financing, total critical violations and total technical

debt grow by less than code volume relative to controls, so both fall per line of code. Appendix G discusses the relationship between these measures.

Growth and quality follow different trajectories before financing. If lower critical-violation density continued the same gradual improvement in firm capabilities that supported earlier growth, we would expect quality to improve before the round as well. Headcount and code volume diverge before financing, while the decline in critical violations emerges around investor entry. This separation in timing supports a change in development practices beyond the continuation of pre-existing growth. We test whether pre-existing growth explains the decline by matching treated firms to controls with similar pre-financing headcount growth in Section 4.1.

Rapid expansion also creates engineering challenges. More developers must coordinate changes to an increasingly complex product, and onboarding new engineers places demands on the existing team. Quality need not improve automatically as scale increases. The decline in violation density during this expansion is consistent with firms adopting practices that help them manage a larger development effort.

## 3.7 Capital and Governance: Mechanism Evidence

The results thus far show that first institutional VC entry is followed by lower violation density as firms expand. We next examine whether these improvements reflect the resources financing provides, the recruitment of senior technical talent, or changes in governance and development practices. The explanations imply different roles for investors. Additional capital can finance engineers, better tools, and technical-debt remediation; if these resources account for the gains, the improvement need not depend on how investors organize oversight. A governance channel instead predicts that quality responds to the monitoring and professionalization accompanying investment—including board oversight, operating advice, and more formal development procedures. We examine these predictions using investor structure and the composition of post-financing hiring.

We first exploit variation in investor count. In larger syndicates, each investor bears the private cost of monitoring while sharing its benefits with more co-investors (Lerner, 1994; Casamatta and Haritchabalet, 2007; Kang, Li, and Oh, 2022). If these dispersed incentives weaken oversight, firms financed by larger syndicates should experience smaller quality improvements. To test this prediction, we classify a round as a large syndicate if it has more than four investors, the treated-firm median. This yields 120 treated firms with large syndicates among the 260 with nonmissing counts. We interact this indicator with treatment and estimate:

$$M_{i,c,t} = \beta_1 VC_{i,c,t} + \beta_2 VC_{i,c,t} \times \text{Syndicate}_i + \beta_3 \text{Post}_{c,t} \times \text{Syndicate}_i + \chi_{c,i} + \delta_{c,t} + \varepsilon_{i,c,t}, \tag{8}$$

where $\beta_1$ is the estimate for the reference group and $\beta_1 + \beta_2$ is the estimate for rounds with large

syndicates. A positive $\beta_2$ indicates smaller quality improvements when more investors share the round, as predicted by weaker individual monitoring incentives.[5]

Table 8 reports the predicted attenuation in critical violations. In Column (3), the large-syndicate interaction is +3.0 ($p = 0.07$), leaving a net point estimate close to zero for larger syndicates. Thus, firms financed by fewer co-investors experience the larger estimated reduction in critical violations. The technical-debt and code-volume interactions are not statistically significant ($p = 0.20$ and $p = 0.38$, respectively).

Table 8: Investor Structure and Engineering Outcomes

| | **Critical Violations / kLoC** | | | **TD per LoC ($)** | | | **Log(Lines of Code)** | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Treated × Post | −1.85** | −1.87** | −3.11** | −1.57** | −1.59** | −2.25* | 0.23*** | 0.23*** | 0.22*** |
| | (0.86) | (0.87) | (1.42) | (0.77) | (0.77) | (1.22) | (0.05) | (0.05) | (0.07) |
| Post × Angel | | −0.21 | | | −0.23 | | | 0.01 | |
| | | (0.40) | | | (0.38) | | | (0.09) | |
| Treated × Post × Angel | | 1.71 | | | 2.46** | | | 0.00 | |
| | | (1.82) | | | (1.11) | | | (0.24) | |
| Post × Large Syndicate | | | −0.09 | | | −0.20 | | | 0.16*** |
| | | | (0.48) | | | (0.43) | | | (0.05) |
| Treated × Post × Large Syndicate | | | 3.00* | | | 1.74 | | | −0.10 |
| | | | (1.67) | | | (1.38) | | | (0.11) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 487,577 | 487,577 | 487,577 | 487,577 | 487,577 | 487,577 | 487,577 | 487,577 | 487,577 |
| $R^2$ | 0.88 | 0.88 | 0.88 | 0.83 | 0.83 | 0.83 | 0.93 | 0.93 | 0.93 |

This table reports heterogeneity in the treatment effect by investor type. Angel is an indicator equal to one if the firm's first qualifying capital round (Angel, Seed, or Series A) is an Angel (individual) round (four treated firms, three with observations before and after financing). Large Syndicate equals one if the first Seed or Series A treatment round has more than four investors, the treated-firm median. This definition classifies 120 treated firms as large syndicates among the 260 with nonmissing investor counts. The 14 treated firms with missing counts remain in the reference group, together with firms with four or fewer investors. Critical violations are counts per 1,000 non-comment lines; technical debt is calibrated dollars per line; code volume is log(LoC). Columns (1), (4), and (7) reproduce the baseline estimates. Columns (2), (5), and (8) introduce the angel interaction. Columns (3), (6), and (9) introduce the large-syndicate interaction. All specifications include Firm × Cohort and Month × Cohort fixed effects. See Appendix A, Table 19, for variable definitions.

*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

Columns (2), (5), and (8) provide a complementary comparison based on prior angel financing. Institutional VC contracts commonly allocate board seats and formal control rights to investors,

[5]The reference group includes firms at or below the cutoff and firms with missing investor counts, which the specification codes as zero.

whereas angel financing can rely more heavily on informal relationships and advice (Kaplan and Strömberg, 2003; Ibrahim, 2008; Chemmanur and Chen, 2014). This difference motivates examining firms whose first qualifying capital round was an Angel (individual) round, while retaining first Seed or Series A entry as the treatment event. Four treated firms meet this definition; three contribute observations before and after financing to the angel comparison. Their net critical-violation estimate is −0.2, and the technical-debt interaction is +2.5 ($p = 0.02$). The point estimates suggest smaller gains for these firms at subsequent institutional entry, but the estimates change substantially when individual firms are excluded (Appendix M.2). We treat this three-firm comparison as exploratory.

To examine quality at angel entry itself, we estimate a separate DiD centered on angel financing, censoring treated firms when they subsequently receive institutional funding. All four point estimates are positive: critical violations +0.13 ($p = 0.78$), total violations +2.26 ($p = 0.06$), maintainability violations +2.29 ($p = 0.05$), and technical debt +0.44 per line ($p = 0.11$). The critical-violation and technical-debt estimates are not statistically significant. The point estimates differ from the declines at institutional entry, but this comparison does not establish an investor-type difference. It includes only five angel-treated firms, five cohorts and 461 controls (31,678 stacked observations), some of which have other financing histories. Financing stage and amount may also differ. Appendix M.2 reports the intervals and individual-firm exclusions.

### Senior Technical Hiring

An alternative explanation is that VCs improve code quality by facilitating the hiring of senior technical talent. We examine this possibility using two measures constructed from LinkedIn employment histories. *Hired Technical Leader* identifies a CTO, VP of Engineering, or Head of Engineering hire during the first 12 calendar months after financing. *High Senior Ratio* identifies firms whose senior engineering hires as a share of all new hires over that window exceed the treated-firm median of 1/7. We restrict these comparisons to firms with LinkedIn coverage and estimate:

$$M_{i,c,t} = \beta_1 VC_{i,c,t} + \beta_2 VC_{i,c,t} \times H_i + \beta_3 \text{Post}_{c,t} \times H_i + \chi_{c,i} + \delta_{c,t} + \varepsilon_{i,c,t}, \tag{9}$$

where $H_i$ denotes the hiring-composition indicator. If senior technical recruitment accounts for the quality gains, firms that add this talent should improve more, producing a negative interaction. The comparison tests whether the observed pattern fits this prediction; the hiring measures themselves are recorded after financing.

The hiring comparisons do not show larger quality gains among firms that recruit senior technical talent (Table 9). For critical violations, the estimated change is 3.4 per 1,000 lines higher among firms hiring a new technical leader than among firms without such a hire ($p = 0.31$). The estimated change is also 1.6 per 1,000 lines higher among firms with a high senior-hire share than among those with a

low share ($p = 0.69$). Neither difference is statistically significant. Appendix F presents the hiring definitions and interaction confidence intervals.

Table 9: Engineering Quality by First-Year Hiring Composition

| | **Critical Violations** | | **All Violations** | | **TD per Line ($)** | | **Maintainability** | |
|---|---|---|---|---|---|---|---|---|
| | Leader | Senior | Leader | Senior | Leader | Senior | Leader | Senior |
| Treated × Post | −2.87 | −2.01 | −4.16 | −2.51 | −1.50 | −0.88 | −4.07 | −2.74 |
| | (3.13) | (3.49) | (4.05) | (4.47) | (1.98) | (2.47) | (3.85) | (4.35) |
| Post × Hiring Group | 0.20 | 0.03 | −3.33 | −2.22 | −1.44 | −0.74 | −2.43 | −1.89 |
| | (1.29) | (1.25) | (3.25) | (2.56) | (1.31) | (1.25) | (2.79) | (2.25) |
| Treated × Post × Hiring Group | 3.38 | 1.63 | 8.29 | 3.56 | 3.21 | 1.61 | 7.16 | 3.43 |
| | (3.30) | (4.07) | (5.14) | (5.06) | (2.57) | (2.92) | (4.75) | (4.85) |
| Hiring-group effect: base + interaction | 0.51 | −0.38 | 4.12 | 1.04 | 1.71 | 0.73 | 3.09 | 0.69 |
| | (2.00) | (2.51) | (5.07) | (4.35) | (2.28) | (1.94) | (4.54) | (3.94) |
| Firm × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 31,925 | 26,451 | 31,925 | 26,451 | 31,925 | 26,451 | 31,925 | 26,451 |
| **# Firms** | 161 | 159 | 161 | 159 | 161 | 159 | 161 | 159 |
| **# Treated Firms** | 137 | 135 | 137 | 135 | 137 | 135 | 137 | 135 |

Critical, total and maintainability violations are counts per 1,000 lines; technical debt is the calibrated remediation cost in dollars per line. Leader equals one for a new CTO, VP of Engineering or Head of Engineering hired from the cohort event date up to, but excluding, its 12-month anniversary. Senior equals one when senior engineering hires as a share of all new hires over that window exceed the treated-firm median of 1/7. Both indicators use the cohort event date for treated and control firms. The leader and senior comparisons have separate LinkedIn coverage sets, each with 63 cohorts. Cohorts retain both treatment arms after sample restrictions and fixed-effect singleton removal. Group effects sum the treatment and interaction coefficients, using their covariance for the standard error.
*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

We next examine hiring composition as an outcome. For technical and nontechnical leadership, we compare the probability of at least one qualifying hire over two equally long periods: months $[-12, -1]$ and $[0, 11]$ relative to financing. We retain firm–cohort pairs with observations throughout both periods. The software-engineering share and mean seniority of new hires remain quarterly outcomes. Table 10 reports these comparisons, with cohort-by-firm fixed effects and either a period effect or cohort-by-quarter effects.

The software-engineering share of new hires rises by 5.8 percentage points (19.0% of the pre-treatment mean; $p = 0.04$). Changes in the probability of a technical- or nontechnical-leadership hire are not statistically significant ($p = 0.89$ and $p = 0.32$, respectively), nor is the change in the mean

seniority of newly hired nonmanagerial software engineers ($p = 0.26$). Financing therefore shifts recruitment toward engineering.

Table 10: Hiring Composition After First VC Entry

| | **Pre/Post 2×2** | | **Quarterly Panel** | |
|---|---|---|---|---|
| | Tech. Leader (pp) | Non-tech. Leader (pp) | SWE Share (pp) | SWE Seniority |
| | (1) | (2) | (3) | (4) |
| Treated × Post | 0.64 | −6.90 | 5.78** | −0.11 |
| | (5.06) | (6.99) | (2.90) | (0.10) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes |
| Post FE | Yes | Yes | — | — |
| Quarter × Cohort FE | — | — | Yes | Yes |
| **# Obs** | 8,298 | 8,298 | 31,339 | 20,889 |
| **# Firms** | 218 | 218 | 260 | 217 |
| $R^2$ | 0.74 | 0.75 | 0.42 | 0.49 |

This table reports the coefficient on Treated × Post from a DiD specification for four hiring-composition outcomes constructed from LinkedIn position histories. Columns (1) and (2) compare two periods for each firm–cohort pair—pre = relative month ∈ [−12, −1] and post = relative month ∈ [0, 11]—and estimate $Y_{i,c,p} = \beta \cdot VC_{i,c,p} + \chi_{c,i} + \delta_p + \varepsilon_{i,c,p}$. $Y_{i,c,p}$ is the corresponding leadership-hiring indicator. We retain firm–cohort pairs observed in every month of both 12-month periods, exclude fixed-effect singletons, and require both treated and control firms in each cohort. Each leadership comparison contains 218 firms, 112 treated firms and 63 cohorts. Leadership coefficients and standard errors in Columns (1) and (2) are displayed in percentage points. *Tech. Leader* is an indicator equal to one if the firm hired a CTO, VP of Engineering, Head/Director of Engineering, or VP of Technology during the period. *Non-tech. Leader* is the analogous indicator for a nontechnical executive (CEO, COO, CFO, CMO, CPO, CRO, CCO, or a head/VP/director of a non-engineering function).

Columns (3) and (4) estimate the quarterly stacked DiD, $M_{i,c,t} = \beta \cdot VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \varepsilon_{i,c,t}$, at firm–cohort-quarter frequency. *SWE Share* is the share of distinct new hires in the quarter classified as software engineers in nonmanagerial roles; its coefficient and standard error are displayed in percentage points. *SWE Seniority* is the mean LinkedIn seniority (1–7 scale) of those hires. Both ratios are missing when the denominator is zero, and firms without LinkedIn coverage drop from all four columns rather than being coded as zero. See Appendix A, Table 19, for full variable definitions.

The relative change in SWE Share divides the estimate in Column (3) by the arithmetic mean over observed treated pre-treatment firm–cohort–quarters in that column's fitted sample (30.4%). Each quarter receives equal weight.

*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

### Mechanism Synthesis

The mechanism tests are more consistent with governance than with resources or senior hiring. Critical violations decline more in rounds with fewer co-investors, where each investor captures more of the benefit of monitoring. Firms that raised larger rounds or that hired senior technical talent in the year after the round show no statistically significant difference in the decline, although financing shifts recruitment toward engineering. The governance channel also predicts more formal development

procedures. Section 3.8 examines this prediction using the development histories, which record changes in procedures and in contributions from the existing organization, complementing evidence on VC involvement in portfolio-company operations (Gorman and Sahlman, 1989; Hellmann and Puri, 2002; Bernstein, Giroud, and Townsend, 2016).

## 3.8 Changes in the Development Process

We examine changes in development practices using individual code contributions from firms in the main estimation universe. Version-control histories record when engineers submit changes, how they document and integrate them, and which earlier changes last modified subsequently repaired lines. We include firms that commercialize multiple open-source projects and aggregate contributions across the repositories covered for each firm. Coverage varies with the data available for each outcome. Appendix L reports the construction rules, fitted samples, and comparisons estimated on identical observations.

Following VC entry, teams leave a more detailed record of how they develop software. To organize their work, teams can keep numbered records of tasks, reported bugs, and proposed changes. A commit message such as "Fix login error, #123" can then connect a change to the problem or discussion it addresses, helping others reconstruct what was done and why. The share of commit messages containing such references rises by 5.5 percentage points (14.1% of the pre-treatment mean; $p < 0.01$; Table 11). Teams can also collect proposed changes in a pull request, allowing others to discuss them before they are incorporated into the main software product. Appendix Table 38 shows that references to pull requests rise by 6.2 percentage points ($p < 0.01$). References to merges or squashes—two ways of incorporating separately developed changes into the main product—rise by 6.5 percentage points ($p < 0.01$). The records also more frequently identify collaboration: the share of commits explicitly crediting a coauthor rises by 1.3 percentage points (44.4% of the pre-treatment mean; $p < 0.01$; Figure 10, Panel A). Together, these patterns indicate more systematic documentation of the work being undertaken, how changes are incorporated, and who contributed to them.

This greater emphasis on documentation also extends to copyright and licensing compliance. Table 11 reports a 1.9-percentage-point increase in the share of commits carrying a developer sign-off (25.2% of the pre-treatment mean; $p = 0.05$). A sign-off is a named declaration that the developer is entitled to contribute the code, either because it is their own work to provide or because they have permission to contribute someone else's work under the project's license. The procedure originated in the Linux community during litigation over allegations that proprietary code had been incorporated without permission, establishing a documented trail of contributions and assurances about their legitimate use (The Linux Foundation, 2026). Projects can enforce this requirement through automated checks that prevent contributions from being accepted until the declarations are supplied (Dapr Authors, 2026). The increase after financing is consistent with professionalization of the procedures

governing the legal acceptance of code, alongside the more systematic documentation of development work.

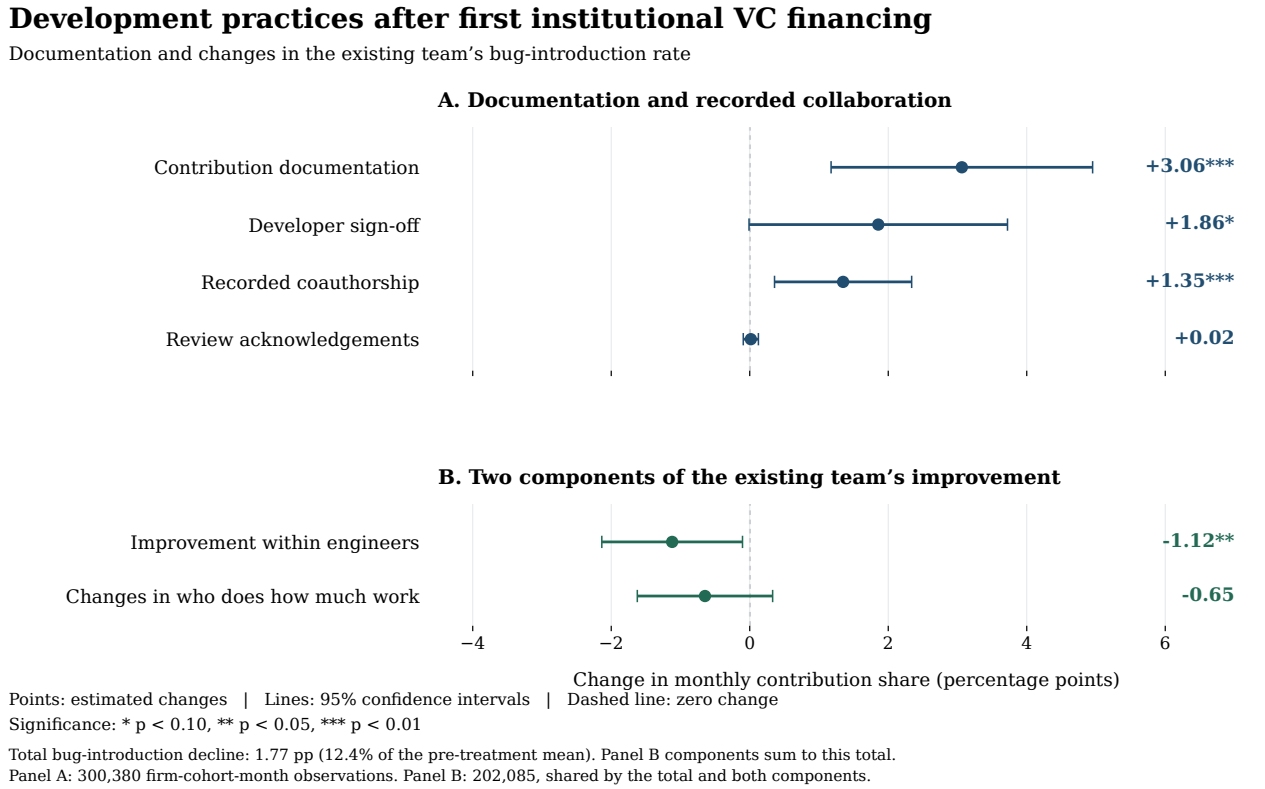



Figure 10: Development Practices and Bug Introduction Among Existing Engineers. Changes after first institutional financing, in percentage points. Panel A reports changes in the share of contributions documenting collaboration or developers' assurances about their right to supply the code. The combined measure includes sign-offs, coauthorship, review, acknowledgement and approval; these markers can overlap. Panel B divides the 1.8-percentage-point decline in bug introduction among engineers active before financing (12.4% of their pre-treatment mean) into improvement within engineers and changes in who does how much work. All comparisons include firms with multiple listed commercialized projects and pool their covered repositories over months $[-12, 12]$. Tables 11 and 41 report construction details and fitted samples. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

Table 11: Engineering Process Measures After First VC Entry

| Measure | Estimate (pp) | $p$-value | # Obs | # Firms | # Treated |
|---|---|---|---|---|---|
| Contribution documentation (combined) | 3.06*** (0.95) | < 0.01 | 300,380 | 371 | 213 |
| Developer sign-offs | 1.86* (0.94) | 0.05 | 300,380 | 371 | 213 |
| Coauthor credits | 1.35*** (0.50) | < 0.01 | 300,380 | 371 | 213 |
| Review acknowledgements | 0.02 (0.05) | 0.77 | 300,380 | 371 | 213 |
| Numbered issue references | 5.51*** (1.89) | < 0.01 | 300,380 | 371 | 213 |
| Headquarters-local weekend commits | −1.93*** (0.71) | < 0.01 | 171,116 | 273 | 199 |

Each row estimates Equation (7), with firm-by-cohort and event-month-by-cohort fixed effects. The estimation sample excludes fixed-effect singletons and cohorts without both treated and control observations. Each outcome is a monthly share of qualifying contributions, pooling numerator and denominator commits across the mapped projects covered for the firm and removing duplicate commits within firms. Firms with multiple listed commercialized projects are eligible.

Contribution documentation consists of standardized lines in commit messages that record review, sign-off, acknowledgement, approval or coauthorship; these markers can overlap. Sign-offs record contributors' declarations about their right to supply the code; coauthorship credits another contributor. The first five rows use the same observations in 89 cohorts. Numbered references match issue or task numbers in messages. The weekend row uses Saturday and Sunday at the company's reported headquarters, applying date-specific time-zone rules; it uses 85 cohorts. Fitted observations, firms and treated firms are reported for each row.

Coefficients and standard errors are percentage points unless otherwise stated. Standard errors in parentheses are clustered by original firm and cohort. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Displayed $p$-values use two decimals; stars use unrounded values.

Following VC entry, developers record a smaller share of code contributions on weekends. We identify Saturdays and Sundays using the local time zone of each company's reported headquarters. Table 11 shows a 1.9-percentage-point decline in the share of weekend commits ($p < 0.01$). Relative to the treated firms' pre-financing weekend share of approximately 13%, this represents a reduction of about 15%.

Weekend contributions are also more frequently linked to subsequent software repairs. We trace later bug fixes back to earlier contributions to the affected code and compare contributions by the same developer within the same software project. Using the same headquarters-local definition of weekends, weekend commits are 0.5 percentage points more likely than weekday commits to be linked to a bug fix within the following 730 days ($p < 0.01$; Appendix Table 40). Together, these findings are consistent with a more structured weekly development schedule after VC entry, with less activity on days when contributions are more often linked to later repairs.

The existing team becomes less likely to introduce bugs.[6] Among developers already active before financing, the probability that a contribution introduces a bug falls by 1.8 percentage points (12.4% of the pre-treatment mean; $p = 0.03$; Table 41). Decomposing this decline (Figure 10,

[6]This measure traces later bug fixes to the earlier contributions that last modified the repaired lines. It differs from SonarQube's Bugs measure in Section 3.4, which counts code patterns likely to cause failures.

Panel B), improvements within the same engineers account for 1.1 points ($p = 0.03$), while changes in who does how much work account for 0.6 points, a component that is not statistically significant ($p = 0.19$). The latter reflects a shift in activity toward engineers with lower bug-introduction rates.

Together, these findings indicate that institutional financing professionalizes how the existing team develops software: firms document copyright and licensing assurances, record collaboration more systematically, and concentrate less activity on weekends, while existing engineers become less likely to introduce bugs.

# 4 Robustness

## 4.1 Matching on Pre-Treatment Growth

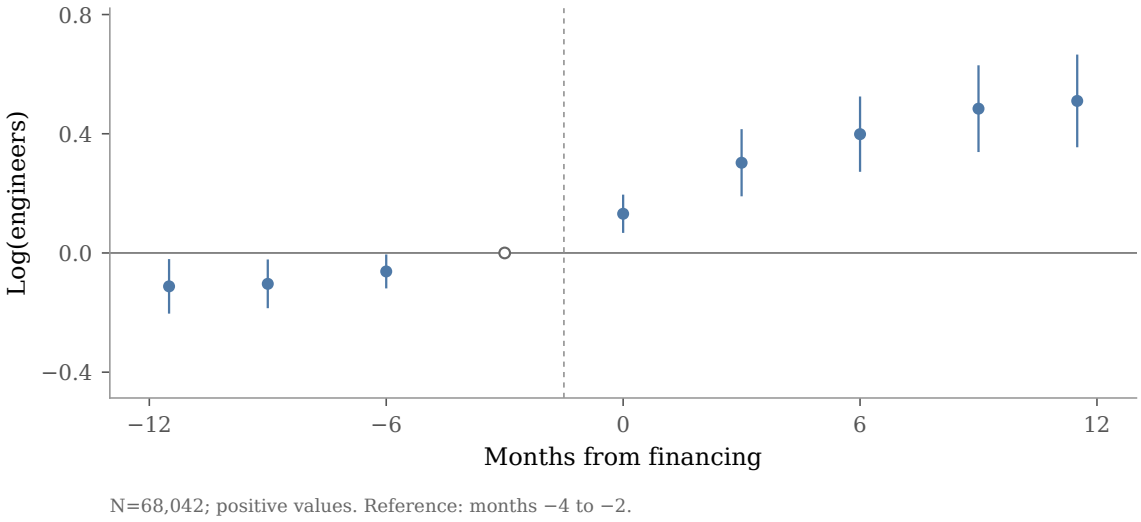
A concern with the baseline comparison is that VC-backed firms are already growing faster before financing, as Figure 8 shows for both headcount and code volume. Faster growth may accompany managerial capabilities or product demand that also improve engineering outcomes. We address this concern by matching treated firms to controls with similar pre-treatment headcount growth. The resulting comparison asks whether quality improves after first VC entry even relative to other fast-growing startups that followed comparable trajectories but remained without institutional financing throughout the event window.

Within each cohort, we retain controls whose pre-treatment headcount growth is close to the average growth of the treated firms. We measure growth as the difference in log average headcount between relative months $[-4, -2]$ and $[-12, -11]$:

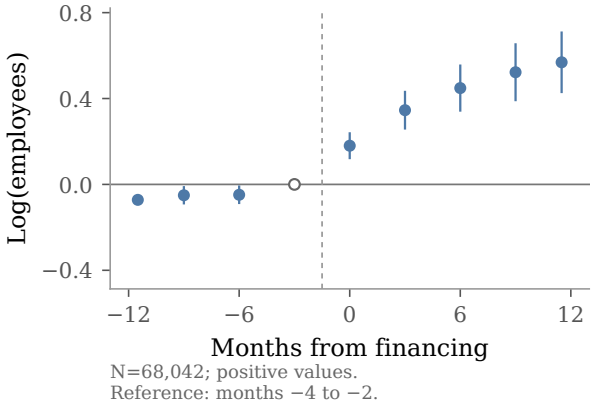
$$G_i^c = \ln\left(\frac{1}{3}\sum_{t=-4}^{-2} \text{Headcount}_{i,c,t}\right) - \ln\left(\frac{1}{2}\sum_{t=-12}^{-11} \text{Headcount}_{i,c,t}\right).$$

Appendix D details the matching procedure. Matching to the treated-group average yields controls that resemble the typical newly financed startup in the cohort, while preserving a broader comparison group than one-to-one matching.

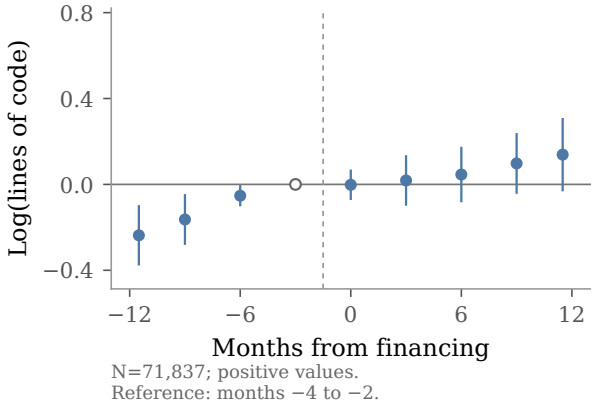
Matching on total-headcount growth improves the comparability of the growth trajectories, although pre-financing differences remain in engineering headcount and code volume (Figure 11). We therefore examine alternative matches based on engineering-headcount growth and on combinations of growth measures that include code volume (Appendices D and E). Matching directly on engineering-headcount growth yields a similar post-financing path: by the end of the event window, engineering-headcount growth is approximately 51 percentage points greater than in controls under the primary match and 53 percentage points greater under the engineering-growth match (Figure 16).




(a) Log(Software Engineers)




(b) Log(Total Employees)




(c) Log(Lines of Code)

Figure 11: Firm and Product Scaling (Matched Sample). Matched on pre-financing headcount growth within cohorts. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

The DiD coefficients likewise show continued expansion after financing. In the matched sample, total-headcount growth is approximately 44 percentage points greater than in controls, and software-engineering-headcount growth is approximately 42 percentage points greater (both $p < 0.01$; Table 12). Thus, the post-financing expansion persists when controls are selected for similar prior headcount growth. We next examine whether the engineering-quality improvements also persist in this comparison.

Table 12: Difference-in-Differences Estimates for Firm Scaling (Matched Sample)

| | **Log(Total Headcount)** | | | **Log(SWE Headcount)** | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Treated × Post | 0.44*** | 0.46*** | 0.54*** | 0.42*** | 0.46*** | 0.43*** |
| | (0.05) | (0.07) | (0.08) | (0.06) | (0.08) | (0.07) |
| Post × High Violations (=1) | | 0.00 | | | 0.03 | |
| | | (0.02) | | | (0.02) | |
| Treated × Post × High Violations (=1) | | −0.04 | | | −0.08 | |
| | | (0.10) | | | (0.11) | |
| Post × Large Round | | | 0.15*** | | | 0.10* |
| | | | (0.04) | | | (0.06) |
| Treated × Post × Large Round | | | −0.33*** | | | −0.08 |
| | | | (0.12) | | | (0.12) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 68,042 | 68,042 | 68,042 | 68,042 | 68,042 | 68,042 |
| $R^2$ | > 0.99 | > 0.99 | > 0.99 | > 0.99 | > 0.99 | > 0.99 |

This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Treated × Post indicator. The outcomes are log(Headcount) and log(SWE), using positive values of each measure. The sample is restricted to treated firms and control firms matched on pre-treatment headcount growth. See Appendix D for details on the matching procedure. See Table 3 for the definitions of the High Violations and Large Round dummies.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Improvements in software quality remain similar after matching firms on headcount growth before financing. Critical violations decline by 2.32 per 1,000 lines (14.4% of the pre-treatment mean; $p = 0.06$; Table 13, Column (1)). Total violations decline by 4.8 per 1,000 lines (13.1% of the pre-treatment mean; $p = 0.04$; Column (4)). Maintainability violations decline by 4.5 per 1,000 lines (12.9% of the pre-treatment mean; $p = 0.03$; Column (7)). The matching analysis covers 110 of the 274 treated firms, mainly because many firms lack the pre-financing data required for matching. For these same 110 firms, critical violations decline by 2.57 per 1,000 lines with all eligible controls (16.0% of the pre-treatment mean; $p = 0.08$) and by 2.32 with matched controls (Appendix Table 55).

Table 13: Difference-in-Differences Estimates for Engineering Quality (Matched Sample)

| | **Critical Violations** | | | **Violations (All)** | | | **Maintainability Violations** | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Treated × Post | −2.32* | 0.49 | −2.76 | −4.80** | −1.64 | −4.72* | −4.46** | −1.46 | −4.50* |
| | (1.22) | (0.83) | (1.71) | (2.37) | (1.21) | (2.44) | (2.11) | (1.16) | (2.31) |
| Post × High Violations (=1) | | −1.80*** | | | −4.58*** | | | −4.16*** | |
| | | (0.45) | | | (1.67) | | | (1.48) | |
| Treated × Post × High Violations (=1) | | −5.57* | | | −6.56 | | | −6.19 | |
| | | (3.23) | | | (4.55) | | | (4.23) | |
| Post × Large Round | | | −0.42 | | | −2.48* | | | −2.33* |
| | | | (0.51) | | | (1.33) | | | (1.22) |
| Treated × Post × Large Round | | | 1.35 | | | 1.67 | | | 1.85 |
| | | | (2.91) | | | (4.63) | | | (4.22) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 71,837 | 71,837 | 71,837 | 71,837 | 71,837 | 71,837 | 71,837 | 71,837 | 71,837 |
| $R^2$ | 0.88 | 0.88 | 0.88 | 0.87 | 0.87 | 0.87 | 0.86 | 0.87 | 0.86 |

This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Treated × Post indicator. The sample is restricted to treated firms and control firms matched on pre-treatment headcount growth. Critical Violations, Violations (All), and Maintainability Violations are measured per 1,000 lines of code. See Appendix D for details on the matching procedure and Appendix A, Table 19, for variable definitions.

See Table 3 for the definitions of the High Violations and Large Round dummies.

Parenthetical percentages use estimates and arithmetic means over observed matched treated firm–cohort–months −12 to −1 in the fitted estimation sample; each observed month receives equal weight. The means are 16.09 critical violations, 36.62 total violations and 34.54 maintainability violations per 1,000 lines.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Technical debt declines by $1.29 per line and the Technical Debt Ratio by 4.7 percentage points in the matched sample (Table 14). Each decline equals 8.7% of its outcome's pre-treatment mean. These estimates retain much of their baseline magnitude but are less precise, with standard errors roughly 28% larger than in the full sample. Neither estimate is statistically significant ($p = 0.19$ for each).

Table 14: Difference-in-Differences Estimates for Technical Debt (Matched Sample)

| | **Technical Debt per LoC** | | | **Technical Debt Ratio** | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Treated × Post | −1.29 | −0.09 | −0.93 | −4.70 | −0.34 | −3.39 |
| | (0.98) | (0.59) | (1.07) | (3.57) | (2.15) | (3.91) |
| Post × High Violations (=1) | | −1.58** | | | −5.74** | |
| | | (0.77) | | | (2.80) | |
| Treated × Post × High Violations (=1) | | −2.47 | | | −8.97 | |
| | | (2.12) | | | (7.70) | |
| Post × Large Round | | | −0.88 | | | −3.21 |
| | | | (0.54) | | | (1.95) |
| Treated × Post × Large Round | | | −0.18 | | | −0.65 |
| | | | (1.89) | | | (6.87) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 71,837 | 71,837 | 71,837 | 71,837 | 71,837 | 71,837 |
| $R^2$ | 0.82 | 0.82 | 0.82 | 0.82 | 0.82 | 0.82 |

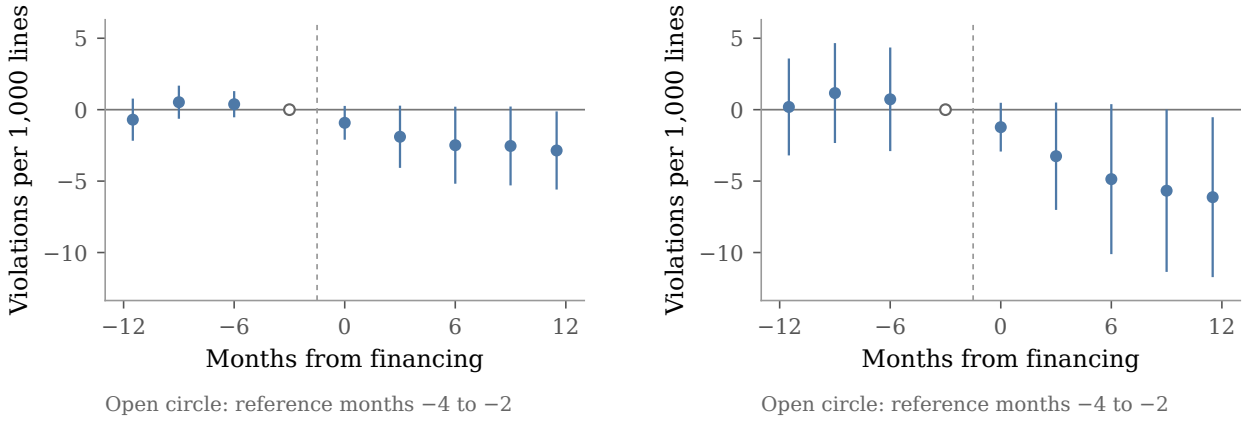
This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$ on firms matched by pre-treatment headcount growth. Technical Debt per Line of Code is calibrated dollars per line; the Technical Debt Ratio is calibrated technical debt as a percentage of software assets. See Section 2.5 and Appendix D.

See Table 3 for the definitions of the High Violations and Large Round dummies.

Parenthetical percentages use estimates and arithmetic means over observed matched treated firm–cohort–months −12 to −1 in the fitted estimation sample; each observed month receives equal weight. The means are $14.81 per line and 53.8% for the ratio.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

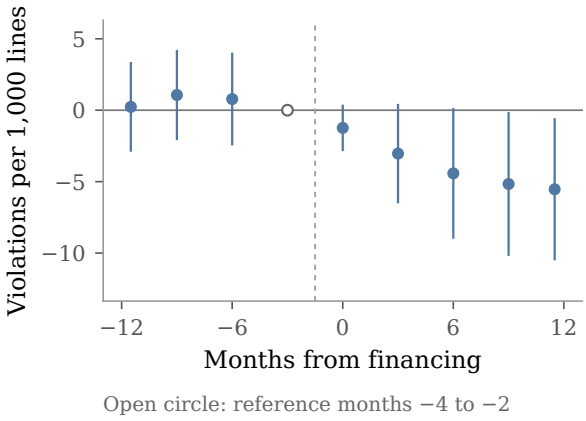
The matched event studies for critical, total, and maintainability violations (Figure 12) and technical debt per line and the Technical Debt Ratio (Figure 13) show a timing pattern similar to the baseline results. Pre-investment coefficients display no systematic divergence, and the post-financing estimates indicate lower violation densities and technical debt. Confidence intervals are wider in the smaller matched sample.




(a) Critical Violations per 1,000 LoC

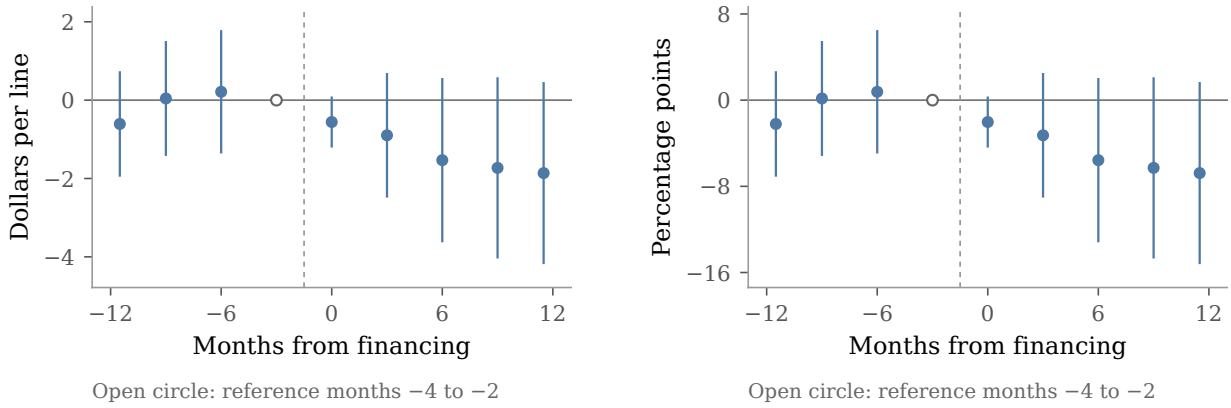



(b) Violations (All) per 1,000 LoC




(c) Maintainability Violations per 1,000 LoC

Figure 12: Engineering Quality (Matched Sample). Matched on pre-financing headcount growth within cohorts. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.




(a) Technical Debt per Line of Code

(b) Technical Debt Ratio

Figure 13: Technical Debt (Matched Sample). Matched on pre-financing headcount growth within cohorts. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

To assess whether the findings depend on how we select comparable controls, we compare estimates across 72 matching choices (Simonsohn, Simmons, and Nelson, 2020). We vary the growth measures used to select controls, whether controls are matched to the treated-group average or to individual firms, and how closely their prior growth must align. Across all 72 specifications, every critical-violation, total-violation, and maintainability coefficient is negative; 85%, 74%, and 79%, respectively, reach the 10% significance level, and 26%, 61%, and 61% reach the 5% level. Technical-debt coefficients are negative in 68 of the 72 specifications. None reaches the 5% level, and six reach the 10% level. Appendix E reports the specification curves and full details.

Overall, the matching analysis supports the robustness of the main quality findings. The declines in critical, total and maintainability violations remain, and technical debt retains most of its estimated decline, although that estimate is not statistically significant. The improvement therefore persists in a comparison that substantially narrows the observed differences in pre-financing growth, making those differences an unlikely explanation for the baseline quality pattern.

## 4.2 Later Financing Rounds

We next compare engineering quality around first entry with quality around later financing. First entry can introduce investor oversight and operating practices that persist into subsequent rounds (Hellmann and Puri, 2002; Kaplan and Strömberg, 2003). Later financing also supplies resources for engineers and tools. Comparing the responses shows whether the observed decline recurs at the later stages examined here.

We construct the later-round comparison by re-centering the stacked difference-in-differences

design on each firm's first completed Series B-or-later round recorded in PitchBook (Series B, C, D, and subsequent rounds). As in the main analysis, each cohort covers the 25 months from $e = -12$ through $e = +12$ relative to financing. For each Series B+ cohort, controls are firms with no Seed or Series A round through the end of the event window.

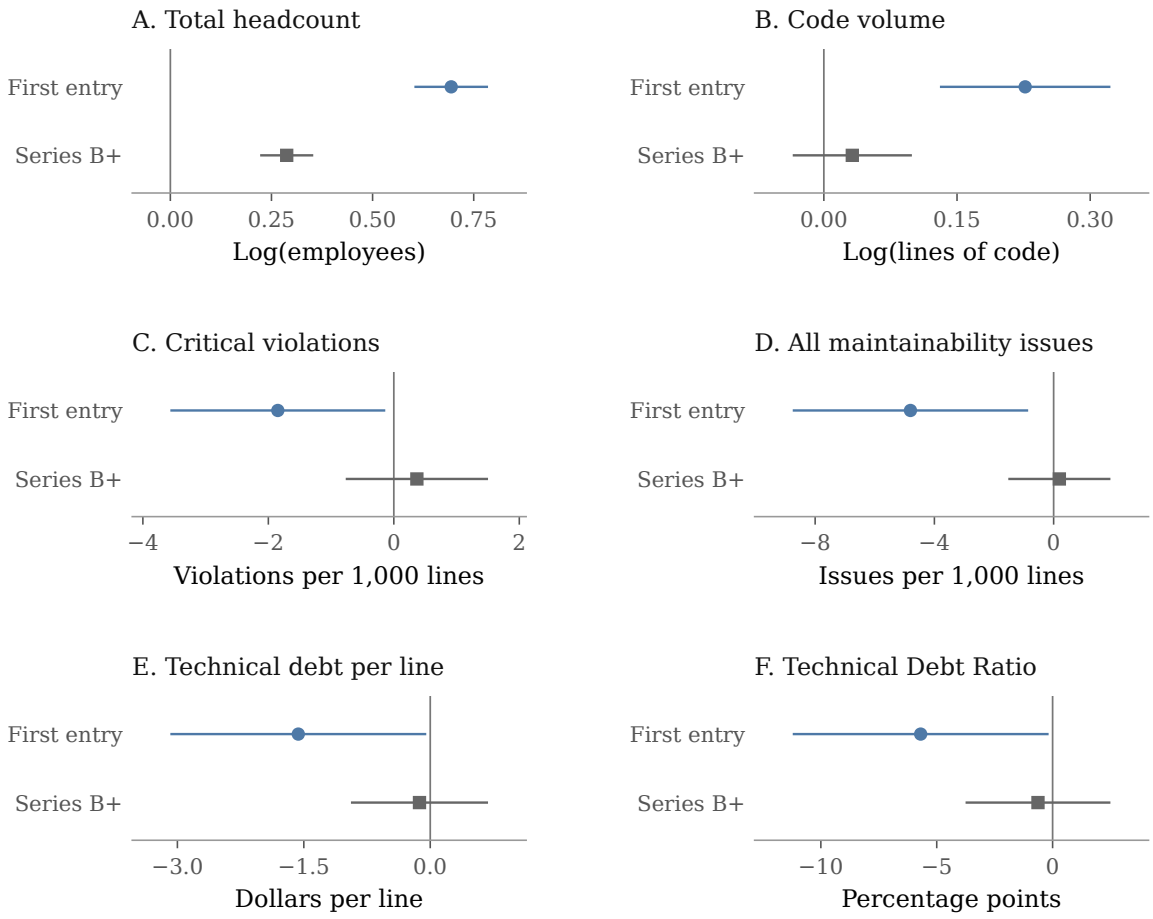
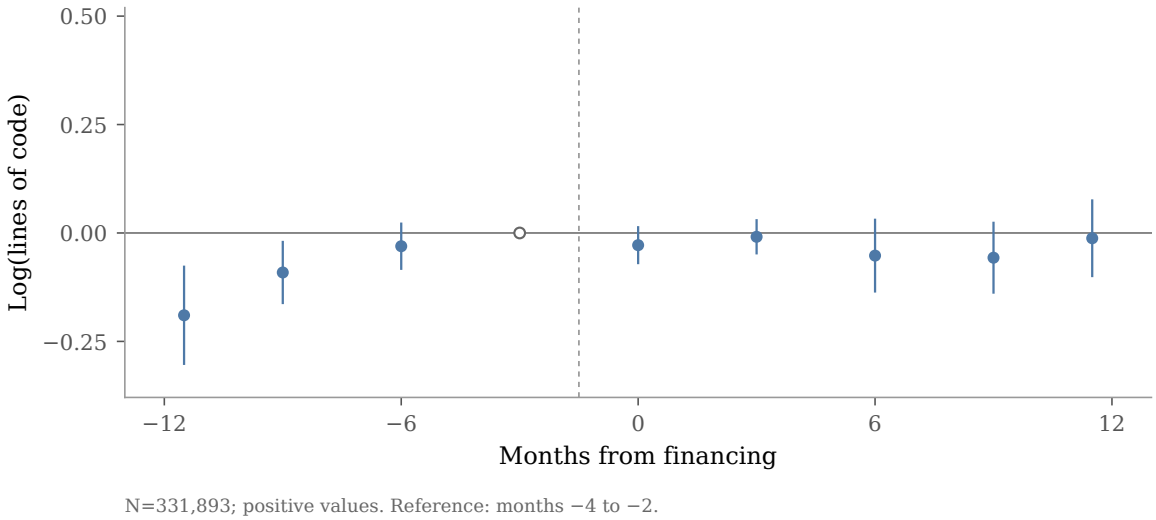



Figure 14: First Institutional Entry and Later Financing Rounds. Separate estimates within 12 months of first entry (circles) and the first Series B-or-later round (squares). Controls remain without a completed Seed or Series A round through each window. Units appear below each panel. Table 15 tests the differences between the two estimates for the quality outcomes. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

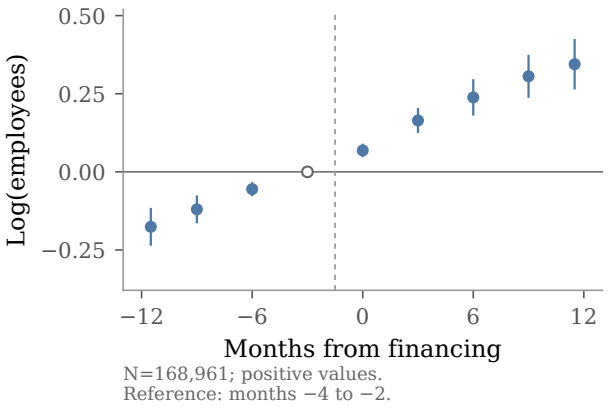
We reestimate Equation (7), with the treatment indicator equal to one for later-round recipients from their first Series B+ round onward. We retain cohort × firm and cohort × month fixed effects and cluster standard errors by cohort and firm. The outcomes are the same engineering-quality measures, together with code volume, total headcount, and software-engineering headcount.

Figure 14 compares the first-entry and later-round estimates on the same scale within each outcome. Headcount rises in both comparisons, whereas the later-round quality estimates are closer

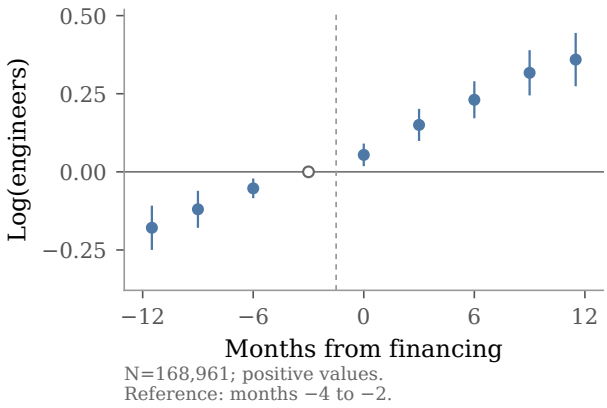
to zero. Figure 15 plots the later-round event studies for code volume in Panel (a), total headcount in Panel (b), and software-engineering headcount in Panel (c).




(a) Log(Lines of Code)




(b) Log(Total Employees)




(c) Log(Software Engineers)

Figure 15: Firm and Product Scaling (Series B+ rounds). Controls remain without a completed Seed or Series A round through the event window. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

Panels (b) and (c) show substantial headcount expansion after later financing, including a rise in the number of software engineers. Code volume in Panel (a), however, shows no statistically significant change ($p = 0.34$).

We find code-quality improvements at first VC entry, but not around Series B+ financing (Figure 14; Table 15). Later-round recipients show no statistically significant improvement relative to firms that remain without a Seed or Series A round through the event window. For critical violations,

the first-entry estimate is 2.2 violations per 1,000 lines lower than the later-round estimate (17.9% of the first-entry pre-treatment mean; $p < 0.01$). The difference between stages is also statistically significant for every other quality measure that improves at first entry (largest $p = 0.03$). Appendix J reports the full estimates and event studies.

Table 15: First Entry versus Later-Round Coefficients

| **Outcome** | **Baseline** (1) | **Series B+** (2) | **Difference** (1)−(2) | **SE** |
|---|---|---|---|---|
| Critical Violations | −1.85** | 0.37 | −2.22*** | (0.80) |
| Violations (All) | −4.74** | 0.13 | −4.87** | (1.94) |
| Maintainability Violations (All) | −4.80** | 0.19 | −5.00** | (1.91) |
| Maintainability Issues (High) | −1.84** | 0.38 | −2.22*** | (0.79) |
| Maintainability Issues (Low) | −0.73 | −0.39 | −0.34 | (0.29) |
| Security Hotspots | −1.18 | 0.07 | −1.26 | (1.09) |
| Bugs | −0.01 | −0.13 | 0.13 | (0.14) |
| Reliability Issues (High) | −0.01 | −0.01 | 0.00 | (0.01) |
| Reliability Issues (Low) | −0.12 | −0.15 | 0.03 | (0.16) |
| Technical Debt | −1.57** | −0.13 | −1.44** | (0.68) |
| Technical Debt Ratio | −5.69** | −0.46 | −5.23** | (2.46) |

Columns (1) and (2) report the Treated × Post coefficient for each outcome from the baseline Seed/Series A specification (Tables 3, 4, 5, 6, and 7) and from the Series B+ placebo specification (Tables 28, 29, 30, 31, and 32), which also report their standard errors. The high- and low-severity maintainability and reliability rows count issues per 1,000 non-comment lines using the alternative classification documented in Appendix A; they are not breakdowns of Maintainability Violations (All) or Bugs, which use the traditional classification. Table 47 compares the two classifications on identical observations. The difference (1)−(2) is estimated in a single regression that stacks the two samples with sample-specific Cohort × Firm and Month × Cohort fixed effects. Because many firms appear in both samples, the two estimates are correlated; the standard error of the difference, clustered by cohort and firm, accounts for this covariance. The Technical Debt Ratio is 40/11 times debt per line on the same observations with positive lines of code; its coefficients are expressed in percentage points. Red text identifies the later-round estimates. * p<0.1, ** p<0.05, *** p<0.01.

The difference between the first-entry and later-round estimates is consistent with professionalization at first institutional entry. Section 3.8 supports this interpretation directly: firms record collaboration more systematically, and existing engineers become less likely to introduce bugs.

## 4.3 Lifecycle Stage and Commercialization

The first institutional round may also coincide with commercialization. As firms begin serving paying customers, they may adopt more demanding engineering practices. If the initial transition to revenue

generation explains our findings, quality gains should be concentrated among firms making that transition.

We test this prediction using commercial status at financing: 183 of the 274 treated firms (67%) were already generating revenue at first entry, and the estimated quality gains are not larger among the remaining pre-commercial firms.

Table 16 shows that several main outcomes improve among firms already generating revenue. Within this group, total violations decline by 5.92 per 1,000 lines, technical debt by $2.23 per line, and the Technical Debt Ratio by 8.1 percentage points ($p = 0.04$ for each). Critical violations in this group decline by 1.90 per 1,000 lines ($p = 0.11$); among pre-commercial firms, they decline by 1.74 ($p = 0.07$).

Table 16: Treatment Effects by Commercial Stage at Entry: Subsample Estimates

| | **Critical Violations** | **Violations** | **Technical Debt** | **Technical Debt Ratio** |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| *Panel A: Treated firms already generating revenue at entry (183 firms)* | | | | |
| Treated × Post | −1.90 | −5.92** | −2.23** | −8.11** |
| | (1.20) | (2.90) | (1.07) | (3.91) |
| **# Obs** | 390,179 | 390,179 | 390,179 | 390,179 |
| *Panel B: Treated firms pre-commercial at entry (91 firms)* | | | | |
| Treated × Post | −1.74* | −2.40 | −0.24 | −0.89 |
| | (0.95) | (1.45) | (0.54) | (1.96) |
| **# Obs** | 297,099 | 297,099 | 297,099 | 297,099 |
| Cohort × Firm FE | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes |

Each panel estimates Equation (7) on treated firms with the stated commercial stage and the controls in their cohorts. Commercial stage is the PitchBook business status at the deal closest to treatment, matched within three months. Firms already generating revenue have the status Generating Revenue, Generating Revenue/Not Profitable, or Profitable. Cohorts left with only treated or only control firms are dropped. Critical Violations and Violations are counts per 1,000 lines of code; Technical Debt is calibrated dollars per line; the Technical Debt Ratio is calibrated technical debt as a percentage of software assets. Joint lead tests show no evidence of differential pre-treatment trends (smallest $p = 0.34$; not statistically significant).

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Table 17 directly tests whether the response differs by commercial stage. The pre-commercial interaction is not statistically significant for critical and total violations ($p = 0.92$ and $p = 0.27$, respectively). For both debt measures, however, it is positive, indicating a smaller decline among pre-commercial firms ($p = 0.08$ for each). Excluding firms with more than five repositories yields similar

estimates. The point estimates therefore do not show the predicted concentration of gains among firms entering commercialization; for technical debt, the direction of the difference is the opposite.

Table 17: Commercial Stage at Entry: Triple-Difference Estimates

| | **Critical Violations** | **Violations** | **Technical Debt** | **Technical Debt Ratio** |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| *Panel A: Main sample* | | | | |
| Treated × Post | −1.90 | −5.92** | −2.23** | −8.11** |
| | (1.20) | (2.90) | (1.07) | (3.90) |
| Treated × Post × Pre-commercial | 0.14 | 3.49 | 1.97* | 7.18* |
| | (1.50) | (3.16) | (1.13) | (4.11) |
| Implied effect, pre-commercial firms | −1.76* | −2.43* | −0.26 | −0.93 |
| | (0.95) | (1.45) | (0.54) | (1.96) |
| **# Obs** | 487,577 | 487,577 | 487,577 | 487,577 |
| *Panel B: At most five repositories per firm* | | | | |
| Treated × Post | −1.91 | −6.01** | −2.34** | −8.51** |
| | (1.23) | (2.99) | (1.11) | (4.02) |
| Treated × Post × Pre-commercial | 0.12 | 3.56 | 2.10* | 7.64* |
| | (1.55) | (3.26) | (1.17) | (4.24) |
| **# Obs** | 447,156 | 447,156 | 447,156 | 447,156 |
| Cohort × Firm FE | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes |

This table estimates $M_{i,c,t} = \beta_1 VC_{i,c,t} + \beta_3 \left(VC_{i,c,t} \times \text{Pre-commercial}_i\right) + \chi_{c,i} + \delta_{c,t} + \varepsilon_{i,c,t}$ on the full estimation sample, retaining every eligible control firm. Firms already generating revenue at entry are the omitted category: $\beta_1$ (Treated × Post) is the treatment effect for these firms, and $\beta_3$ is the incremental effect for pre-commercial firms. Pre-commercial equals one for treated firms not yet generating revenue at the financing deal closest to the treatment date (PitchBook deal business status, matched within three months) and zero for treated firms already generating revenue; it is identically zero for control firms, whose commercial stage at the cohort baseline is unobservable, so the specification estimates treatment-effect heterogeneity across treated firms on the common control pool rather than a fully saturated triple difference. The level and Post interactions of the indicator are absorbed by the fixed effects or collinear with the triple term and are not separately identified. The implied effect for pre-commercial firms is $\beta_1 + \beta_3$; its standard error comes from the equivalent reparameterization with pre-commercial firms as the base group. Joint tests show no evidence of differential pre-treatment leads by commercial stage for any of the four outcomes (smallest $p = 0.78$; not statistically significant). Panel B drops firms with more than five scanned repositories before aggregation. Critical and total violations are counts per 1,000 non-comment lines of code; technical debt is calibrated dollars per line, and the Technical Debt Ratio is calibrated debt as a percentage of software replacement cost.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Commercialization can develop gradually even after a firm begins generating revenue. We therefore complement the stage comparison with direct measures of how the software asset changes around financing. Table 18 uses developers' descriptions of code changes (commit messages) and the

names and folder locations of the files they alter to track references to billing, payments, subscriptions, trials, and upgrades. We also measure changes to files whose names or folder locations refer to commercial features, payment processing, enterprise features, or growth, while Panel C records the scale and breadth of development. These comparisons cover firms with one listed commercialized project, as described in the table notes.

Table 18: Commercialization Measures in the Software Asset

| Measurement | Dependent Variable | Treated × Post | $p$-value |
|---|---|---|---|
| *Panel A: Commit-message language* | | | |
| | Any commercialization term (pp) | 0.70 | 0.34 |
| | | (0.74) | |
| | Billing / payments / metering (pp) | 0.01 | 0.95 |
| | | (0.11) | |
| | Subscription / trial / upgrade (pp) | 0.65 | 0.37 |
| | | (0.73) | |
| *Panel B: Names and locations of changed files* | | | |
| | Commercial features (pp) | −0.03 | 0.33 |
| | | (0.03) | |
| | Payment-processing software (pp) | −0.04 | 0.36 |
| | | (0.05) | |
| | Enterprise / service-level agreements (pp) | 0.14 | 0.33 |
| | | (0.14) | |
| | Growth / expansion (pp) | 0.12 | 0.61 |
| | | (0.24) | |
| *Panel C: Development activity* | | | |
| | Log file changes per month | 0.34*** | <0.01 |
| | | (0.07) | |
| | Top-level folders with file changes | 0.88** | 0.02 |
| | | (0.38) | |
| | User-facing interface share of changed files (pp) | 1.77** | 0.04 |
| | | (0.86) | |

This table estimates the baseline specification of Equation (7) on contribution-based outcomes for firms with one listed commercialized project. Panel A uses 4.8 million commit messages (177,676 stacked observations); each outcome is the fraction of a firm's commits in a month carrying the stated marker. Panels B and C use file-change histories from 441 repositories. Panel B measures the fraction of changed files whose names or folder locations match each category. Panel C measures log(file changes), the number of top-level project folders with changed files (including the main project folder), and the fraction of changed files in the user-facing interface. For each repository, we take the natural logarithm of its positive monthly file-change count, then average these values across the firm's repositories within each month. The activity regression uses 236,646 observations. Joint tests do not reject zero pre-treatment coefficients for the reported measures. Coefficients and standard errors for share outcomes, marked pp, are reported in percentage points. These outcomes derive from commit histories and contain no SonarQube input.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

The monetization markers show no statistically significant shift after financing, while development activity expands substantially. The growth rate in monthly file changes is approximately 34 percentage points greater than in controls ($p < 0.01$). Developers change files in 0.88 additional top-level project folders, and the share of changed files in the user-facing interface rises by 1.8 percentage

points ($p = 0.02$ and $p = 0.04$, respectively). The increase in development therefore extends across the software asset without a detectable shift toward the measured monetization features.

Employment records show a similar expansion in engineering capacity. The sales-related share of headcount rises by 1.5 percentage points ($p = 0.02$), compared with 6.6 percentage points for software engineering ($p < 0.01$). Firms build commercial capacity, but the larger compositional shift is toward engineering. Appendix K reports the employment decomposition and additional development diagnostics.

The stage, codebase, and employment tests do not support a narrow lifecycle explanation in which quality improves because firms make the initial transition to selling a product. The gains are present among firms already generating revenue and accompany broad development activity, with a larger employment shift toward engineering than toward sales. These findings complement the process evidence by showing that the engineering improvement extends beyond the firms and activities most directly associated with the onset of commercialization.

## 4.4 Additional Robustness Tests

We next examine whether the quality improvement depends on the normalization of engineering outcomes, the months surrounding the financing event, or particular firms and cohorts in the stacked sample. These tests address distinct features of the design, while supplementary outcomes show which dimensions of engineering practice improve and which do not.

**Density versus levels.** Violations per line can fall because a codebase grows, not only because violations decline. We therefore study firms in the bottom quartile of pre-financing code growth (Appendix H). After financing, their code volume grows no faster than that of controls; the difference, $-0.02$ log points, is not statistically significant ($p = 0.73$). Their critical violations nevertheless decline by 2.85 per 1,000 lines ($p = 0.06$), more than the 1.85 decline in the full sample. Appendix G connects the density results to changes in issue counts and total technical debt.

**Excluding months around financing.** To check whether the gains depend on changes concentrated around the deal, we reestimate the baseline after excluding the financing month and either one or two months on each side. Quality improvements persist beyond the months immediately surrounding financing (Table 43). Under both exclusions, critical violations, total violations, and technical debt decline (largest $p = 0.04$), with estimates modestly larger in magnitude than the baseline.

**Observation coverage and repository composition.** We next examine whether changes in observation coverage or repository composition account for the quality gains. One comparison requires observations in every month of the 25-month event window. Another follows the pre-financing repos-

itories, retaining only months when all remain observed. We also hold each pre-financing repository's share of the firm's lines at its value in the month before financing, while allowing quality within repositories to change. Critical violations and technical debt decline under each comparison (Tables 49 and 51). The declines also persist in comparisons of pre- and post-financing averages on complete windows. When we exclude firms with more than five scanned repositories, critical violations decline by 1.87 per 1,000 lines and technical debt by $1.63 per line ($p = 0.03$ and $p = 0.04$, respectively).

**Alternative event windows.** Quality improvements persist across shorter and longer observation periods (Table 43). To check whether the findings depend on the observation period, we replace the baseline of 12 months on each side of financing with 6, 9, 18, or 24 months. Critical violations, total violations, and technical debt decline in every window (largest $p = 0.06$).

**Individual financing cohorts.** To check whether a particular financing cohort drives the results, we reestimate the baseline 99 times, omitting one cohort each time. Critical violations, total violations, and technical debt decline in every case (Table 43).

**Repeated use of control firms.** The same firm can serve as a control for several financing cohorts. To check whether repeated use drives the findings, we weight control observations inversely by the number of cohorts in which each firm appears. Critical violations, total violations, and technical debt decline under this weighting (largest $p = 0.04$; Table 43).

**Programming-language exclusions.** To check whether a single programming language accounts for the gains, we successively exclude firms whose primary language is Go, JavaScript, C/C++, Java, or Python—the five most common languages among treated firms. The estimated changes in critical violations, total violations, and technical debt remain negative in every remaining sample. Critical-violation declines are significant at the 5% level in four of the five comparisons.

**Additional SonarQube outcomes.** A supplementary screen of 42 SonarQube outcomes clarifies the scope of the result. The 42 outcomes include the maintainability measure reported above. The growth rate in duplicated lines is approximately 29 percentage points greater than in controls (multiplicity-adjusted $q < 0.01$), and duplicated code blocks rise by 0.9 per 1,000 lines, although the latter is not statistically significant after adjustment ($q = 0.11$). No security density measure, including vulnerabilities per 1,000 lines, is statistically significant. Improved rule compliance and maintainability thus coexist with increased duplication as firms scale.

**Process documentation and repository allocation.** We also examine whether firms begin consistently mentioning testing and related development practices in their commit messages. The monthly

probability of beginning a sustained pattern of these references rises by 0.2 percentage points from a 0.5% baseline—a relative increase of approximately 47% ($p = 0.04$). The estimate is no longer statistically significant after adjustment for the four process measures tested (Holm-adjusted $p = 0.18$). Separately, neither the commit share of the flagship repository—the firm's largest repository before financing—nor within-firm commit concentration shows a statistically significant change ($p = 0.32$ and $p = 0.63$, respectively). The allocation results provide no evidence that the improvement simply reflects a shift of development toward the flagship repository. Appendices M.5 and M.6 report the sample restrictions and full results.

# 5 Conclusion

Venture capital professionalization reaches the engineering practices through which startups build their products. Using longitudinal codebase data on 569 open-core technology firms, we find that critical violations fall by 1.85 per 1,000 lines and technical debt falls by $1.57 per line after first VC entry, approximately 15% and 12% of their treated pre-financing means. These improvements accompany substantial expansion of codebases and engineering teams. The critical-violation decline persists when we compare firms with similar prior growth, require complete observation windows, and hold the covered repositories fixed. Quality gains also extend to firms already generating revenue.

The contribution histories show how this change reaches the existing organization. Teams more frequently document copyright and licensing assurances, collaboration, and references to numbered development records, and they make a smaller share of contributions on weekends. Among engineers already active before financing, the probability of introducing a bug falls by 12.4% of its pre-treatment mean, and improvements within the same engineers account for nearly two-thirds of this decline. The quality improvement is larger around first entry than around later financing rounds, placing these changes at the beginning of the VC relationship. Together, the codebase and contribution records connect improved engineering quality to the procedures through which existing engineers build and maintain software.

Our dollar-denominated measure of technical debt makes the estimated maintenance liability embedded in software comparable with other business costs. Combining this measure with development histories provides a way to study how firms accumulate and manage software liabilities alongside intangible assets. The findings connect VC value creation to the maintenance of a technology firm's core product, with applications to software-intensive firm valuation and M&A due diligence.

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# Appendix

# A Variable Definitions

Table 19: Data and Variable Definitions

| Variable | Description |
|---|---|
| *A. Dimension of the data* | |
| Firms | Open-core technology companies that commercialize open-source repositories and for which we observe longitudinal code histories, funding events, and employment data. The universe comprises 649 unique firms identified through OSSRank, an analytics platform for the open-source ecosystem; 621 enter the firm–month panel, 569 of them with covered codebase months. |
| Repositories | Core product repositories associated with each firm. The firm–month panel draws on 703 repositories hosted on GitHub or GitLab platforms over the period 2010–2024. A repository functions as the central storage location for the code underlying a firm's core product. |
| Month | Monthly observations covering the period 2010–2024. |
| Unit of observation | The underlying data contain one observation per firm–month. The estimation panel is a firm–cohort–month because a firm can appear in more than one treatment stack. |
| Firm–month aggregation | We sum lines of code, remediation effort and issue counts by severity across each firm's repositories within a month. Ratio measures, including the Technical Debt Ratio, are recomputed from the summed components rather than averaged across repositories. |
| *B. Dependent variables (Engineering quality metrics)* | |

*Continued on next page*

*Table 19 (continued)*

| Variable | Description |
|---|---|
| Critical Violations | Number of SonarQube rule violations at the critical severity level per 1,000 non-comment lines of code in month $t$. These severe issues can threaten system stability, security, or future maintainability. SonarQube also has a small Blocker level above critical; neither this measure nor the high-severity measures under the alternative classification include it. |
| Violations (All) | Number of rule violations across all severity levels per 1,000 non-comment lines of code. SonarQube flags code patterns associated with reliability, security, and maintainability concerns. |
| Maintainability Violations (All) | Number of maintainability-related rule violations across all severities per 1,000 non-comment lines of code. These rules flag code patterns that may make software harder to understand, modify, or extend. |
| Maintainability Issues (High) | Number of maintainability issues classified as high severity by SonarQube per 1,000 non-comment lines of code. |
| Maintainability Issues (Low) | Number of maintainability issues classified as low severity by SonarQube per 1,000 non-comment lines of code. |
| kLoC | One thousand non-comment lines of source code. Issue densities use this unit; technical debt per LoC uses one line. |
| Lines of code (LoC) | Non-comment lines of source code summed across the firm's covered repositories in month $t$. |
| SonarQube Security Hotspots | Number of security-sensitive code segments flagged for manual review per 1,000 non-comment lines of code. A hotspot requires human assessment and is not a confirmed vulnerability. |
| Bugs | Number of code patterns classified by SonarQube as likely to cause incorrect or unexpected behavior, per 1,000 lines of code. |
| Reliability Issues (High) | Number of reliability issues classified as high severity by SonarQube per 1,000 non-comment lines of code. The classification indicates a risk of incorrect behavior, rather than an observed application failure. |

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*Table 19 (continued)*

| Variable | Description |
|---|---|
| Reliability Issues (Low) | Number of reliability issues classified as low severity by SonarQube per 1,000 non-comment lines of code. |
| Technical Debt per Line of Code | Estimated dollar value of remediation effort per line of code. Let $w$ denote the wage per hour ($55/hour) and let the SQALE Index be SonarQube's remediation effort for maintainability issues (in hours). Let $\ell$ denote the development time per line (in hours) and set $\ell = 0.5$ (30 minutes), SonarQube's default. We approximate redevelopment time as $\ell \times$ LoC hours and scale the raw SQALE Index by $\kappa$ to obtain calibrated remediation effort. We choose $\kappa$ so that median calibrated debt equals a target share $\varphi$ of median replacement-cost software assets (we set $\varphi = 0.20$): $$\kappa = \varphi \times \frac{\text{Median}(\ell \times \text{LoC})}{\text{Median}(\text{SQALE Index})}.$$ We then define $$\text{Technical Debt}_{i,t} = \kappa \times \text{SQALE Index}_{i,t} \times w,$$ and normalize by lines of code: $$\text{Technical Debt per LoC}_{i,t} = \frac{\text{Technical Debt}_{i,t}}{\text{LoC}_{i,t}}.$$ |
| Technical Debt Ratio | Calibrated Technical Debt divided by Software Assets, multiplied by 100, where Software Assets is the replacement cost of the codebase ($55 per hour times 30 minutes per line times lines of code). Equivalently, $100\kappa$ times raw remediation hours divided by estimated redevelopment hours; the wage cancels. The ratio is a percentage of replacement cost, and changes are reported in percentage points. For the same observations with positive lines of code, it equals $40/11$ times Technical Debt per LoC; both measures share the same missingness rule and describe one outcome in two units. |

*Continued on next page*

*Table 19 (continued)*

| Variable | Description |
|---|---|
| Technical Debt (level) | Calibrated dollar value of the remediation burden in month $t$, defined as $\kappa \times \text{SQALE Index}_{i,t} \times w$. |
| $\kappa$ (Technical-debt calibration) | Scaling constant that maps SonarQube's raw SQALE Index into dollar-denominated technical debt. It is chosen so that median calibrated debt equals 20% of median replacement-cost software assets ($\varphi = 0.20$). The main-sample calibration is $\kappa = 68.60$ and is held fixed across specifications using that sample, including the paired comparison of SonarQube's traditional and alternative classification systems in Table 47. |
| *C. Main explanatory variables* | |
| $VC_{i,c,t}$ (Treated × Post) | Indicator equal to 1 for a treated firm in cohort $c$ beginning in the month of its first completed Seed round, or first completed Series A round if no Seed is recorded. Rumored, failed or cancelled, announced or in-progress, upcoming, and postponed deals are excluded when assigning the event. |
| Treated | Indicator equal to 1 for a firm whose first qualifying financing defines cohort $c$, and 0 for that cohort's controls. A firm can serve as a control in an earlier cohort before entering its own treated cohort. |
| Post | Dummy equal to 1 for months at or after the first VC deal for treated firms (and the corresponding aligned event-time months for controls in each cohort), and 0 otherwise. |
| High Violations | Dummy equal to 1 if the firm's pre-treatment critical violation density (at $t = -1$) exceeds the median across baseline firm–cohort observations of firms that appear as treated somewhere in the sample, and 0 otherwise. A firm can contribute baseline observations in more than one cohort. |

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*Table 19 (continued)*

| Variable | Description |
|---|---|
| Large Codebase | Indicator equal to one if log code volume at event month $-1$ exceeds the median across baseline firm–cohort observations of firms treated somewhere in the sample. The comparison uses the same baseline convention as High Violations. |
| Large Round | Dummy equal to 1 if the VC deal size (first Seed or Series A round amount, in USD) exceeds the treated-firm median, and 0 otherwise. Used to compare quality changes across financing amounts. |
| Angel First Round | Dummy equal to 1 if the firm's first qualifying capital round among Angel (individual), Seed, and Series A rounds is an Angel (individual) round, and 0 otherwise. Used to compare quality changes at first Seed or Series A entry according to whether the firm previously received angel financing. |
| Large Syndicate (median split) | Indicator equal to one if the first Seed or Series A treatment round has more than four recorded investors, and zero otherwise. Four is the median among treated firms with a recorded investor count; missing counts are coded zero. |
| Founder Technical Leadership | Dummy equal to 1 if any member of the founding team (identified via LinkedIn job titles containing "founder" or "co-founder" and active at the firm within six months of first VC financing) simultaneously held a technical leadership role (CTO, VP of Engineering, Head of Engineering, or Director of Engineering) at the same firm around the deal date. Captures technical leadership within the founding team around financing. |
| Prior Executive Experience | Dummy equal to 1 if any founder of the firm held a prior CEO, Chief, Founder, or President role at a different company before the first VC deal date (excluding assistant and vice president roles). Captures prior senior leadership experience within the founding team. |

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*Table 19 (continued)*

| Variable | Description |
|---|---|
| Series B+ placebo treatment | In the Series B+ placebo sample of mature firms, dummy equal to 1 in months at or after a Series B or later funding round (conditional on observing earlier Seed/Series A history), and 0 otherwise. Treated firms already have institutional backing; controls remain without a first Seed or Series A round through the end of the event window. |
| Already generating revenue | Dummy equal to 1 if the firm was already generating revenue when venture capital arrived, based on the PitchBook business status recorded at the financing deal that defines treatment (matched within ±3 months of the treatment date) and taking the value *Generating Revenue*, *Generating Revenue/Not Profitable*, or *Profitable*; 0 otherwise. Defined for treated firms. |
| Pre-commercial | Complement of *Already generating revenue* for treated firms: dummy equal to 1 if the firm was not yet generating revenue at the treatment deal, and 0 otherwise. Used to test whether the effect of VC entry differs by the firm's commercial stage at entry. |
| *D. Other covariates and constructed variables* | |
| Total firm headcount | Total number of employees associated with the firm based on LinkedIn data in month $t$. |
| Software engineer headcount | Number of employees with software-engineering roles (based on LinkedIn job titles) in month $t$. |
| Senior engineering hires | Number of new hires in the stated 12- or 24-month post-event window whose LinkedIn job titles indicate senior engineering roles: senior engineer/developer, staff engineer, principal engineer, architect, engineering director/VP, engineering manager, tech lead, or CTO. |
| Senior hire ratio | Ratio of senior engineering hires to total new hires in the stated post-event window; the first-year hiring comparisons use 12 calendar months. Missing for firms with no new hires in the window. |

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*Table 19 (continued)*

| Variable | Description |
|---|---|
| High Senior Ratio | Indicator equal to one if the ratio of senior engineering hires to all new hires over the first 12 calendar months after the cohort event exceeds the treated-firm median of $1/7$. The endpoint 12 months after the event is excluded. Missing hiring measures are excluded from the interaction regressions. |
| Hired technical leader | Indicator equal to one if the firm hires a new CTO, VP of Engineering or Head of Engineering during [event, event+12 months). Both arms use the cohort event. Hiring requires a recorded start date and no position at the firm before the event. |
| Technical-leader hiring outcome | Indicator equal to one if a firm hires a CTO, VP of Engineering, Head or Director of Engineering, or VP of Technology during the pre-financing period $[-12, -1]$ or post-financing period $[0, 11]$. The direct comparison retains firm–cohort pairs observed in every month of both 12-month periods. This outcome is defined separately in each period; the first-year hiring moderator above instead classifies firms over the post-event year. |
| Nontechnical-leader hiring outcome | Analogous period-specific indicator for hiring a nontechnical executive (chief executive, operating, financial, marketing, product, revenue, or commercial officer, or a head, vice president, or director of a non-engineering function). |
| SWE Share | Nonmanagerial software engineers as a share of distinct new hires in a firm–cohort–quarter, based on LinkedIn role classifications. Missing when no hires are recorded. SWE denotes software engineering. |
| SWE Seniority | Mean LinkedIn seniority, on a scale from 1 to 7, among newly hired nonmanagerial software engineers in a firm–cohort–quarter. Missing when no classified engineering hires are recorded. |
| $X_{i,c,t}$ | Additional regressors in the heterogeneity specifications, including Post and Treated $\times$ Post interactions with baseline quality, financing size, and other stated moderators. The baseline regression omits $X_{i,c,t}$. |

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*Table 19 (continued)*

| Variable | Description |
|---|---|
| Log LoC | Natural logarithm of positive non-comment lines of code summed across covered repositories. Nonpositive observations are excluded. |
| Log total headcount | Natural logarithm of positive total firm headcount. Nonpositive observations are excluded. |
| Log software engineer headcount | Natural logarithm of positive software-engineer headcount. Nonpositive observations are excluded. |
| Log critical violations | Natural logarithm of positive critical violations. Nonpositive observations are excluded. |
| Log total violations | Natural logarithm of positive total violations. Nonpositive observations are excluded. |
| Log technical debt in dollars | Natural logarithm of positive technical debt in dollars. Nonpositive observations are excluded. |
| Log maintainability issues | Natural logarithm of positive maintainability-issue counts. Nonpositive observations are excluded. |
| Log duplicated lines | Natural logarithm of positive duplicated lines. Nonpositive observations are excluded. |
| Log file-change activity | Natural logarithm of positive file-change activity, transformed at the repository–month level before averaging within firm–month. Nonpositive observations are excluded. |

*Continued on next page*

*Table 19 (continued)*

| Variable | Description |
|---|---|
| Pre-treatment headcount growth | Difference in log average headcount between months $t \in [-4, -2]$ and $t \in [-12, -11]$ relative to the VC event month for firm $i$ in cohort $c$. Used in Mahalanobis-distance matching to select controls with similar observed pre-treatment growth: $$G_i^c = \ln\left(\frac{1}{3}\sum_{t=-4}^{-2} \text{Headcount}_{i,c,t}\right) - \ln\left(\frac{1}{2}\sum_{t=-12}^{-11} \text{Headcount}_{i,c,t}\right).$$ |
| Pre-treatment SWE headcount growth | Analogous pre-treatment growth measure constructed using software engineer headcount instead of total headcount over the same pre-treatment windows. |
| Pre-treatment software growth | Analogous pre-treatment growth measure constructed using lines of code (LoC) over the same pre-treatment windows. |
| Software Assets | Estimated replacement cost of the codebase in month $t$. Software Assets represent a coarse approximation of the capital required to re-develop the codebase from scratch, defined using a replacement cost method: $$\text{Software Assets}_{i,t} = w\ell\,\text{LoC}_{i,t},$$ where $\ell = 0.5$ hours per line and $w = \$55$ per hour. This is a replacement-cost proxy, not observed expenditure. |
| Event-time indicators | Set of dummy variables for each quarter or month $t \in [-12, +12]$ relative to the VC event date. Used to estimate dynamic difference-in-differences (event-study) coefficients and assess pre-treatment outcome paths. |
| Cohort × Firm FE ($\chi_{c,i}$) | Firm-by-cohort fixed effects that absorb time-invariant firm characteristics within each stacked cohort. |

*Continued on next page*

*Table 19 (continued)*

| Variable | Description |
|---|---|
| Month × Cohort FE ($\delta_{c,t}$) | Month-by-cohort fixed effects that control for common time-varying shocks affecting all firms within a cohort at month $t$. |
| Cohort eligibility | After each sample restriction, cohorts containing only treated firms or only control firms are excluded because they provide no identifying variation. |
| *E. Contribution-history outcomes* | |
| Contribution documentation share | Fraction of a firm's monthly commits containing Reviewed-by, Signed-off-by, Acked-by, Approved-by or Co-authored-by. Component shares count each marker separately and can overlap. |
| Numbered-reference share | Fraction of a firm's monthly commits whose messages contain a numbered issue or task reference. Explicit pull-request URLs and external ticket keys are measured separately. |
| Headquarters-local weekend share | Fraction of a firm's monthly commits recorded on Saturday or Sunday after converting contribution timestamps to the local time zone of the company's reported headquarters, including date-specific daylight-saving rules. |
| Pre-financing engineer | Engineer first active before the firm's financing event; control engineers are classified relative to their cohort event. The classification is based on repository contribution histories. |
| Bug-introduction rate | Share of contributions introducing bugs. We identify these contributions by tracing later bug fixes back to earlier changes to the affected code. After excluding unclassified and bot authors, we keep the first attribution link for each repository and introducing commit in the data's row order, then apply the lag restriction of 0–24 calendar months from the contribution to the fix. Selection follows row order rather than an earliest-fix rule. Fixed-engineer checks instead count a commit if any recorded fix qualifies. The commit-level timing regression uses 730 days. Zero contribution denominators yield missing rates. |

*Continued on next page*

*Table 19 (continued)*

| Variable | Description |
|---|---|
| Existing-engineer decomposition | Additive decomposition of the change in the bug-introduction rate for engineers active before financing. The within-engineer component captures changes in individual rates; the composition component captures changes in their shares of contribution activity. Both components use the same firm–month observations as the estimated total change. |
| Post-entry newcomer gap | We subtract the bug-introduction rate for engineers active before financing from newcomers' rate. We compare this difference between treated and control firms in post-event months when both groups contributed, using cohort-by-event-month fixed effects. Newcomers first contribute at or after the relevant cohort event; no pre-event newcomer rate enters this comparison. |

*Note:* Engineering-quality metrics are measured using SonarQube Community Edition, which checks code against more than 9,000 language-specific rules. Severity-specific maintainability and reliability measures use the alternative classification described in Appendix M.4; they are not breakdowns of Maintainability Violations (All) or Bugs, which use the traditional classification. Technical Debt uses the SQALE (Software Quality Assessment based on Lifecycle Expectations) methodology. The universe comprises 649 open-core technology firms identified through OSSRank. The 2010–2024 firm–month panel contains 621 firms. The company–repository mapping links 703 repositories to 609 firms, 569 of which contribute at least one covered codebase month. The stacked difference-in-differences sample contains 484 unique firms and 99 event cohorts. Among these firms, 274 appear as treated and 412 as controls; the counts overlap because a not-yet-treated firm can serve as a control before its own financing event. Outcomes are observed over $t \in [-12, +12]$ around the first qualifying financing.

# B Language-Specific Analysis Configurations

To reconstruct engineering quality over time, we analyze historical source code using SonarQube Community Edition[7] and its language analyzers, supplemented by the community *sonar-cxx* plugin for C and C++. Historical repositories often lack working build environments or compiled dependencies. We use the following language-specific settings to reduce reliance on those artifacts and accommodate large codebases.

- **C/C++:** The *sonar-cxx* plugin parses source text without rebuilding each historical project. It constructs a representation of the code's syntax and measures features such as branching complexity and duplication. The available checks depend on the plugin's parsing and rule coverage.
- **Java:** We analyze source code when compiled class files and dependency libraries are unavailable. This limits checks that require compiled information.
- **JavaScript/TypeScript:** The analysis disables bundle detection and the HTML analyzer.
- **Rust:** We analyze source code without the optional Clippy checks, which require an external toolchain.

The scan checks code against thousands of language-specific rules that reflect widely used software-quality frameworks and industry practice (e.g., ISO/IEC 25010). Three common Java rule violations illustrate what the tool measures. (i) Commented-out code makes it harder to understand what the software actually does and whether the implementation is complete. (ii) Lingering `TODO` markers are placeholders for unfinished work that often persist and embed a hidden backlog inside the code. (iii) Overly complex functions (high cognitive complexity) are harder to read, test, and modify, raising the cost of future changes and increasing the risk of mistakes.[8]

[7] SonarQube Community Edition is the freely available, open-source distribution used for all of our scans.
[8] For additional examples of common Java issues highlighted by SonarSource, see SonarSource, "Top issues in Java projects".

# C Stacked Difference-in-Differences

**Design.** The objective is to estimate how first institutional VC entry changes engineering quality when firms receive financing at different dates. Standard two-way fixed-effects regressions can use already-treated firms as controls, producing misleading estimates when treatment effects vary across cohorts or evolve over time. We therefore use the cohort-stacking approach in Gormley and Matsa (2011), constructing a separate comparison group that remains untreated throughout each cohort's event window. This appendix defines the cohort-event causal contrasts, states the assumptions under which they are identified, and relates them to the stacked DiD regression used in the paper.

**Cohorts and comparison firms.** Let $E_i$ denote the calendar month of firm $i$'s first completed Seed round, or its first completed Series A round if no Seed is recorded. Set $E_i = \infty$ for firms without a qualifying financing. We exclude firms with a later-stage financing but no observed Seed or Series A history because their first institutional round cannot be dated.

For cohort month $c$, define the treated and eligible control sets as

$$\mathcal{T}_c = \{i : E_i = c\}, \qquad \mathcal{J}_c = \{j : E_j > c + L\},$$

where $L = 12$ is the event-window half-width. Event time is $e = t - c$, with $e \in [-L, L]$; treatment begins at $e = 0$. The control set includes never-funded firms and firms funded after the window closes.

We stack the available observations for $\mathcal{T}_c \cup \mathcal{J}_c$ over the event window. At construction, a cohort must contain at least one treated observation at every observed event month and an eligible control group. Formally, let $\mathcal{E}_c$ denote the observed event months and let $n^0_{T,c,e}$ and $n^0_{C,c,e}$ count treated and control observations before estimation-sample filters. The construction rule retains

$$C^0_L = \left\{ c : \min_{e \in \mathcal{E}_c} n^0_{T,c,e} \geq 1, \quad \sum_{e \in \mathcal{E}_c} n^0_{C,c,e} \geq 1 \right\}.$$

After each estimation-sample restriction, we drop any cohort left without both treated and control firms; $C_L \subseteq C^0_L$ denotes the retained cohorts. The sample is unbalanced because each firm need not have a recorded scan in every month. A firm can also appear as a control in several earlier cohorts before entering its own treated cohort, which is why the stacked regression count exceeds the number of underlying firm–months.

**Potential outcomes and identification.** Let $Y^g_{it}$ denote the outcome for firm $i$ in calendar month $t$ if its first financing occurs in month $g$, and let $Y^\infty_{it}$ denote the outcome without financing. The observed outcome is $Y_{it} = Y^{E_i}_{it}$. This notation assumes that a firm's outcome is unaffected by the financing of

other sample firms. The cohort-event average treatment effect on the treated is

$$\theta_{c,e} = \mathbb{E}\left[Y^{c}_{i,c+e} - Y^{\infty}_{i,c+e} \mid E_i = c\right].$$

A causal interpretation requires the following assumptions for the firms represented in each comparison.

**Assumption PT (Parallel Trends).** Absent financing, treated and control firms would have the same expected outcome change. For each post-treatment month $e \geq 0$ and pre-treatment reference month $\bar{e} < 0$,

$$\mathbb{E}\left[Y^{\infty}_{i,c+e} - Y^{\infty}_{i,c+\bar{e}} \mid E_i = c\right] = \mathbb{E}\left[Y^{\infty}_{i,c+e} - Y^{\infty}_{i,c+\bar{e}} \mid i \in \mathcal{J}_c\right].$$

**Assumption NA (No Anticipation).** Financing does not affect outcomes before the recorded event month:

$$Y^{g}_{it} = Y^{\infty}_{it} \quad \text{for } t < g.$$

We use the donut exclusions in Appendix M.1 to examine sensitivity to observations adjacent to the recorded event month. The monthly comparisons in Table 56 omit month $-4$, allowing a separate estimate for month $-1$.

Under PT and NA, the population difference-in-differences contrast identifies $\theta_{c,e}$. To see this, write the observed change for treated firms and subtract the change for their cohort controls:

$$\begin{aligned}
\theta^{\text{DID}}_{c,e} &= \mathbb{E}[Y_{i,c+e} - Y_{i,c+\bar{e}} \mid E_i = c] - \mathbb{E}[Y_{i,c+e} - Y_{i,c+\bar{e}} \mid i \in \mathcal{J}_c] \\
&= \mathbb{E}[Y^{c}_{i,c+e} - Y^{\infty}_{i,c+\bar{e}} \mid E_i = c] - \mathbb{E}[Y^{\infty}_{i,c+e} - Y^{\infty}_{i,c+\bar{e}} \mid i \in \mathcal{J}_c] \\
&= \mathbb{E}[Y^{c}_{i,c+e} - Y^{\infty}_{i,c+e} \mid E_i = c] \\
&\quad + \mathbb{E}[Y^{\infty}_{i,c+e} - Y^{\infty}_{i,c+\bar{e}} \mid E_i = c] - \mathbb{E}[Y^{\infty}_{i,c+e} - Y^{\infty}_{i,c+\bar{e}} \mid i \in \mathcal{J}_c] \\
&= \theta_{c,e}.
\end{aligned}$$

The second line uses no anticipation and the fact that controls remain untreated throughout the window. The third adds and subtracts the treated firms' counterfactual post-financing outcome, separating their treatment effect from the difference in untreated changes. Parallel trends makes the latter difference zero. For the paired comparisons in Table 49, we apply PT to the firms observed at both the reference month and the comparison month.

**Aggregation and regression weights.** A treated-firm-weighted summary of the cohort-event effects would be

$$\tau_e = \sum_{c \in C_L} \omega_c \theta_{c,e}, \qquad \omega_c = \frac{|\mathcal{T}_c|}{\sum_{c' \in C_L} |\mathcal{T}_{c'}|}, \qquad \tau_{\text{post}} = \frac{1}{L+1} \sum_{e=0}^{L} \tau_e,$$

where $C_L$ denotes the retained cohorts. This aggregation weights cohorts by treated-firm counts and post-financing months equally. The stacked least-squares specification below uses weights determined by the available comparisons.

In the stacked estimation sample, define $D_{ice} = \mathbf{1}\{E_i = c\}\mathbf{1}\{e \geq 0\}$. The baseline specification is

$$Y_{ice} = \beta D_{ice} + \alpha_{ic} + \lambda_{ce} + \varepsilon_{ice}.$$

Here $Y_{ice}$ corresponds to $M_{i,c,t}$ in Equation (7), with $t = c + e$; $D_{ice}$ corresponds to $VC_{i,c,t}$, and $\alpha_{ic}$ and $\lambda_{ce}$ correspond to $\chi_{c,i}$ and $\delta_{c,t}$. The fixed effects absorb firm differences and common monthly shocks separately within each cohort. Let $\widetilde{D}_{ice}$ be the residual from projecting $D_{ice}$ on these fixed effects, and let $O$ denote the observed estimation rows. The estimated coefficient satisfies

$$\widehat{\beta} = \frac{\sum_{(i,c,e) \in O} \widetilde{D}_{ice} Y_{ice}}{\sum_{(i,c,e) \in O} \widetilde{D}_{ice}^2}.$$

Writing $\tau_{ice} = Y^c_{i,c+e} - Y^\infty_{i,c+e}$ for a treated-post row and using NA gives the exact decomposition

$$\widehat{\beta} = \frac{\sum_O \widetilde{D}_{ice} Y^\infty_{i,c+e}}{\sum_O \widetilde{D}_{ice}^2} + \sum_{\substack{(i,c,e) \in O \\ D_{ice}=1}} w_{ice} \tau_{ice}, \qquad w_{ice} = \frac{\widetilde{D}_{ice}}{\sum_O \widetilde{D}_{ice}^2}.$$

The weights sum to one because the fixed-effect projection satisfies $\sum_O \widetilde{D}_{ice} D_{ice} = \sum_O \widetilde{D}_{ice}^2$. A sufficient conditional-mean condition for the expected untreated-outcome term to vanish is

$$\mathbb{E}\left[Y^\infty_{i,c+e} \mid O, \{E_j\}\right] = a_{ic} + \lambda_{ce} \quad \text{on the retained rows.}$$

This condition puts the expected untreated path, conditional on the complete observation history and financing dates, in the span of the fitted fixed effects. More generally, the required restriction is that its inner product with $\widetilde{D}$ is zero. It specifies the weighted comparison underlying the stacked DiD estimator, separately from PT for a population cohort-event contrast.

The available rows determine the weights in the DiD coefficient. With balanced cohorts and equal window lengths, cohort weights are proportional to $|\mathcal{T}_c||\mathcal{J}_c|/(|\mathcal{T}_c|+|\mathcal{J}_c|)$. In the fitted unbalanced baseline, 97 treated-post row weights are negative; their total absolute mass is less than 0.01% of the

net weight. The largest absolute row weight is 0.06% of the net weight. Concentration across original firms, measured as $10{,}000 \sum_i (\sum_{c,e} w_{ice})^2$, is 41.3; this diagnostic sums each firm's weights across its repeated cohort appearances before computing concentration. The 487,577 fitted observations follow removal of 18 fixed-effect singletons from the source stack. We report $\widehat{\beta}$ as the DiD coefficient and cluster standard errors by cohort and original firm to account for cohort shocks and firms reused across stacks.

For comparison, we also retain firm–cohort units observed in all 25 months and subtract each firm's pre-period mean from its post-period mean. Cohorts receive weights proportional to their retained treated-firm counts, and the same total weight is assigned to controls within each cohort. The resulting aggregate has nonnegative weights and fixes each firm's observation history. A second comparison differences each post observation from that firm's month $-1$ value, gives each post month equal weight, and fixes cohort weights at baseline treated-firm counts. Both preserve firm-and-cohort clustering. Table 49 reports these estimators and their respective supported populations. Appendix M.1 reports sensitivity to control reuse.

**Event-study implementation.** The quarterly event study replaces $D_{ice}$ with interactions between cohort treatment status and event-quarter indicators:

$$Y_{ice} = \sum_{q \in Q,\, q \neq 0} \beta_q \mathbf{1}\{E_i = c\}\mathbf{1}\{q(e) = q\} + \alpha_{ic} + \lambda_{ce} + \varepsilon_{ice},$$

where $Q$ is the set of observed event-quarter bins. The omitted bin $q = 0$ contains months $\{-4, -3, -2\}$. The next bin, $q = 1$, contains months $\{-1, 0, 1\}$ and combines one pre-treatment month with two months at or after financing. It therefore averages over the onset of treatment; later bins are entirely post-treatment. The coefficients are normalized relative to the omitted pre-treatment bin and inherit the stacked regression's weighting. The plots and joint tests describe the fitted pre-treatment coefficients. For a lead $q < 0$, let $r^q_{ice}$ be the residual from projecting its treated-quarter indicator on the fixed effects and all other event-quarter interactions. Under the conditional untreated-outcome restriction above and no anticipation,

$$\mathbb{E}[\widehat{\beta}_q \mid O, \{E_j\}] = \frac{\sum_{O: D_{ice}=1} r^q_{ice}\, \mathbb{E}[\tau_{ice} \mid O, \{E_j\}]}{\sum_O (r^q_{ice})^2}.$$

Zero-centered leads therefore also require this weighted post-treatment-effect term to vanish. A sufficient restriction is that the post-treatment effect vector lies in the span of the other included event interactions and fixed effects. With unbalanced coverage, heterogeneous responses can enter the fitted leads through this projection. The static treated-post weight diagnostics describe a different regressor and do not verify this event-study restriction.

# D Matching on Pre-Treatment Growth and Quality Trends

We match within financing cohorts on growth measured before the round. For firm $i$ in cohort $c$, let $H_{i,c,e}$ denote headcount at event month $e$. The matching feature is

$$G_i^c = \ln\left(\frac{H_{i,c,-4} + H_{i,c,-3} + H_{i,c,-2}}{3}\right) - \ln\left(\frac{H_{i,c,-12} + H_{i,c,-11}}{2}\right).$$

This feature compares average headcount near the beginning and end of the pre-financing period, excluding the month immediately before the round to reduce sensitivity to deal-adjacent hiring. Sensitivity specifications add analogously constructed growth in software-engineering headcount and lines of code, making $G_i^c$ a vector.

Using the notation of Appendix C, the candidate controls satisfy $E_j > c + 12$. We require an observation at $e = -1$ and nonmissing matching features for both treated firms and candidate controls. This eligibility step retains 112 treated firms and 206 candidate controls in 70 cohorts. The distance cutoff and subsequent requirement for both arms retain 110 treated firms and 196 controls in 68 cohorts; the preferred matched regression has 71,837 observations. Table 55 compares the eligible and final treated populations before and after selecting controls. Let $\bar{G}_c^{\mathcal{T}}$ be the mean feature vector among eligible treated firms. In the primary matched specification, the distance between a candidate control and this treated-group mean is

$$d_c(j) = \sqrt{(G_j^c - \bar{G}_c^{\mathcal{T}})' \widehat{\Sigma}_c^{-1} (G_j^c - \bar{G}_c^{\mathcal{T}})},$$

where $\widehat{\Sigma}_c$ is the sample covariance of the candidate-control vectors together with the treated-group mean. With headcount growth as the sole feature, this is the absolute growth difference divided by the corresponding sample standard deviation.

We retain controls with $d_c(j) \leq 1$, ranked by distance, and keep the available event-window observations for eligible treated firms and selected controls. Each control enters at most once within a cohort but can appear in several cohorts, and we discard any cohort left without both groups. Matching to the treated-group mean retains more eligible controls than individual one-to-one matching. In the fitted primary population, the mean treated-minus-control headcount-growth gap is 0.06 log points, averaging within-cohort gaps with weights proportional to fitted treated-firm counts. The distance cutoff therefore narrows the comparison without imposing equality of the two groups' mean growth.

Appendix E examines whether this result depends on the particular definition of a comparable firm. We vary the distance cutoff from 0.5 to 2.0, add growth in code volume and software-engineering headcount, and replace matching to the treated-group mean with individual nearest-neighbor matching at $K \in \{1, 3\}$, selecting controls sequentially without replacement within a cohort. These alternatives change both the dimensions along which firms are matched and how tightly their trajectories must

align, allowing us to assess whether the quality improvement persists across plausible comparison groups.

### Matching on Engineering-Headcount Growth

We also match on pre-financing engineering-headcount growth alone, using the same within-cohort procedure, pre-treatment windows and distance cutoff as the primary match. Figure 16 compares engineering-headcount trajectories under matching on total-headcount growth (Panel A) and engineering-headcount growth (Panel B). By the end of the event window, the growth rate in engineering headcount is approximately 51 percentage points greater than in controls under the primary match and 53 percentage points greater under the engineering-growth match (both $p < 0.01$).

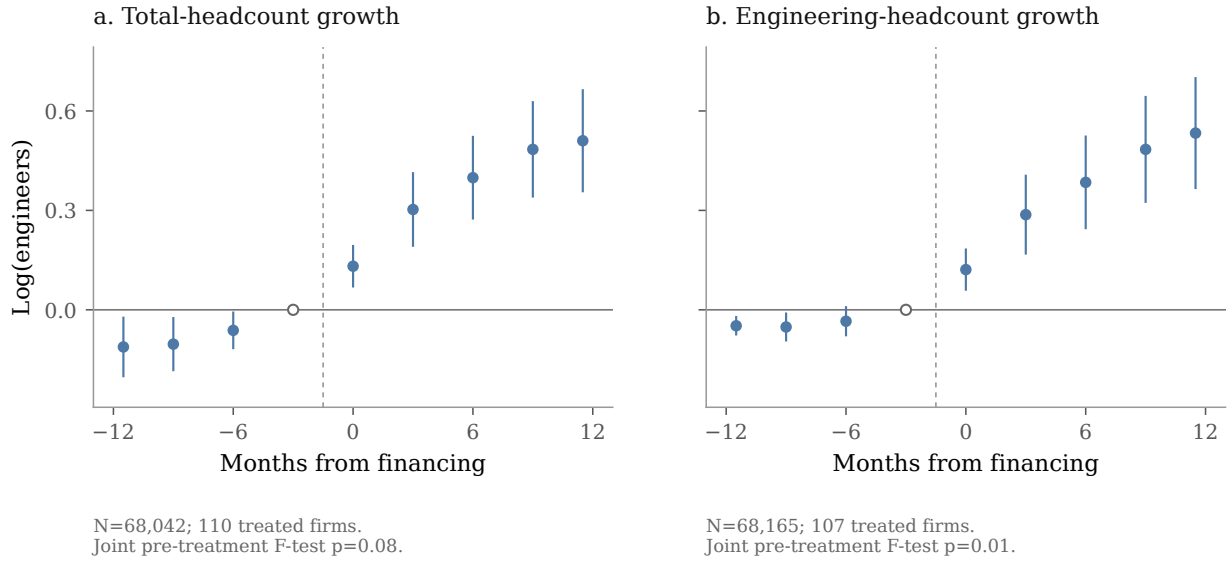



Figure 16: Engineering Headcount under Alternative Growth Matches. The outcome is the natural logarithm of positive software-engineering headcount. Panel A uses the primary match on total-headcount growth (110 treated firms in 68 cohorts; 68,042 stacked observations); Panel B matches on engineering-headcount growth (107 treated firms in 67 cohorts; 68,165 stacked observations). Joint tests of the three pre-treatment coefficients give $F(3, 67) = 2.31$ ($p = 0.08$) and $F(3, 66) = 3.93$ ($p = 0.01$), respectively. Both panels use the same axis scales. Controls are selected within financing cohorts using growth between months $[-12, -11]$ and $[-4, -2]$, group-mean matching and a Mahalanobis distance cutoff of one. Regressions include firm-by-cohort and month-by-cohort fixed effects. Quarterly estimates are relative to months $-4$ to $-2$; the zero bin spans months $-1$ to $+1$. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

### Matching on Pre-Financing Violation-Density Changes

The baseline pre-financing point estimates are noisy but may suggest an earlier decline in violation density relative to controls. We therefore examine whether the post-financing decline persists after matching controls on the pre-financing change in total violation density, measured as violations per 1,000 non-comment lines without taking logs. The matching feature is the mean density in months $[-4, -2]$ minus its mean in months $[-12, -11]$, requiring all five observations and an observation at month $-1$. Controls are selected once within each financing cohort using the treated-group mean and a Mahalanobis distance cutoff of one. Figure 17 compares the baseline event study with this additional match. After matching, the pre-financing estimates are closer to zero and the post-financing decline persists.

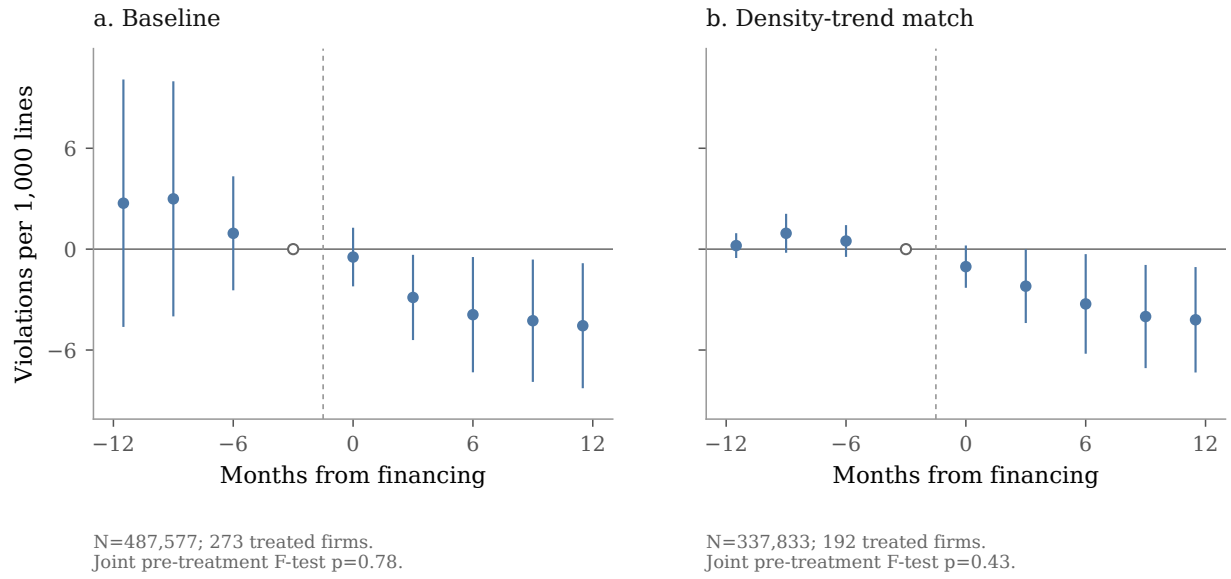



Figure 17: Total Violation Density under Pre-Financing Trend Matching. The outcome is total violations per 1,000 non-comment lines, without logs. Panel A reports the baseline (273 treated firms in 99 cohorts; 487,577 stacked observations). Panel B matches on the pre-financing density change (192 treated firms in 89 cohorts; 337,833 stacked observations). The panels use identical axis scales. Joint tests of the three pre-treatment coefficients give $F(3, 98) = 0.35$ ($p = 0.78$) and $F(3, 88) = 0.91$ ($p = 0.43$), respectively. Neither joint test is statistically significant. Regressions include firm-by-cohort and month-by-cohort fixed effects. Quarterly estimates are relative to months $-4$ to $-2$; the zero bin spans months $-1$ to $+1$. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

### Matching on Pre-Financing Maintainability-Density Changes

We match on the change in average maintainability issues per 1,000 non-comment lines between months $[-12, -11]$ and months $[-4, -2]$, under the traditional classification used in Figure 4. We

require all five monthly observations and an observation at month −1. Within each financing cohort, we select controls within a Mahalanobis distance of one from the treated-group mean and keep membership fixed throughout the event window. The data requirements leave 195 eligible treated firms; two more are excluded because their cohorts have no controls within the cutoff. The resulting sample contains 193 treated firms in 90 cohorts. Figure 18 compares the baseline and matched event studies on identical axes.

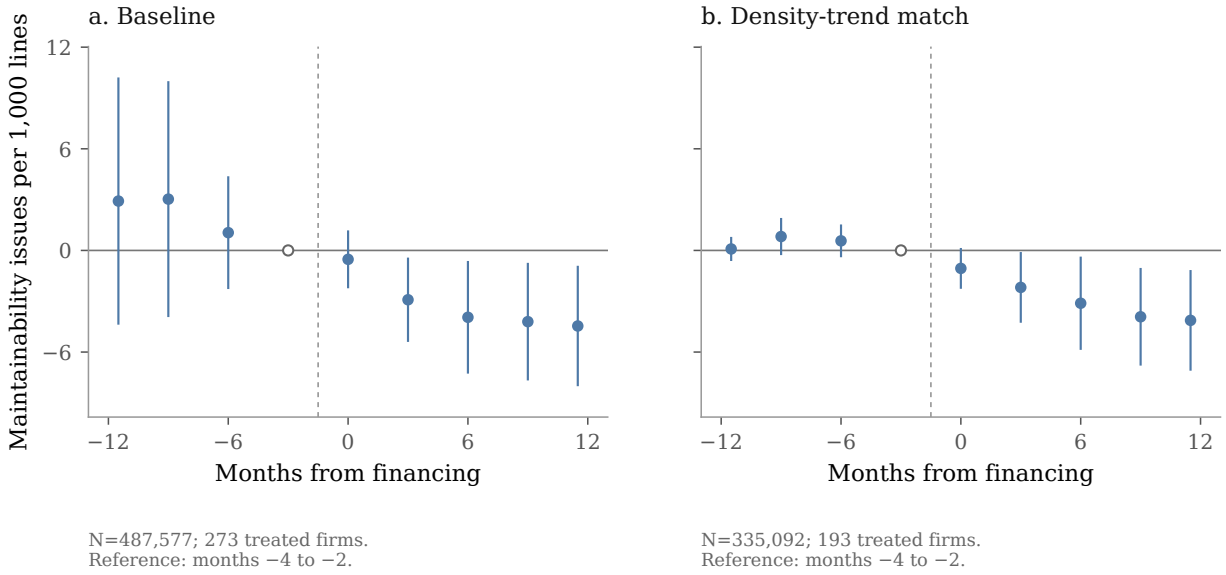



Figure 18: Maintainability Density under Pre-Financing Trend Matching. The outcome is maintainability issues per 1,000 non-comment lines under the traditional classification, without logs. Panel A reports the baseline (273 treated firms in 99 cohorts; 487,577 stacked observations). Panel B matches on the pre-financing density change (193 treated firms in 90 cohorts; 335,092 stacked observations). The panels use identical axis scales. The open marker denotes the omitted months −4 to −2, plotted at month −3. Regressions include firm-by-cohort and month-by-cohort fixed effects. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

# E Matching Sensitivity Analysis

We examine whether the matched engineering-quality estimates depend on how controls are selected. We construct a specification curve (Simonsohn, Simmons, and Nelson, 2020) over 72 configurations that vary three choices. First, the matching features include growth in total headcount alone, headcount plus lines of code, headcount plus software-engineering headcount, or all three measures. Second, we match controls either to the cohort's treated-group mean (centroid matching) or to individual treated

firms, selecting one or up to three nearest neighbors. Third, the maximum Mahalanobis distance, or caliper, ranges from 0.5 to 2.0 ($\delta \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$). Within a cohort, each control is used at most once; it can appear in different cohorts. We re-estimate Equation (7) on the matched sample for each configuration.

Figures 19–22 present the specification curves for our four core engineering-quality outcomes. In each figure, the top panel plots the point estimate and 95% confidence interval for every specification, grouped by feature set, matching mode, and caliper in the same order across outcomes. The dashed gray line marks the baseline (unmatched) estimate and its 95% confidence band (gray shading); the solid gray line marks zero. The orange diamond identifies our primary matched specification (centroid matching on headcount growth, caliper $\delta = 1$). The bottom panel displays an indicator matrix identifying the feature set, matching mode, and caliper active in each specification.

The direction of the quality estimates is stable across matching choices. Every critical-violation, total-violation, and maintainability-violation coefficient is negative, as are 68 of the 72 technical-debt coefficients. At the 5% level, 26% of critical-violation specifications, 61% of total-violation specifications, and 61% of maintainability-violation specifications reject the null; the corresponding shares at the 10% level are 85%, 74%, and 79%. No technical-debt specification reaches the 5% level, although six reach the 10% level.

Matching substantially reduces the sample: the primary specification retains approximately 72,000 of 488,000 baseline observations, and some individual nearest-neighbor configurations retain fewer than 10,000. Despite this contraction, the median matched-to-baseline coefficient ratio is 1.38 for critical violations, 0.90 for total violations, and 0.85 for maintainability violations. The stability of these magnitudes, alongside wider confidence intervals, indicates that reduced statistical power contributes to the loss of significance. By contrast, technical debt is less stable, with a median ratio of 0.63 and no estimate significant at the 5% level.

Eligibility varies with the matching features. To distinguish this population change from control selection, we also retain the same 37 treated firms in 21 cohorts across all 72 configurations, require complete 25-month observations for both arms, and fix cohort weights at treated-firm counts. For each firm we compare the average of months 0–12 with that of months −12 to −1. Critical-violation estimates are negative in all 72 comparisons, ranging from −4.6 to −0.5 per 1,000 lines. Total-violation estimates range from −16.7 to +3.0 (46 negative), maintainability from −14.9 to +1.8 (59 negative), and debt from −7.10 to +1.09 dollars per line (20 negative). This fixed-population comparison uses nonnegative aggregation weights and a smaller treated population than the stacked DiD specification curves above.

Table 20 reports selected treated and control firms, cohorts and monthly rows for every configuration, distinguishing these membership counts from fitted observations after singleton deletion.

Table 20: Sample Sizes across the 72 Matching Configurations

| Features | Matching | Caliper | Treated | Controls | Cohorts | Rows |
|---|---|---|---|---|---|---|
| Headcount + SWE | Group mean | 0.50 | 65 | 141 | 41 | 13,330 |
| Headcount + SWE | Group mean | 0.75 | 79 | 161 | 49 | 23,850 |
| Headcount + SWE | Group mean | 1.00 | 96 | 181 | 61 | 35,229 |
| Headcount + SWE | Group mean | 1.25 | 102 | 191 | 63 | 46,891 |
| Headcount + SWE | Group mean | 1.50 | 103 | 193 | 64 | 56,775 |
| Headcount + SWE | Group mean | 2.00 | 107 | 196 | 66 | 82,131 |
| Headcount + SWE | One neighbor | 0.50 | 87 | 52 | 50 | 3,864 |
| Headcount + SWE | One neighbor | 0.75 | 92 | 57 | 54 | 4,239 |
| Headcount + SWE | One neighbor | 1.00 | 104 | 69 | 64 | 4,885 |
| Headcount + SWE | One neighbor | 1.25 | 106 | 71 | 65 | 5,035 |
| Headcount + SWE | One neighbor | 1.50 | 107 | 71 | 66 | 5,134 |
| Headcount + SWE | One neighbor | 2.00 | 107 | 72 | 66 | 5,209 |
| Headcount + SWE | Three neighbors | 0.50 | 87 | 95 | 50 | 6,419 |
| Headcount + SWE | Three neighbors | 0.75 | 92 | 103 | 54 | 7,194 |
| Headcount + SWE | Three neighbors | 1.00 | 104 | 114 | 64 | 8,340 |
| Headcount + SWE | Three neighbors | 1.25 | 106 | 123 | 65 | 9,065 |
| Headcount + SWE | Three neighbors | 1.50 | 107 | 126 | 66 | 9,464 |
| Headcount + SWE | Three neighbors | 2.00 | 107 | 128 | 66 | 9,989 |
| Headcount + code | Group mean | 0.50 | 78 | 151 | 48 | 19,668 |
| Headcount + code | Group mean | 0.75 | 97 | 176 | 58 | 35,204 |
| Headcount + code | Group mean | 1.00 | 103 | 189 | 62 | 47,487 |
| Headcount + code | Group mean | 1.25 | 108 | 195 | 66 | 60,053 |
| Headcount + code | Group mean | 1.50 | 108 | 198 | 66 | 72,707 |
| Headcount + code | Group mean | 2.00 | 109 | 201 | 67 | 97,785 |
| Headcount + code | One neighbor | 0.50 | 98 | 62 | 56 | 4,477 |
| Headcount + code | One neighbor | 0.75 | 103 | 63 | 61 | 4,802 |
| Headcount + code | One neighbor | 1.00 | 105 | 65 | 63 | 4,977 |
| Headcount + code | One neighbor | 1.25 | 108 | 68 | 66 | 5,201 |
| Headcount + code | One neighbor | 1.50 | 108 | 71 | 66 | 5,276 |
| Headcount + code | One neighbor | 2.00 | 109 | 74 | 67 | 5,376 |
| Headcount + code | Three neighbors | 0.50 | 98 | 113 | 56 | 7,746 |
| Headcount + code | Three neighbors | 0.75 | 103 | 123 | 61 | 8,655 |
| Headcount + code | Three neighbors | 1.00 | 105 | 126 | 63 | 9,155 |
| Headcount + code | Three neighbors | 1.25 | 108 | 129 | 66 | 9,677 |
| Headcount + code | Three neighbors | 1.50 | 108 | 133 | 66 | 9,927 |
| Headcount + code | Three neighbors | 2.00 | 109 | 136 | 67 | 10,352 |
| All three | Group mean | 0.50 | 47 | 93 | 28 | 5,624 |
| All three | Group mean | 0.75 | 62 | 138 | 36 | 13,090 |
| All three | Group mean | 1.00 | 70 | 164 | 42 | 23,043 |
| All three | Group mean | 1.25 | 87 | 181 | 54 | 32,694 |
| All three | Group mean | 1.50 | 97 | 185 | 60 | 41,756 |
| All three | Group mean | 2.00 | 106 | 194 | 65 | 65,941 |
| All three | One neighbor | 0.50 | 66 | 40 | 36 | 2,736 |
| All three | One neighbor | 0.75 | 78 | 49 | 42 | 3,386 |
| All three | One neighbor | 1.00 | 88 | 55 | 49 | 3,907 |
| All three | One neighbor | 1.25 | 97 | 63 | 58 | 4,482 |
| All three | One neighbor | 1.50 | 101 | 66 | 62 | 4,756 |
| All three | One neighbor | 2.00 | 106 | 68 | 65 | 5,031 |
| All three | Three neighbors | 0.50 | 66 | 78 | 36 | 4,267 |

*Table continued*

| Features | Matching | Caliper | Treated | Controls | Cohorts | Rows |
|---|---|---|---|---|---|---|
| All three | Three neighbors | 0.75 | 78 | 91 | 42 | 5,317 |
| All three | Three neighbors | 1.00 | 88 | 107 | 49 | 6,488 |
| All three | Three neighbors | 1.25 | 97 | 114 | 58 | 7,586 |
| All three | Three neighbors | 1.50 | 101 | 123 | 62 | 8,360 |
| All three | Three neighbors | 2.00 | 106 | 128 | 65 | 9,510 |
| Headcount | Group mean | 0.50 | 109 | 183 | 67 | 42,188 |
| Headcount | Group mean | 0.75 | 109 | 193 | 67 | 58,222 |
| Headcount | Group mean | 1.00 | 110 | 196 | 68 | 71,837 |
| Headcount | Group mean | 1.25 | 110 | 200 | 68 | 84,865 |
| Headcount | Group mean | 1.50 | 110 | 201 | 68 | 97,090 |
| Headcount | Group mean | 2.00 | 110 | 202 | 68 | 117,394 |
| Headcount | One neighbor | 0.50 | 109 | 70 | 67 | 5,230 |
| Headcount | One neighbor | 0.75 | 109 | 71 | 67 | 5,255 |
| Headcount | One neighbor | 1.00 | 110 | 74 | 68 | 5,379 |
| Headcount | One neighbor | 1.25 | 110 | 74 | 68 | 5,404 |
| Headcount | One neighbor | 1.50 | 110 | 74 | 68 | 5,404 |
| Headcount | One neighbor | 2.00 | 110 | 74 | 68 | 5,429 |
| Headcount | Three neighbors | 0.50 | 109 | 123 | 67 | 9,962 |
| Headcount | Three neighbors | 0.75 | 109 | 123 | 67 | 10,112 |
| Headcount | Three neighbors | 1.00 | 110 | 126 | 68 | 10,336 |
| Headcount | Three neighbors | 1.25 | 110 | 126 | 68 | 10,411 |
| Headcount | Three neighbors | 1.50 | 110 | 126 | 68 | 10,436 |
| Headcount | Three neighbors | 2.00 | 110 | 127 | 68 | 10,636 |

*Note:* Counts describe the selected samples before fixed-effect singleton deletion, after matching and the both-arm cohort rule. Treated and control counts are unique original firms by role and can overlap across cohorts. Rows are firm–cohort–month observations. Features are pre-event growth in total headcount, non-comment lines of code and software-engineering headcount (SWE). Group mean matches to the cohort's average treated profile; neighbor methods select one or up to three controls per treated firm without replacement within cohort. Calipers are Mahalanobis-distance units. The preferred configuration is headcount, group mean, caliper one; its selected and fitted row counts both equal 71,837. The separate common-population comparison fixes 37 treated firms and 21 cohorts across configurations.

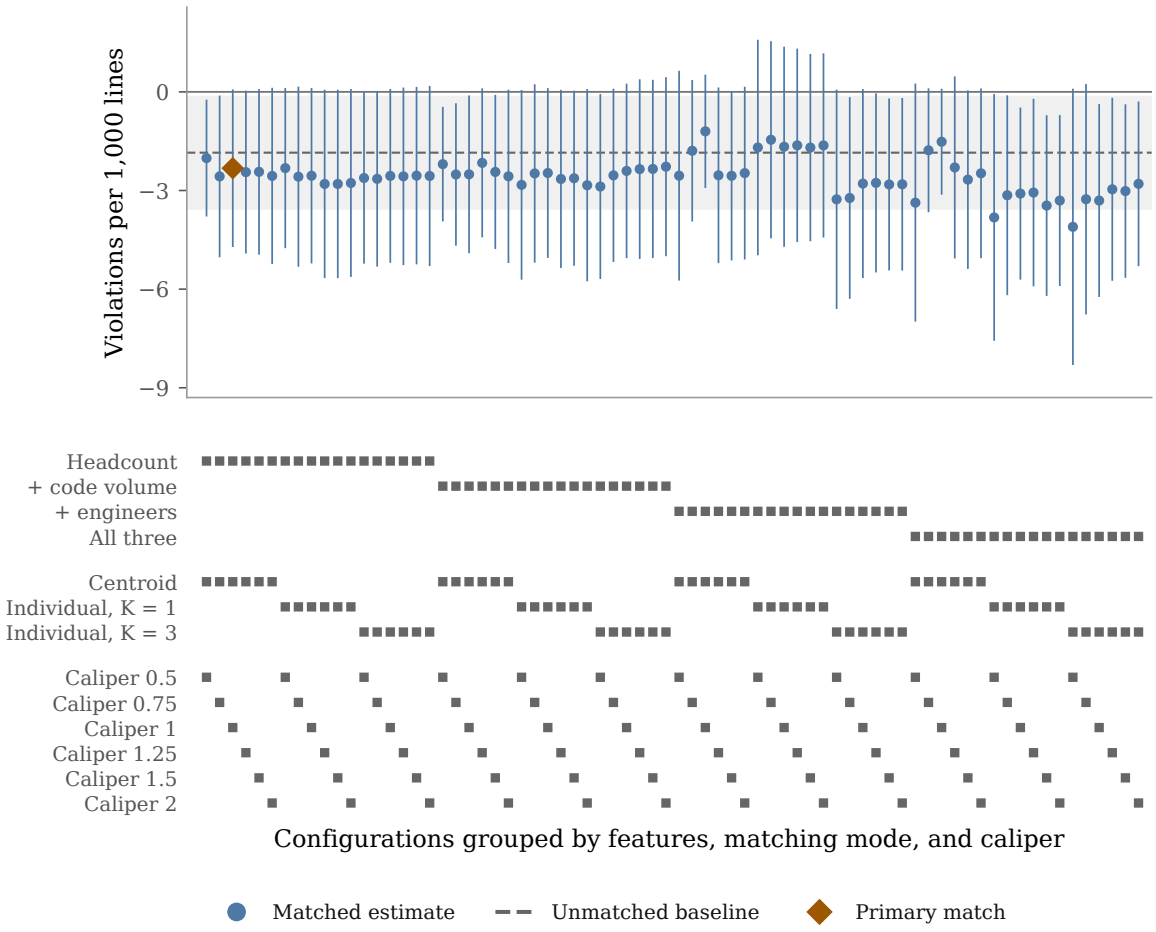



Figure 19: Specification Curve: Critical Violations per 1,000 LoC. Estimates across 72 matching configurations, identified in the lower panel. The dashed line and shaded band show the unmatched estimate and its interval; the orange diamond marks the preferred matched specification. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

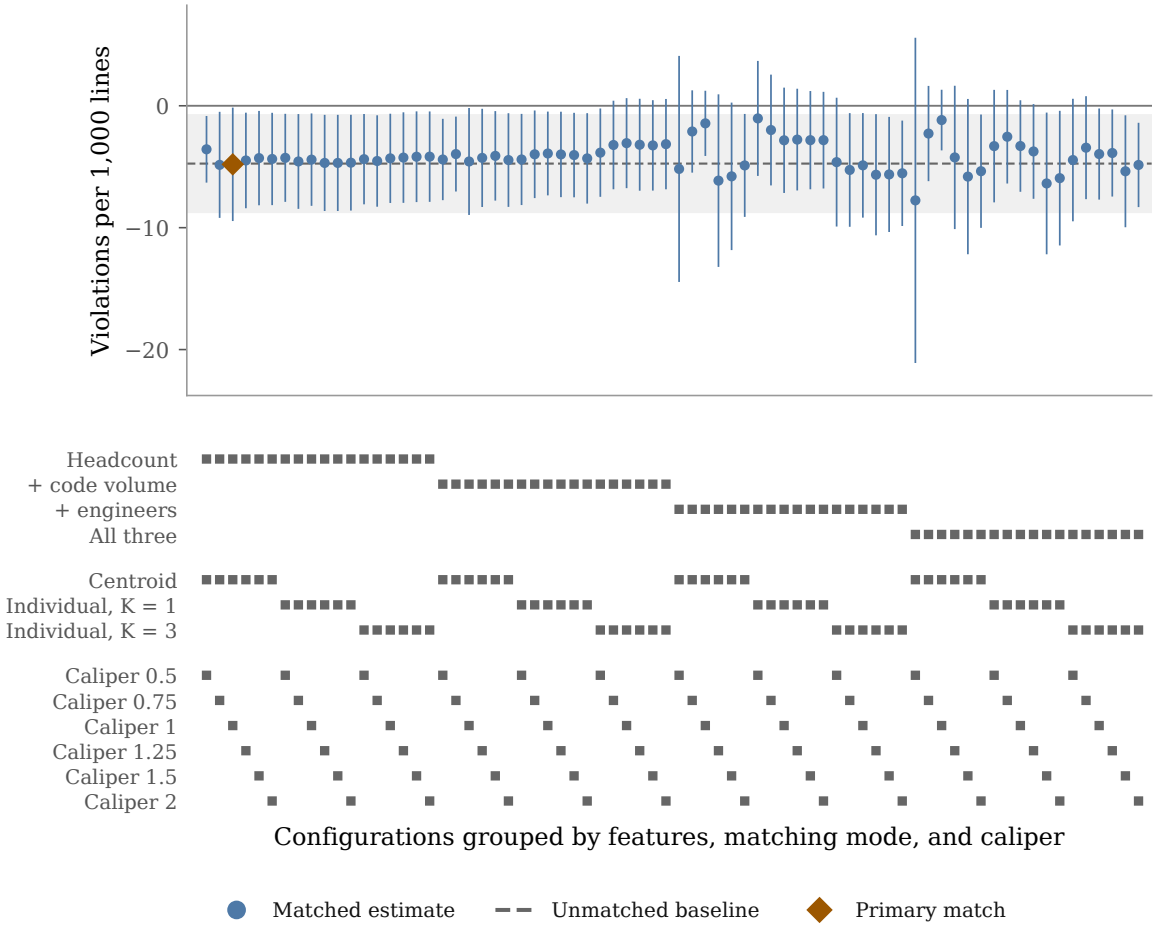



Figure 20: Specification Curve: Violations (All) per 1,000 LoC. See Figure 19 for a description of the layout.

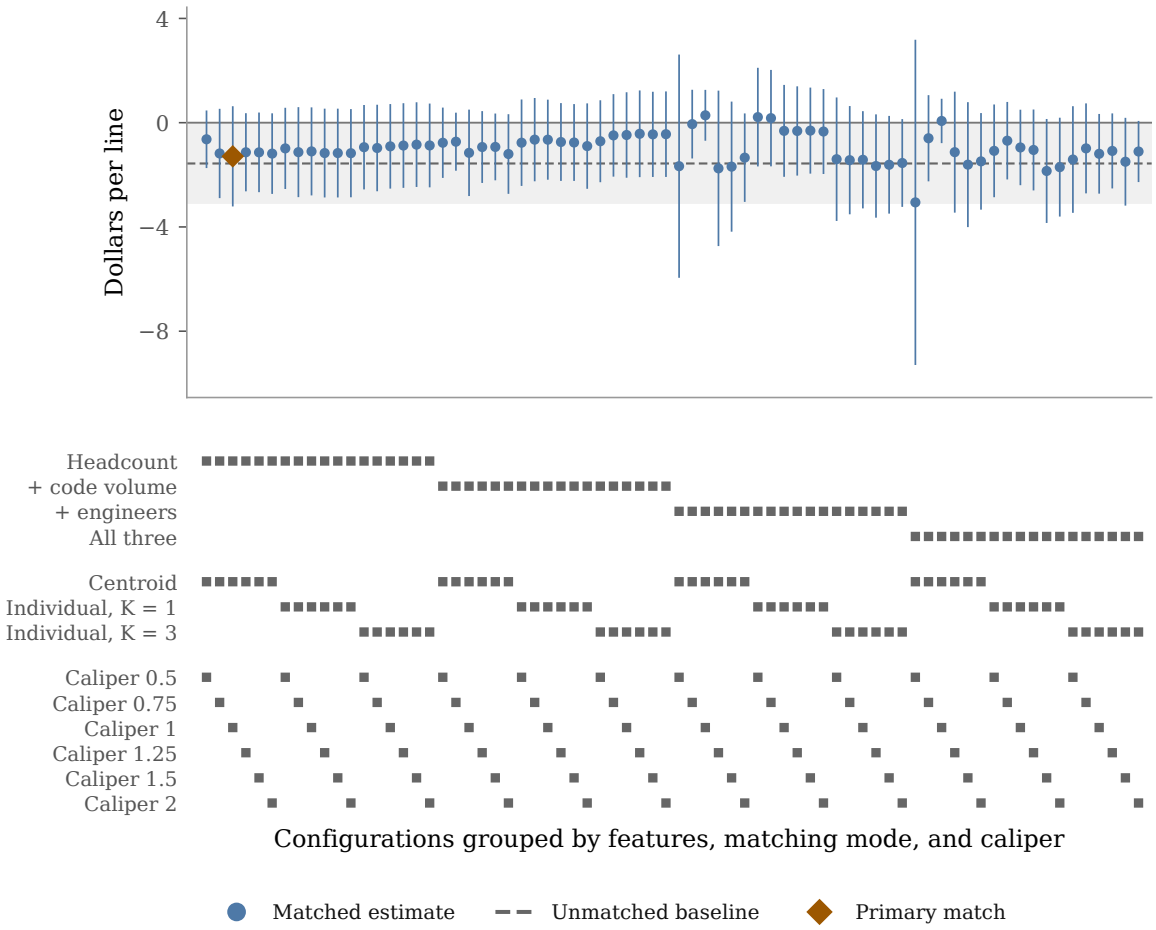



Figure 21: Specification Curve: Technical Debt per Line of Code. See Figure 19 for a description of the layout.

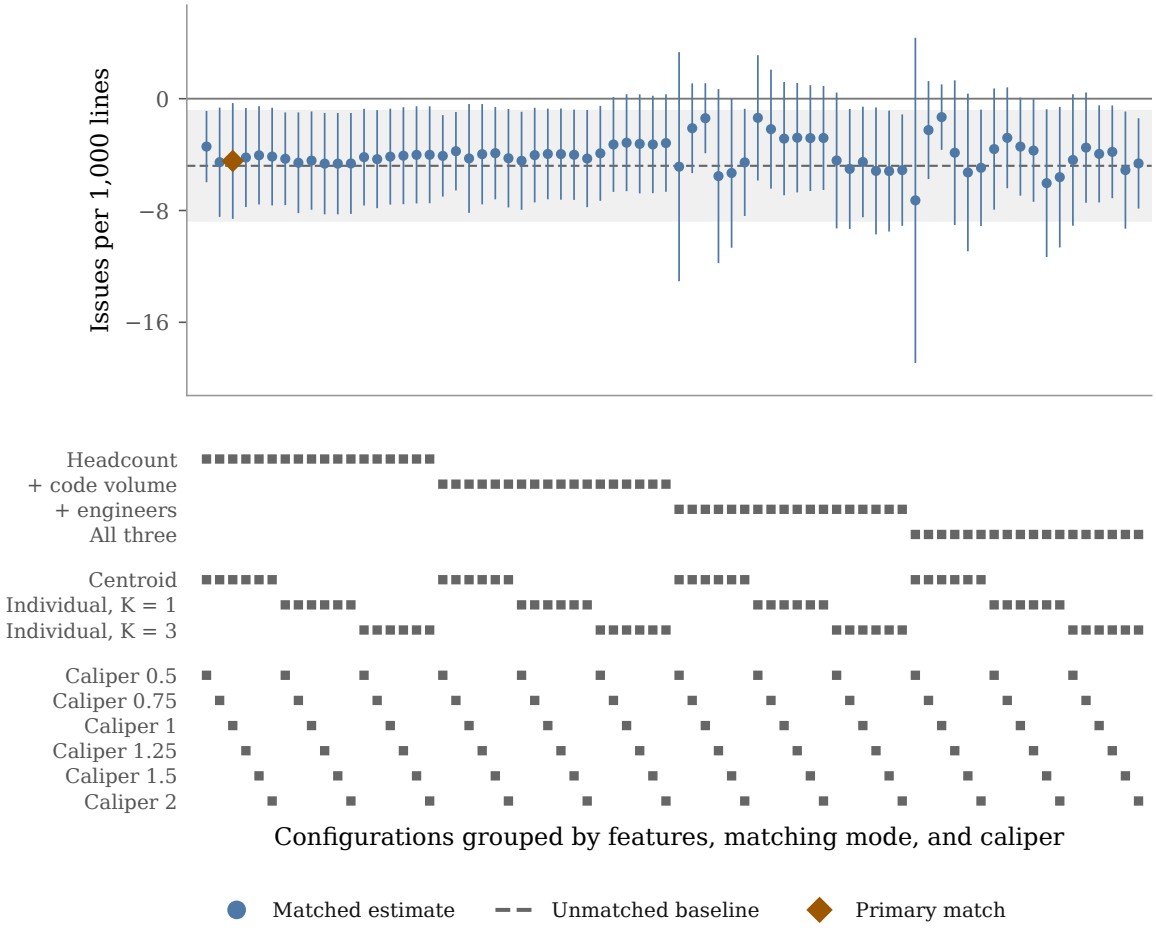



Figure 22: Specification Curve: Maintainability Violations per 1,000 LoC. See Figure 19 for a description of the layout.

## E.1 Outcome Paths under Alternative Comparison Groups

Figures 23–28 compare all 16 outcomes using the unmatched sample and the primary match on pre-financing total-headcount growth. Log-transformed outcomes include only observations with positive values.

The matching feature is the log of mean total headcount in months −4, −3, and −2 minus the log of mean total headcount in months −12 and −11. Within each cohort, controls are retained if their standardized distance from the eligible treated firms' mean growth is at most one. Both comparisons use unit regression weights. Appendix D gives the full eligibility and matching procedure. Panel notes report each regression's actual sample after missing-value and fixed-effect exclusions.

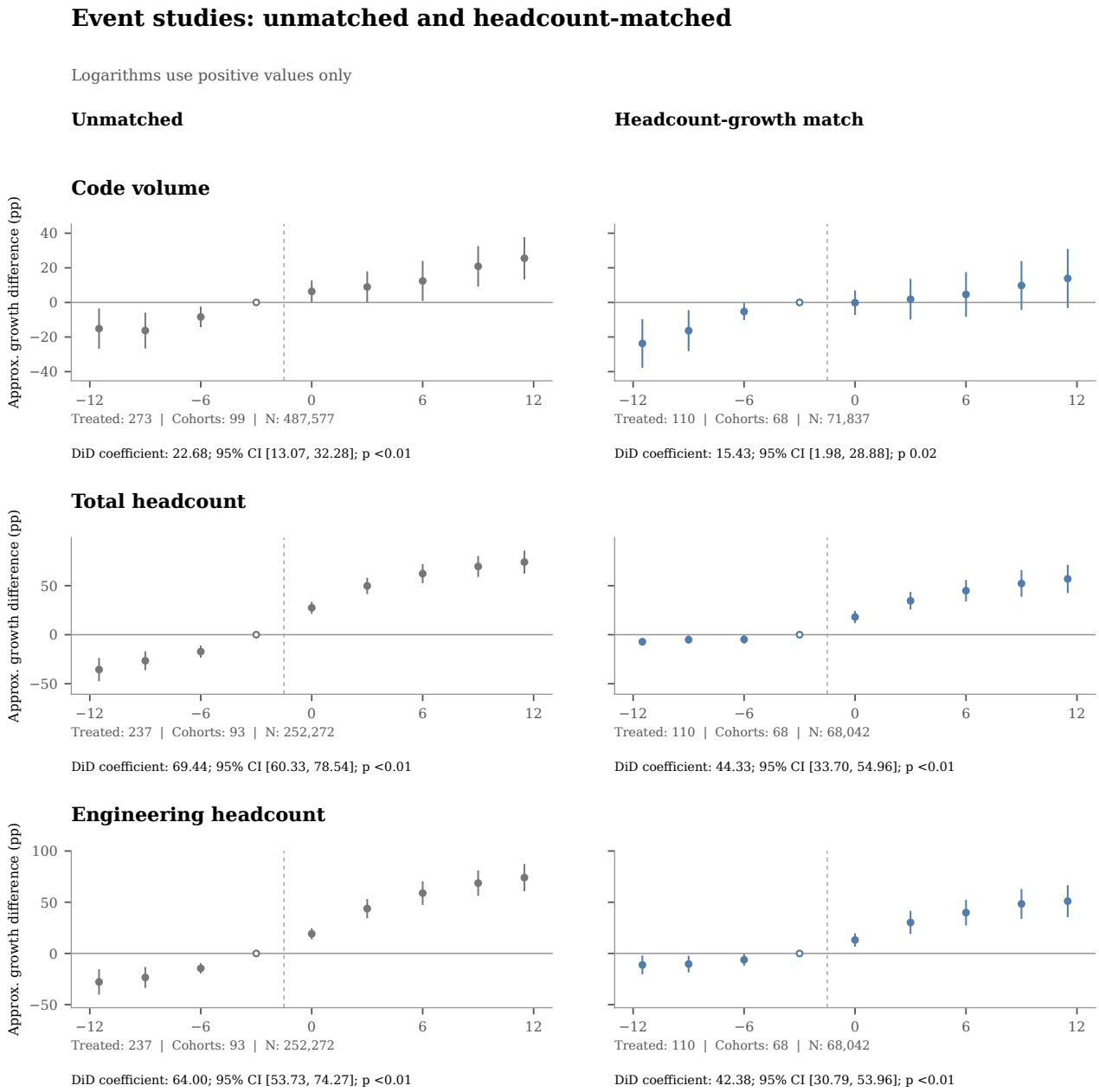



Figure 23: Alternative Comparison Groups: Part 1. Code volume, total headcount, and engineering headcount. Columns report unmatched stacked DiD and headcount-growth matching. Axes match within each row; error bars are 95% confidence intervals with standard errors clustered by cohort and firm. The dashed line marks the end of omitted months −4 through −2. Logged outcomes display approximate percentage-point growth differences relative to controls.

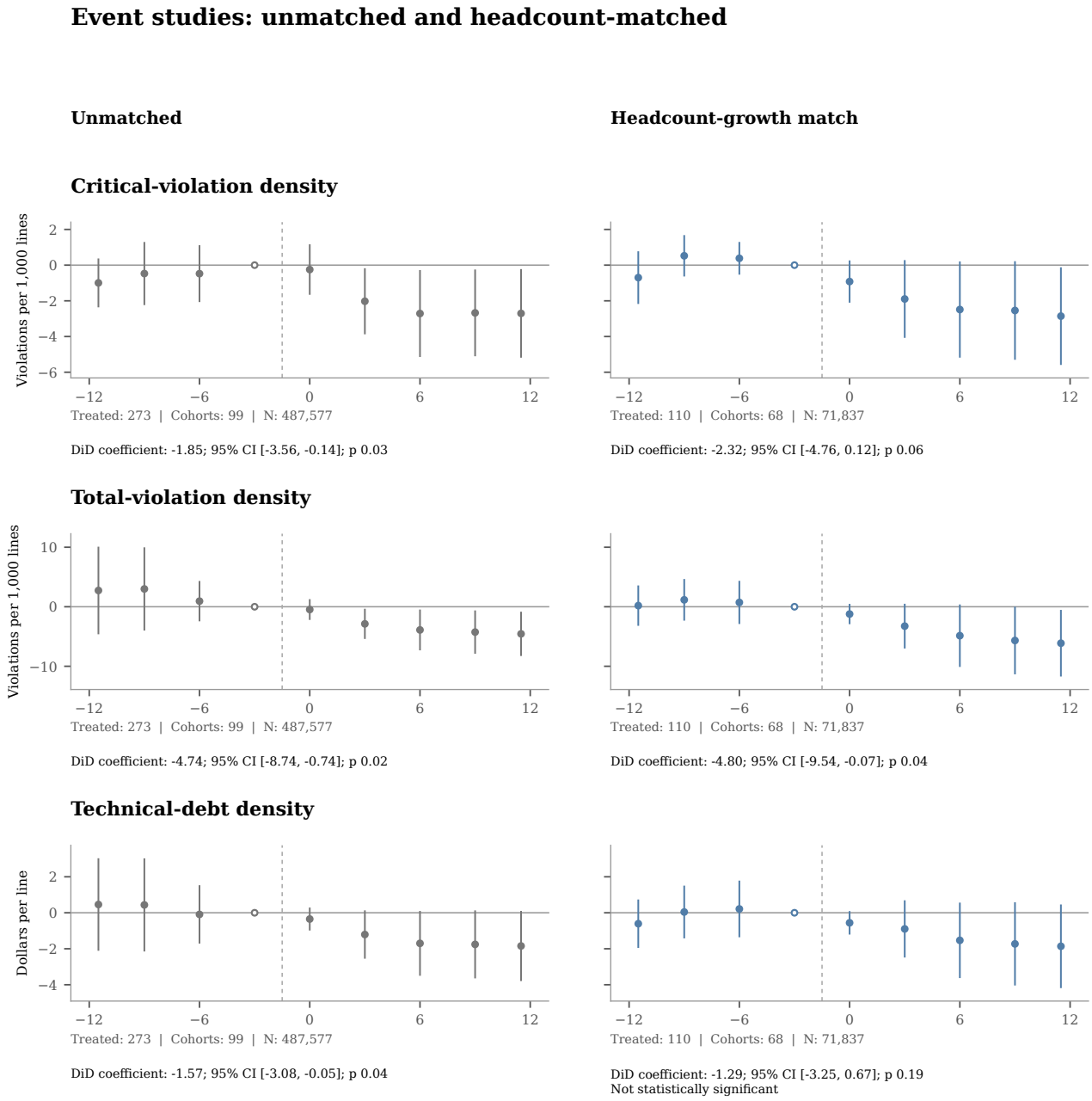



Figure 24: Alternative Comparison Groups: Part 2. Critical-violation density, total-violation density, and technical debt per line. Columns report unmatched stacked DiD and headcount-growth matching. Axes match within each row; error bars are 95% confidence intervals with standard errors clustered by cohort and firm. The dashed line marks the end of omitted months −4 through −2.

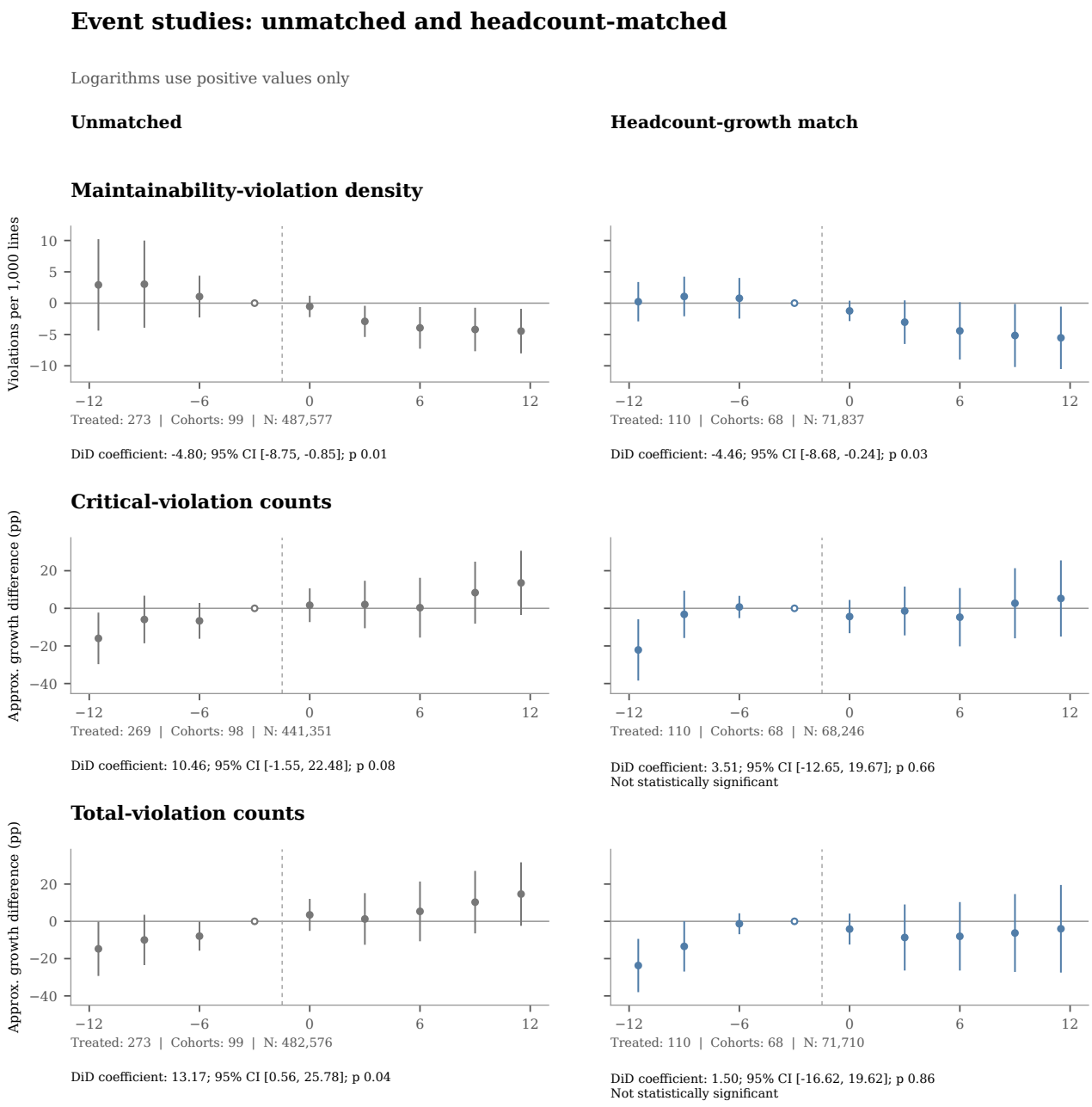



Figure 25: Alternative Comparison Groups: Part 3. Maintainability-violation density, critical-violation counts, and total-violation counts. Columns report unmatched stacked DiD and headcount-growth matching. Axes match within each row; error bars are 95% confidence intervals with standard errors clustered by cohort and firm. The dashed line marks the end of omitted months −4 through −2. Logged outcomes display approximate percentage-point growth differences relative to controls.

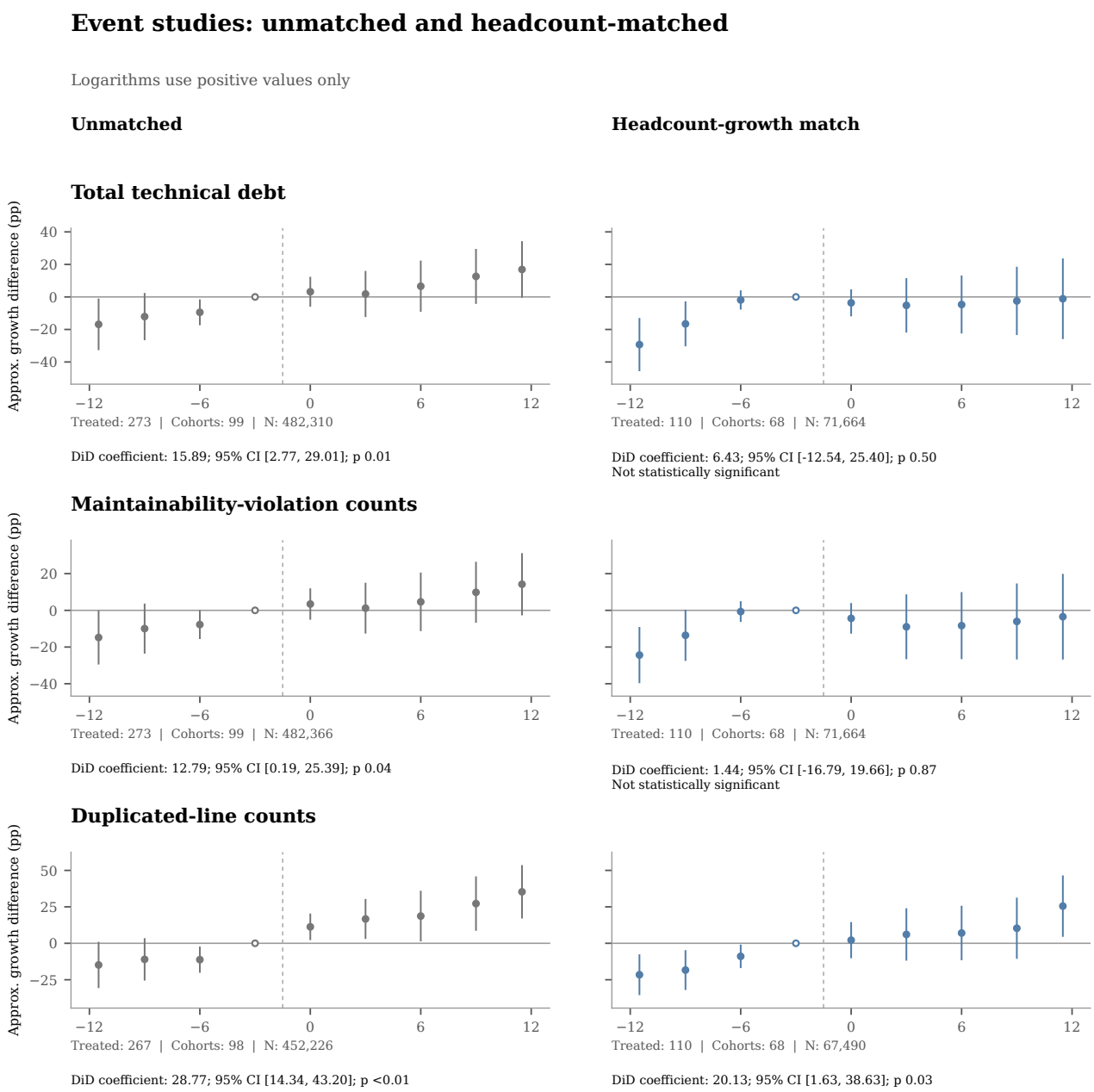



Figure 26: Alternative Comparison Groups: Part 4. Total technical debt, maintainability-violation counts, and duplicated-line counts. Columns report unmatched stacked DiD and headcount-growth matching. Axes match within each row; error bars are 95% confidence intervals with standard errors clustered by cohort and firm. The dashed line marks the end of omitted months −4 through −2. Logged outcomes display approximate percentage-point growth differences relative to controls.

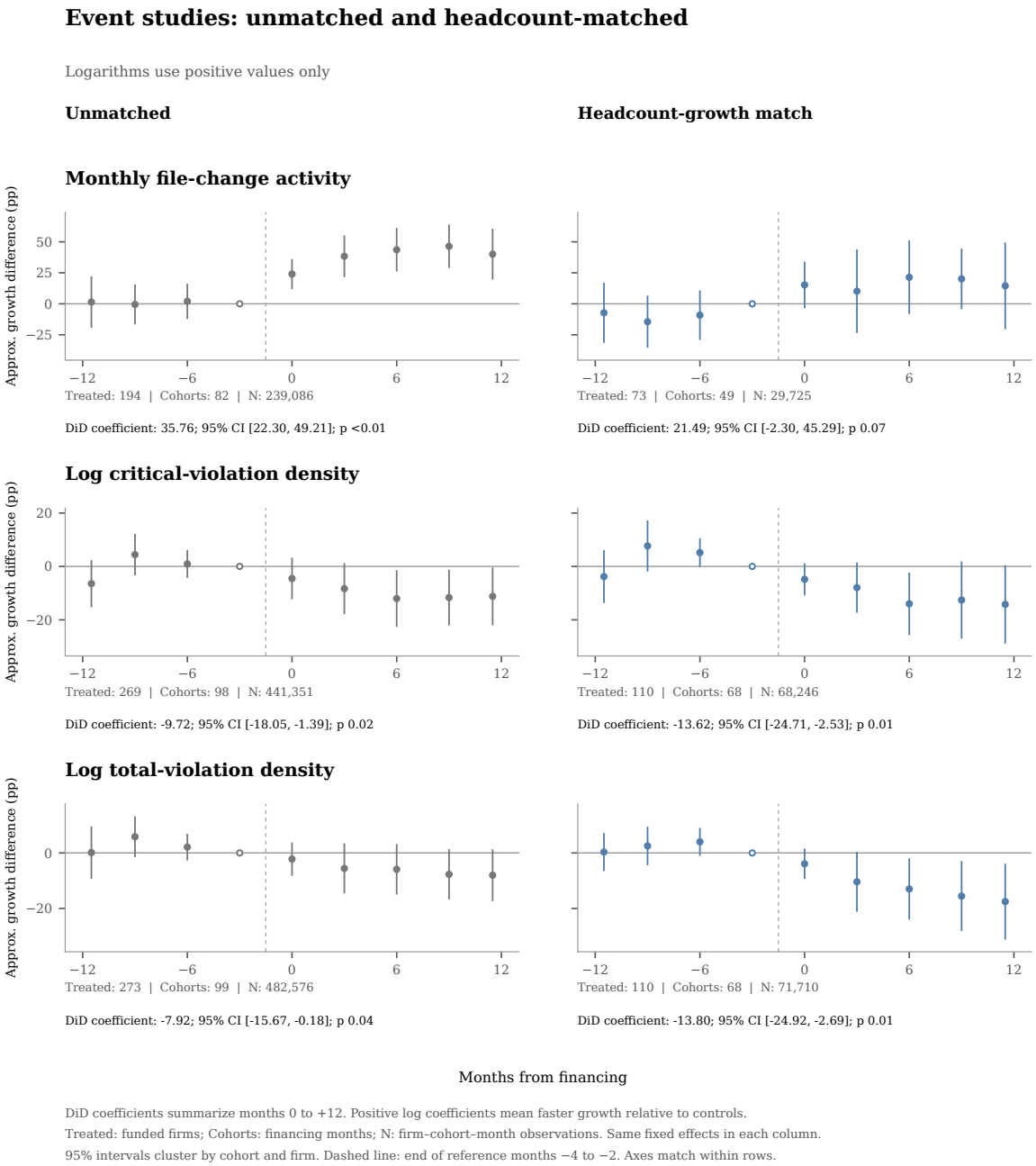



Figure 27: Alternative Comparison Groups: Part 5. Monthly file-change activity and logged critical- and total-violation densities. Columns report unmatched stacked DiD and headcount-growth matching. Axes match within each row; error bars are 95% confidence intervals with standard errors clustered by cohort and firm. The dashed line marks the end of omitted months −4 through −2. Logged outcomes display approximate percentage-point growth differences relative to controls. The file-change comparison uses the main sample with file-activity coverage, distinct from the sample of firms with one listed commercialized project in Table 18.

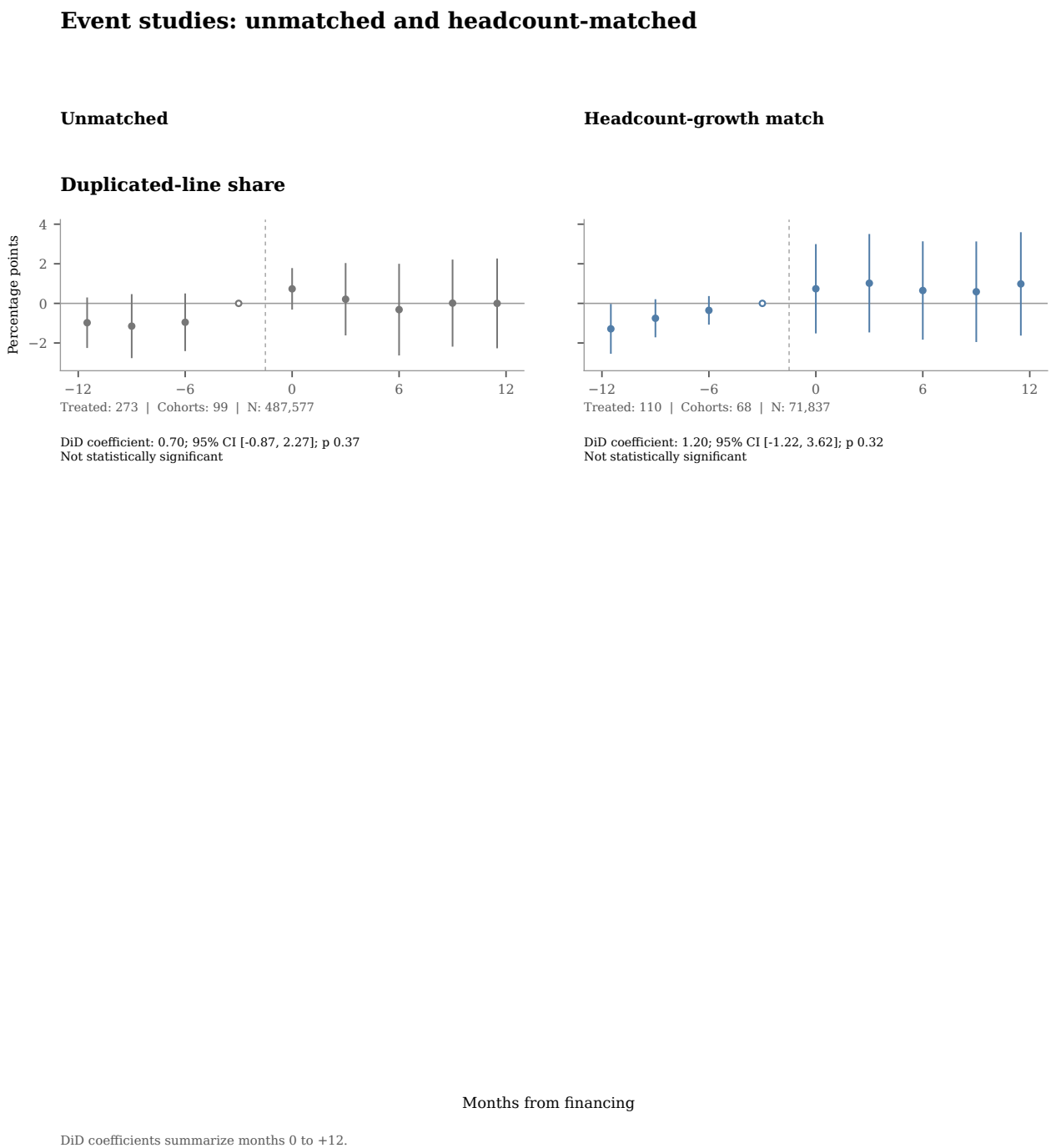



Figure 28: Alternative Comparison Groups: Part 6. Duplicated-line share. Columns report unmatched stacked DiD and headcount-growth matching. Axes match within each row; error bars are 95% confidence intervals with standard errors clustered by cohort and firm. The dashed line marks the end of omitted months −4 through −2.

# F Founding-Team Heterogeneity

We examine whether the engineering response depends on technical leadership in the founding team and founders' prior executive experience. If these leadership roles provide knowledge that VC involvement would otherwise supply, firms whose founders hold such roles should have less scope to improve. A broader change in organizational practices could instead reach firms with either type of founding team. Using LinkedIn career histories, we identify individuals whose titles contain "founder" or "co-founder" and who were active at the firm within six months of first VC entry, then retrieve their prior employment records to construct two indicators.

*Founder Technical Leadership* equals one if any member of the founding team simultaneously held a technical leadership role—CTO, VP of Engineering, Head of Engineering, or Director of Engineering—at the same firm around the deal date.

*Prior Executive Experience* equals one if any founder previously held a CEO, Chief, Founder, or President role at a different company before the first VC deal date. Founders with these prior leadership roles may bring organizational practices from previous firms, leaving less scope for subsequent change.

We interact each indicator with the Treated × Post term, following the same triple-interaction specification used for the pre-treatment quality and deal size splits in the main analysis. The columns for founder characteristics include only firms with the required LinkedIn career histories, and cohorts without both treated and control firms are removed. Each marker is computed at the firm's own first financing and retained across its cohort appearances. Eligible controls with observed founder markers use their later financing dates, at least 13 months after the comparison cohort's event, while never-financed firms have no deal-dated marker. The six-month window used to identify founders can also include roles beginning after the firm's financing date. Tables 21–23 report the full regression output. Each table presents six columns: columns (1)–(3) correspond to one outcome and columns (4)–(6) to a second outcome, with each set cycling through Large Codebase, Founder Technical Leadership and Prior Executive Experience.

Table 21: Founder Heterogeneity: Technical Debt per LoC and Technical Debt Ratio

| | **Technical Debt per LoC** | | | **Technical Debt Ratio** | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Treated × Post | −2.14 | −1.05 | −1.17 | −7.77 | −3.81 | −4.25 |
| | (1.31) | (1.13) | (1.62) | (4.75) | (4.09) | (5.90) |
| Post × Large Codebase (=1) | 0.09 | | | 0.32 | | |
| | (0.42) | | | (1.51) | | |
| Treated × Post × Large Codebase (=1) | 1.29 | | | 4.68 | | |
| | (1.37) | | | (4.99) | | |
| Post × Founder Technical Leadership (=1) | | 0.14 | | | 0.50 | |
| | | (0.82) | | | (2.99) | |
| Treated × Post × Founder Technical Leadership (=1) | | 0.12 | | | 0.44 | |
| | | (1.83) | | | (6.65) | |
| Post × Prior Executive Experience (=1) | | | 0.53 | | | 1.93 |
| | | | (0.93) | | | (3.37) |
| Treated × Post × Prior Executive Experience (=1) | | | 0.35 | | | 1.29 |
| | | | (1.89) | | | (6.86) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 477,107 | 64,803 | 64,803 | 477,107 | 64,803 | 64,803 |
| $R^2$ | 0.83 | 0.78 | 0.78 | 0.83 | 0.78 | 0.78 |

This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$. Technical Debt per LoC is calibrated remediation cost in dollars per line; the Technical Debt Ratio is that debt as a percentage of replacement-cost software assets. Columns (1) and (4) interact with a Large Codebase indicator (above-median lines of code). Columns (2) and (5) interact with a Founder Technical Leadership indicator (founding team includes a simultaneous CTO/VP Eng/Director Eng role). Columns (3) and (6) interact with a Prior Executive Experience indicator (any founder previously held a C-suite or founder role at a prior firm). See Appendix A, Table 19, for full variable definitions.

The employment-based indicators are available only for firms with LinkedIn coverage; those columns run on the covered subsample rather than coding missing markers as zero, so Columns (2), (3), (5), and (6) have substantially smaller samples than Columns (1) and (4). Cohorts left without both treated and control firms after this restriction are removed.

*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

Table 22: Founder Heterogeneity: Violations per kLoC and Critical Violations per kLoC

| | **Violations per kLoC** | | | **Critical Violations per kLoC** | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Treated × Post | −6.35* | −2.92 | −3.49 | −2.35 | −1.98 | −2.75 |
| | (3.42) | (2.23) | (2.85) | (1.49) | (1.63) | (2.21) |
| Post × Large Codebase (=1) | 1.64** | | | 0.53 | | |
| | (0.81) | | | (0.42) | | |
| Treated × Post × Large Codebase (=1) | 4.16 | | | 1.30 | | |
| | (3.54) | | | (1.59) | | |
| Post × Founder Technical Leadership (=1) | | −0.26 | | | 0.38 | |
| | | (1.93) | | | (0.83) | |
| Treated × Post × Founder Technical Leadership (=1) | | −1.14 | | | 0.12 | |
| | | (4.28) | | | (2.73) | |
| Post × Prior Executive Experience (=1) | | | 1.74 | | | −0.50 |
| | | | (1.99) | | | (0.90) |
| Treated × Post × Prior Executive Experience (=1) | | | 0.47 | | | 1.45 |
| | | | (3.71) | | | (2.77) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 477,107 | 64,803 | 64,803 | 477,107 | 64,803 | 64,803 |
| $R^2$ | 0.86 | 0.82 | 0.82 | 0.89 | 0.80 | 0.80 |

This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$. Columns (1) and (4) interact with a Large Codebase indicator. Columns (2) and (5) interact with a Founder Technical Leadership indicator. Columns (3) and (6) interact with a Prior Executive Experience indicator. Violations per kLoC counts total rule violations across all severity levels per 1,000 lines of code. Critical Violations per kLoC counts violations classified as critical per 1,000 lines of code. See Appendix A, Table 19, for full variable definitions.

The employment-based indicators are available only for firms with LinkedIn coverage; those columns run on the covered subsample rather than coding missing markers as zero, so Columns (2), (3), (5), and (6) have substantially smaller samples than Columns (1) and (4). Cohorts left without both treated and control firms after this restriction are removed.

*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

Table 23: Founder Heterogeneity: Bugs per kLoC and Reliability Issues (High) per kLoC

| | **Bugs per kLoC** | | | **Reliability Issues (High) per kLoC** | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Treated × Post | 0.04 | 0.04 | −0.04 | −0.01 | 0.01 | −0.03 |
| | (0.20) | (0.16) | (0.15) | (0.03) | (0.02) | (0.02) |
| Post × Large Codebase (=1) | 0.08 | | | −0.02 | | |
| | (0.08) | | | (0.02) | | |
| Treated × Post × Large Codebase (=1) | −0.08 | | | 0.00 | | |
| | (0.24) | | | (0.03) | | |
| Post × Founder Technical Leadership (=1) | | −0.10 | | | 0.02 | |
| | | (0.30) | | | (0.01) | |
| Treated × Post × Founder Technical Leadership (=1) | | −0.63 | | | −0.02 | |
| | | (0.46) | | | (0.03) | |
| Post × Prior Executive Experience (=1) | | | 0.32 | | | 0.00 |
| | | | (0.30) | | | (0.01) |
| Treated × Post × Prior Executive Experience (=1) | | | −0.24 | | | 0.05 |
| | | | (0.38) | | | (0.03) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 477,107 | 64,803 | 64,803 | 477,107 | 64,803 | 64,803 |
| $R^2$ | 0.90 | 0.91 | 0.91 | 0.51 | 0.76 | 0.76 |

This table reports the coefficient $\beta$ from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$. Columns (1) and (4) interact with a Large Codebase indicator. Columns (2) and (5) interact with a Founder Technical Leadership indicator. Columns (3) and (6) interact with a Prior Executive Experience indicator. Bugs per kLoC counts patterns classified by SonarQube as likely to cause incorrect or unexpected behavior per 1,000 lines of code. Reliability Issues (High) per kLoC captures stability-related rule violations at high severity per 1,000 lines of code. Bugs use the traditional classification; Reliability Issues (High) uses the alternative classification described in Appendix M.4 and is not a subset of Bugs. See Appendix A, Table 19, for full variable definitions.

The employment-based indicators are available only for firms with LinkedIn coverage; those columns run on the covered subsample rather than coding missing markers as zero, so Columns (2), (3), (5), and (6) have substantially smaller samples than Columns (1) and (4). Cohorts left without both treated and control firms after this restriction are removed.

*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

**Interpretation.** Neither the Founder Technical Leadership nor the Prior Executive Experience interaction is statistically significant for any of the six outcomes. For technical debt per line, the Founder Technical Leadership interaction is \$0.12 and the Prior Executive Experience interaction is \$0.35 ($p = 0.94$ and $p = 0.85$, respectively). Thus, neither characteristic shows a systematic association with the engineering response.

The quality gains are not systematically concentrated among firms without these founder leadership indicators. The pattern is consistent with organizational changes—such as formal tracking procedures and coordination across developers—that can benefit teams with different founder leadership roles. Section 3.8 examines the development practices of existing engineers directly.

## Post-Financing Hiring Composition

We compare quality changes by hiring composition during the first year after financing. *Hired Technical Leader* identifies a CTO, VP of Engineering or Head of Engineering hire in the interval from the event through, but excluding, the date 12 calendar months later. *High Senior Ratio* identifies firms whose senior-engineering hires as a share of all new hires in that interval exceed the treated-firm median of $1/7$. Both treated firms and controls are classified using the relevant cohort event. New hires require recorded start dates and no prior position at the firm before that event.

Tables 24 and 25 report the base, interaction and subgroup sum. Missing hiring records are excluded, and cohorts without both treated and control firms are removed. The leader comparison contains 31,925 observations, 161 firms and 137 treated firms; the senior-share comparison contains 26,451 observations, 159 firms and 135 treated firms. Both retain 63 cohorts.

Table 24: First-Year Hiring Composition: Critical and Total Violations

| | **Critical Violations / kLoC** | | **Violations / kLoC** | |
|---|---|---|---|---|
| | Leader | Senior share | Leader | Senior share |
| | (1) | (2) | (3) | (4) |
| Baseline on the same sample, no hiring interactions | −1.39 | −1.13 | −1.27 | −0.90 |
| | (2.22) | (2.28) | (3.49) | (3.58) |
| Treated × Post | −2.87 | −2.01 | −4.16 | −2.51 |
| | (3.13) | (3.49) | (4.05) | (4.47) |
| Post × Hiring indicator | 0.20 | 0.03 | −3.33 | −2.22 |
| | (1.29) | (1.25) | (3.25) | (2.56) |
| Treated × Post × Hiring indicator | 3.38 | 1.63 | 8.29 | 3.56 |
| | (3.30) | (4.07) | (5.14) | (5.06) |
| Hiring-group effect (base + interaction) | 0.51 | −0.38 | 4.12 | 1.04 |
| | (2.00) | (2.51) | (5.07) | (4.35) |
| 95% interval, hiring-group effect | [−3.48, 4.51] | [−5.40, 4.64] | [−6.01, 14.25] | [−7.66, 9.75] |
| Cohort × Firm FE | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes |
| **# Obs** | 31,925 | 26,451 | 31,925 | 26,451 |
| **# Firms** | 161 | 159 | 161 | 159 |
| **# Treated** | 137 | 135 | 137 | 135 |

Critical violations, total violations and maintainability issues are counts per 1,000 non-comment lines; technical debt is calibrated dollars per line. Leader and senior-share indicators are constructed from LinkedIn hiring during [event, event + 12 months), using the cohort event for both arms. Senior share is senior-engineering hires divided by all distinct new hires and is missing when no hires are recorded. Each moderator is estimated on the observations shared by all 12- and 24-month classifications. The baseline omits all hiring interactions and uses the same observations. The hiring-group effect is the sum of the base and interaction; its standard error and 95% interval use their full covariance. Both comparisons retain 63 cohorts.

*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

Table 25: First-Year Hiring Composition: Technical Debt and Maintainability

| | TD per LoC ($) | | Maintainability / kLoC | |
|---|---|---|---|---|
| | Leader | Senior share | Leader | Senior share |
| | (1) | (2) | (3) | (4) |
| Baseline on the same sample, no hiring interactions | −0.41 | −0.11 | −1.49 | −1.16 |
| | (1.61) | (1.65) | (3.26) | (3.34) |
| Treated × Post | −1.50 | −0.88 | −4.07 | −2.74 |
| | (1.98) | (2.47) | (3.85) | (4.35) |
| Post × Hiring indicator | −1.44 | −0.74 | −2.43 | −1.89 |
| | (1.31) | (1.25) | (2.79) | (2.25) |
| Treated × Post × Hiring indicator | 3.21 | 1.61 | 7.16 | 3.43 |
| | (2.57) | (2.92) | (4.75) | (4.85) |
| Hiring-group effect (base + interaction) | 1.71 | 0.73 | 3.09 | 0.69 |
| | (2.28) | (1.94) | (4.54) | (3.94) |
| 95% interval, hiring-group effect | [−2.85, 6.27] | [−3.14, 4.61] | [−5.99, 12.18] | [−7.19, 8.56] |
| Cohort × Firm FE | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes |
| **# Obs** | 31,925 | 26,451 | 31,925 | 26,451 |
| **# Firms** | 161 | 159 | 161 | 159 |
| **# Treated** | 137 | 135 | 137 | 135 |

Critical violations, total violations and maintainability issues are counts per 1,000 non-comment lines; technical debt is calibrated dollars per line. Leader and senior-share indicators are constructed from LinkedIn hiring during [event, event + 12 months), using the cohort event for both arms. Senior share is senior-engineering hires divided by all distinct new hires and is missing when no hires are recorded. Each moderator is estimated on the observations shared by all 12- and 24-month classifications. The baseline omits all hiring interactions and uses the same observations. The hiring-group effect is the sum of the base and interaction; its standard error and 95% interval use their full covariance. Both comparisons retain 63 cohorts.

*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

The point estimates do not show larger quality improvements among firms that hire technical leaders or a high share of senior engineers. The hiring interactions are positive across all four outcomes, and none is statistically significant. The estimates for firms without these hiring indicators are negative and are also not statistically significant. Tables 24 and 25 report the group effects and their confidence intervals. Section 3.8 examines the development records of existing engineers directly.

# G Levels versus Density

The main results concern the quality of software production, measured by violations per unit of code. This normalization is analogous to a manufacturing defect rate: a producer can make more units and accumulate more defects in total while reducing the share of output that is defective. In software, each addition or modification to a codebase can introduce a new issue, so violation density helps distinguish the quality of engineering practices from the amount of code produced. We examine that distinction by estimating the baseline specification on total critical violations, total violations, and total technical debt, transformed as log(Critical), log(Violations), log(Debt), respectively.

Table 26 compares the density and stock estimates. Critical violations fall by 1.85 per 1,000 lines of code ($p = 0.03$; Column (1)), while the transformed critical-violation count has a positive coefficient of 0.10 ($p = 0.08$; Column (3)). Technical debt falls by \$1.57 per line ($p = 0.04$; Column (2)). The growth rate in total technical debt in dollars is approximately 16 percentage points greater than in controls ($p = 0.01$; Column (4)). The growth rate in total violations is approximately 13 percentage points greater than in controls ($p = 0.04$; not tabulated). These findings show why counts and densities answer different economic questions: the accumulated burden can increase as the product grows even when each unit of software carries less debt.

Table 26: Critical Violations and Technical Debt: Per-Line Measures and Totals

| | **Density** | | **Log totals** | |
|---|---|---|---|---|
| | Critical Violations / kLoC | TD per LoC (\$) | Critical Violations | TD (\$) |
| | (1) | (2) | (3) | (4) |
| Treated × Post | −1.85** | −1.57** | 0.10* | 0.16** |
| | (0.86) | (0.77) | (0.06) | (0.07) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes |
| **# Obs** | 487,577 | 487,577 | 441,351 | 482,310 |
| $R^2$ | 0.88 | 0.83 | 0.94 | 0.92 |

This table compares the stacked DiD estimates for critical-violation density and technical debt per line with estimates for total critical violations and total technical debt. Columns (1)–(2) reproduce the baseline per-LoC results from Tables 3 and 7. Columns (3)–(4) estimate the same specification using log(Critical) and log(Debt). The log specifications use only positive values. All specifications include Firm × Cohort and Month × Cohort fixed effects. See Appendix A, Table 19, for full variable definitions.

*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

A constant violation rate provides a useful benchmark for these magnitudes. If violation counts expanded proportionately with code, their growth rates would approximately match the estimated code-volume growth rate. The growth rate in code volume is approximately 23 percentage points greater

than in controls. The growth rates in critical violations, total violations, and total technical debt are below that benchmark. Using the same fitted observations as each outcome, we estimate code volume coefficients of 0.20 for critical violations ($N = 441,351$), 0.21 for total violations ($N = 482,576$), and 0.21 for technical debt in dollars ($N = 482,310$). The comparison is consistent with firms adding code to a higher standard as they expand, and it explains how significant density improvements can coexist with no statistically significant reduction in absolute critical-violation counts.[9]

The count-density comparison motivates a further check among firms with low pre-financing codebase growth, reported in Appendix H. Appendix I presents event studies for issue counts and total technical debt, while the contribution-level analysis in Section 3.8 provides separate evidence that contributions from people active before financing are less often linked to later fixes.

# H Decomposing the Quality Improvement

We examine whether engineering quality improves among firms whose codebases had previously grown slowly. We then separate changes in violations per line into changes in the number of violations and changes in codebase size.

### Pre-Treatment Codebase Growth

For each financing cohort, we measure codebase growth by comparing average log lines of code in months $-12$ through $-7$ with months $-6$ through $-1$. A firm can enter several cohort comparisons, first as a control and later as a treated firm. Among these observations for firms that eventually receive financing, we identify the lowest-growth quartile. We retain a firm if it falls in this bottom quartile in any cohort comparison, including an earlier comparison in which it serves as a control, and retain the controls in its cohorts. Thus, a firm qualifies based on low growth before at least one cohort's financing date, which need not be its own financing date. All growth measurements used for selection precede the relevant cohort's financing date. Across these observations, pre- and post-financing growth have a correlation of 0.57, and 62% of observations in the lowest pre-financing growth quartile remain in the lowest quartile afterward.

Table 27 shows that the critical-violation estimate strengthens in this subsample. Compared with 1.85 in the full sample, critical violations decline by 2.85 per 1,000 lines of code ($p = 0.06$). Technical debt per line declines by \$1.64 ($p = 0.05$), and total violations by 3.7 per 1,000 lines ($p = 0.04$). Joint tests do not reject zero pre-treatment coefficients for these outcomes. The log-code-volume coefficient, which measures growth relative to controls, is $-0.02$ ($p = 0.73$; not statistically significant). The treated firms' median change in their own mean log code volume from the pre- to

[9] The transformations follow Table 19. The scale outcomes have different pre-financing trajectories, and lower aggregate density can also reflect remediation, deletion, or changes in code composition.

the post-financing period is +0.13, approximately 13% growth in their own codebases on this fitted sample. The comparison therefore documents quality gains among firms selected using low prior growth while allowing their codebases to expand.

The larger critical-violation estimate also has an economic interpretation. Firms whose codebases expand slowly may have more scope to devote engineering effort to existing software, including remediation and improvements in development procedures. Alternatively, the full-sample estimate may combine these gains with the additional coordination demands of rapid feature development. The low-growth result is consistent with both possibilities: improved engineering discipline need not depend on producing large amounts of additional code.

Table 27: Engineering Quality Among Firms with Low Pre-Treatment Codebase Growth

| | **Critical Violations / kLoC** | | **TD per LoC** | |
|---|---|---|---|---|
| | Full Sample | Low Pre-Treatment Growth (bottom quartile) | Full Sample | Low Pre-Treatment Growth (bottom quartile) |
| | (1) | (2) | (3) | (4) |
| Treated × Post | −1.85** | −2.85* | −1.57** | −1.64* |
| | (0.86) | (1.53) | (0.77) | (0.83) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes |
| **# Obs** | 487,577 | 361,439 | 487,577 | 361,439 |

This table compares the baseline per-LoC estimates (Columns 1 and 3) with estimates for treated firms that fall in the bottom quartile of codebase growth in at least one cohort comparison, including an earlier comparison in which they serve as controls (Columns 2 and 4). Growth compares average log lines of code in months $[-12, -7]$ with months $[-6, -1]$ before the relevant cohort event. We retain all control firms in the selected firms' cohorts. In this subsample, the coefficient for log(LoC) is −0.02 and is not statistically significant ($p = 0.73$); this null does not imply constant code volume. Cohorts left with only treated or only control firms after the restriction are dropped. Total violations per kLoC in the same subsample: −3.7 ($p = 0.04$). Joint tests do not reject zero pre-treatment coefficients for the reported outcomes. See Appendix A, Table 19, for variable definitions.

*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

### Log-Density Decomposition

We also estimate the baseline specification on log-density measures. The critical-violation outcome is log(Critical) − log(LoC), and the total-violation outcome is log(Violations) − log(LoC). The critical-violation coefficient is −0.10 ($p = 0.02$; $N = 441,351$), and the total-violation coefficient is −0.08 ($p = 0.04$; $N = 482,576$). The corresponding joint pre-treatment tests give $p = 0.05$ for critical violations and $p = 0.15$ for total violations; the total-violation joint test is not statistically significant. The log specifications include only positive counts. The critical log-density specification therefore has weaker pre-trend support than the baseline density measure.

The critical-violation estimate decomposes into a count coefficient of +0.10 ($p = 0.08$) and a

code-volume coefficient of +0.20 ($p < 0.01$). The identity makes the source of the transformed density decline explicit: the estimated count change is smaller than the estimated change in code volume. The single-repository subsample yields critical and total log-density coefficients of −0.11 ($N = 264,961$) and −0.11 ($N = 291,179$), respectively. This decomposition describes the accounting relationship between the two components, while the low-growth comparison above shows that the density decline also occurs among firms selected using low prior code growth.

# I Level Outcomes: Event-Study Figures

This appendix presents event-study estimates for total code violations and total technical debt. We take logarithms of total violations, critical violations, total maintainability issues, and technical debt in dollars. Observations with zero values are excluded. These measures describe changes in the total burden as the codebase grows; the per-line measures in the main text describe its density. Before financing, the estimated differences between treated and control firms generally rise toward zero, indicating that their total violations and technical debt were already growing at different rates.

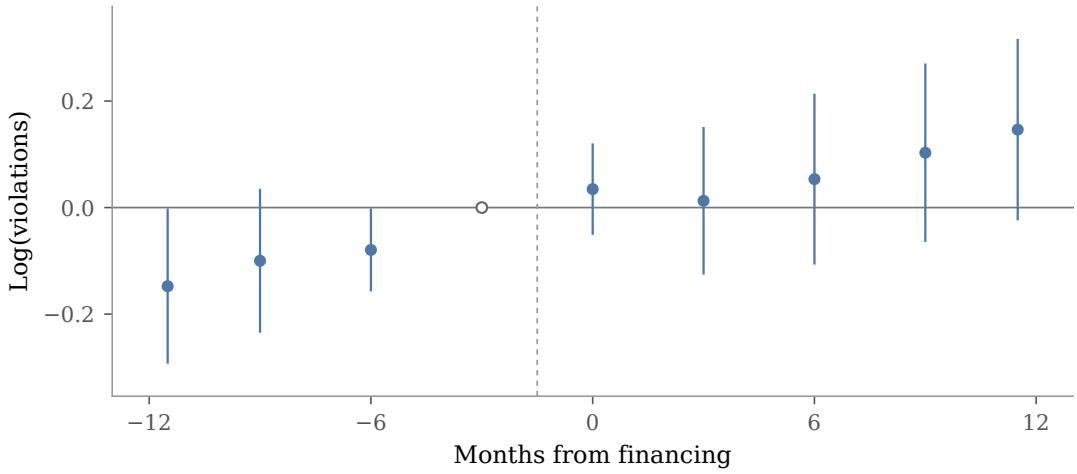
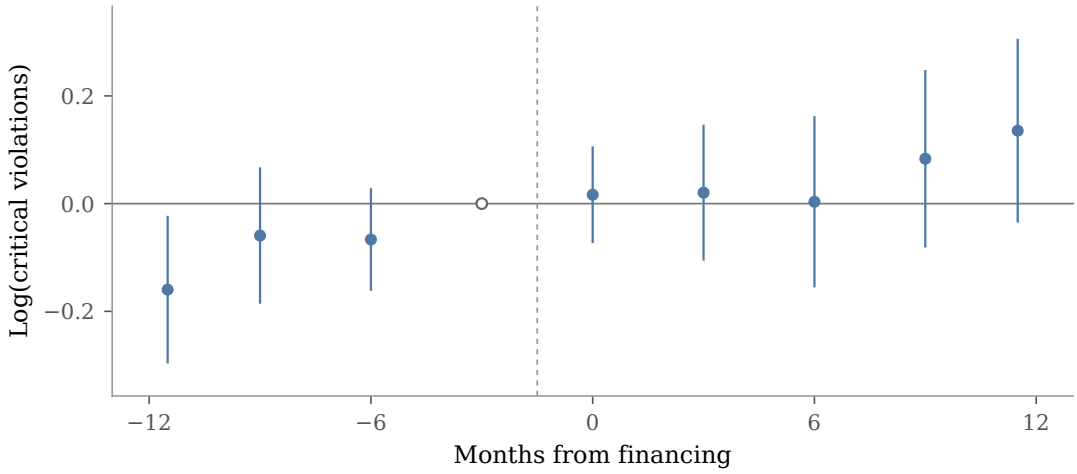
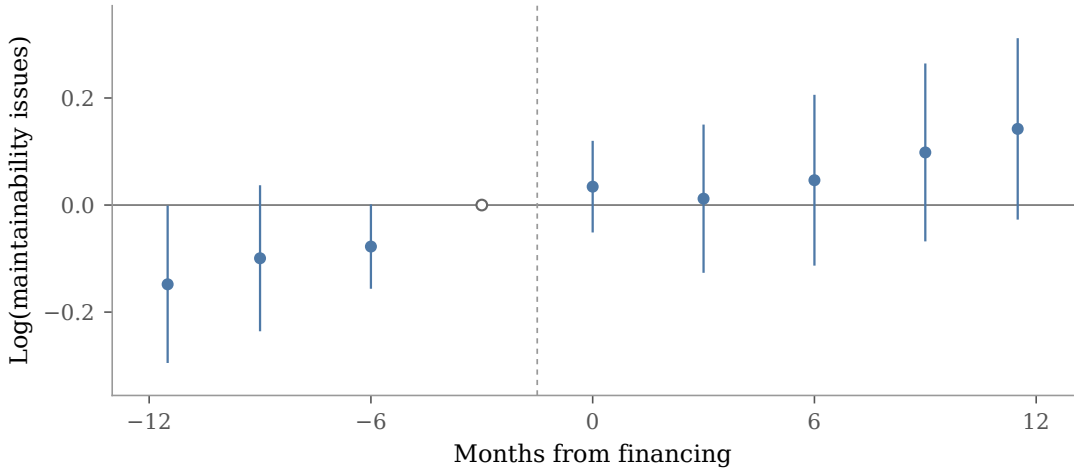



Figure 29: Log(Total Violations). Positive-value sample, $N = 482,576$. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.



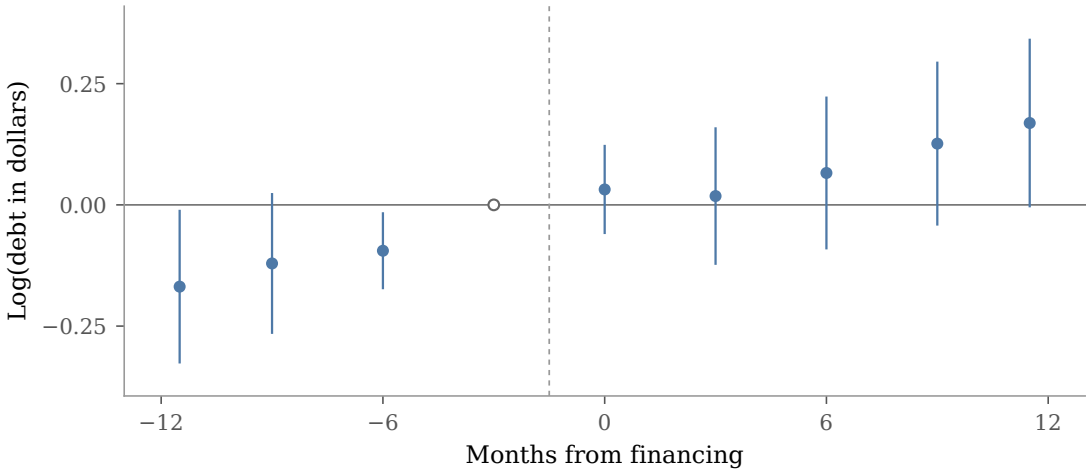

Positive values; N=441,351. Reference: months −4 to −2.

Figure 30: Log(Critical Violations). Positive-value sample, $N = 441,351$. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.




Positive values; N=482,366. Reference: months −4 to −2.

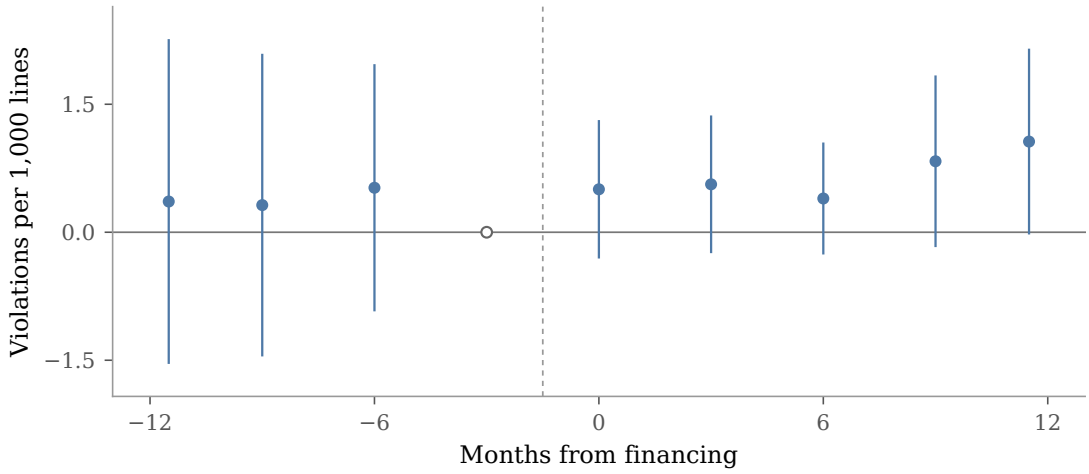
Figure 31: Log(Total Maintainability Issues). Positive-value sample, $N = 482,366$. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.




Positive values; N=482,310. Reference: months −4 to −2.

Figure 32: Log(Technical Debt $). Positive-value sample, $N = 482,310$. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

# J Series B+ Placebo Estimates: Later Financing Rounds




Open circle: reference months −4 to −2

Figure 33: Later Financing: Critical Rule Violations per 1,000 Lines of Code. Controls remain without a completed Seed or Series A round through the event window. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

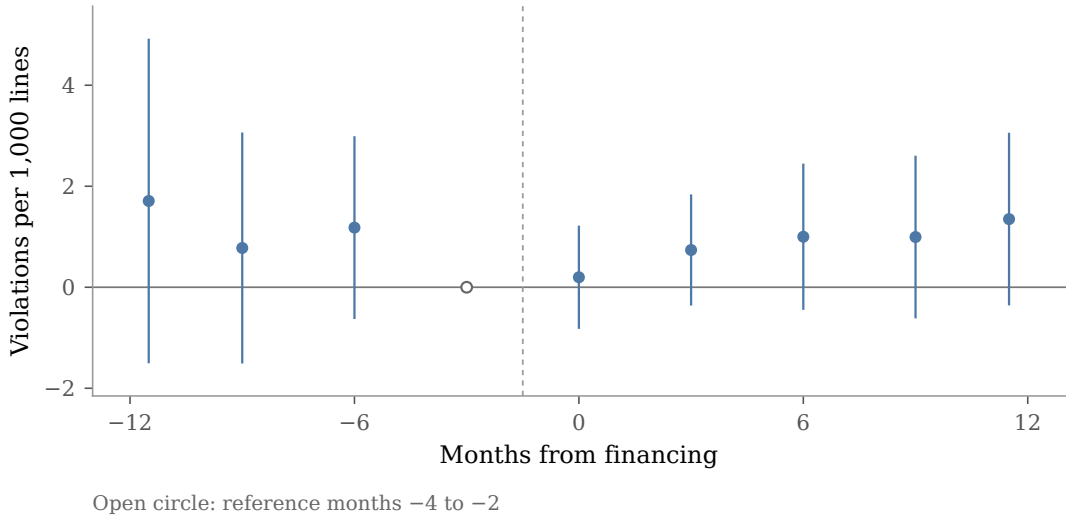



Figure 34: Later Financing: Rule Violations (All Severities) per 1,000 Lines of Code. Controls remain without a completed Seed or Series A round through the event window. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

Table 28: Series B+ Placebo Difference-in-Differences Estimates for Critical Violations and All Violations

| | Critical Violations | | | Violations | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Treated × Post | 0.37 | 0.30* | 0.51* | 0.13 | −0.26 | −0.07 |
| | (0.57) | (0.16) | (0.27) | (0.96) | (0.54) | (0.73) |
| Post × High Violations (=1) | | −1.45*** | | | −2.05* | |
| | | (0.37) | | | (1.06) | |
| Treated × Post × High Violations (=1) | | 0.29 | | | 1.02 | |
| | | (1.17) | | | (1.88) | |
| Post × Large Round | | | −0.29 | | | −1.38 |
| | | | (0.46) | | | (0.85) |
| Treated × Post × Large Round | | | −0.10 | | | 1.52 |
| | | | (1.29) | | | (1.92) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 331,893 | 330,508 | 331,893 | 331,893 | 330,508 | 331,893 |
| $R^2$ | 0.87 | 0.87 | 0.87 | 0.87 | 0.87 | 0.87 |

This table reports Series B+ placebo estimates from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Series B+ Treated × Post indicator. *Critical Violations* count severe SonarQube rule violations per 1,000 non-comment lines of code, including risky code structures with a potentially substantial impact on reliability, security, or maintainability. *Violations (All)* denote total deviations from engineering best practices across all severity levels per 1,000 lines of code. See Appendix A, Table 19, for full variable definitions.

High Violations is defined from critical-violation density in the month before the placebo event. Large Round refers to the amount raised at the earlier Seed or Series A treatment round. The quality cutoff uses baseline firm–cohort observations of firms that appear as placebo-treated; the round-size cutoff uses one observation per placebo-treated firm.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Table 29: Series B+ Placebo Difference-in-Differences Estimates for Maintainability Issues

| | **Maintainability Violations (All)** | | | **Maintainability Issues (High)** | | | **Maintainability Issues (Low)** | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Treated × Post | 0.19 | −0.28 | −0.06 | 0.38 | 0.30* | 0.53* | −0.39 | −0.20 | −0.58 |
| | (0.86) | (0.53) | (0.69) | (0.57) | (0.16) | (0.27) | (0.47) | (0.26) | (0.49) |
| Post × High Violations (=1) | | −1.92* | | | −1.46*** | | | 0.28 | |
| | | (1.06) | | | (0.37) | | | (0.72) | |
| Treated × Post × High Violations (=1) | | 1.17 | | | 0.30 | | | −0.42 | |
| | | (1.64) | | | (1.17) | | | (0.81) | |
| Post × Large Round | | | −1.47* | | | −0.29 | | | −0.86* |
| | | | (0.83) | | | (0.46) | | | (0.46) |
| Treated × Post × Large Round | | | 1.70 | | | −0.13 | | | 1.10** |
| | | | (1.59) | | | (1.29) | | | (0.53) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 331,893 | 330,508 | 331,893 | 331,893 | 330,508 | 331,893 | 331,893 | 330,508 | 331,893 |
| $R^2$ | 0.86 | 0.86 | 0.86 | 0.87 | 0.87 | 0.87 | 0.84 | 0.85 | 0.84 |

This table reports Series B+ placebo estimates from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Series B+ Treated × Post indicator. Maintainability Violations (All) capture total maintainability-related rule violations per 1,000 lines of code under the traditional classification, as in Table 4. High-Severity Maintainability Issues identify maintainability problems assigned high impact, while Low-Severity Maintainability Issues capture more minor structural or readability deficiencies; both use the alternative classification described in Appendix M.4 and are not a breakdown of the traditional overall count. See Appendix A, Table 19, for full variable definitions.

High Violations is defined from critical-violation density in the month before the placebo event. Large Round refers to the amount raised at the earlier Seed or Series A treatment round. The quality cutoff uses baseline firm–cohort observations of firms that appear as placebo-treated; the round-size cutoff uses one observation per placebo-treated firm.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Table 30: Series B+ Placebo Difference-in-Differences Estimates for Security Hotspots

| | Security Hotspots | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Treated × Post | 0.07 | 0.04 | 0.09 |
| | (0.08) | (0.13) | (0.09) |
| Post × High Violations (=1) | | 0.08 | |
| | | (0.11) | |
| Treated × Post × High Violations (=1) | | 0.07 | |
| | | (0.12) | |
| Post × Large Round | | | −0.11 |
| | | | (0.11) |
| Treated × Post × Large Round | | | 0.06 |
| | | | (0.12) |
| Cohort × Firm FE | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes |
| **# Obs** | 331,893 | 330,508 | 331,893 |
| $R^2$ | 0.70 | 0.70 | 0.70 |

This table reports Series B+ placebo estimates from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Series B+ Treated × Post indicator. *Security Hotspots* measure security-sensitive code segments requiring manual audit per 1,000 lines of code. See Appendix A, Table 19, for full variable definitions.

High Violations is defined from critical-violation density in the month before the placebo event. Large Round refers to the amount raised at the earlier Seed or Series A treatment round. The quality cutoff uses baseline firm–cohort observations of firms that appear as placebo-treated; the round-size cutoff uses one observation per placebo-treated firm.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Table 31: Series B+ Placebo Difference-in-Differences Estimates for Bugs and Reliability Issues

| | **Bugs** | | | **Reliability Issues (High)** | | | **Reliability Issues (Low)** | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Treated × Post | −0.13 | −0.05 | −0.06 | −0.01 | −0.01 | −0.02 | −0.15 | −0.20 | −0.15 |
| | (0.16) | (0.06) | (0.10) | (0.01) | (0.01) | (0.01) | (0.13) | (0.19) | (0.13) |
| Post × High Violations (=1) | | −0.19** | | | 0.00 | | | −0.02 | |
| | | (0.09) | | | (0.02) | | | (0.14) | |
| Treated × Post × High Violations (=1) | | −0.16 | | | 0.00 | | | 0.11 | |
| | | (0.33) | | | (0.02) | | | (0.22) | |
| Post × Large Round | | | 0.08 | | | 0.00 | | | −0.04 |
| | | | (0.07) | | | (0.01) | | | (0.10) |
| Treated × Post × Large Round | | | −0.22 | | | 0.02 | | | 0.03 |
| | | | (0.38) | | | (0.02) | | | (0.21) |
| Cohort × Firm FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **# Obs** | 331,893 | 330,508 | 331,893 | 331,893 | 330,508 | 331,893 | 331,893 | 330,508 | 331,893 |
| $R^2$ | 0.91 | 0.91 | 0.91 | 0.50 | 0.50 | 0.50 | 0.89 | 0.89 | 0.89 |

This table reports Series B+ placebo estimates from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Series B+ Treated × Post indicator. *Bugs* count, per 1,000 lines of code, patterns that SonarQube classifies as likely to cause incorrect or unexpected behavior, such as invalid operations or failures to handle an input. *High-* and *Low-Severity Reliability Issues* capture stability-related rule violations at the stated severity. Bugs use the traditional classification; the reliability measures use the alternative classification described in Appendix M.4 and are not a breakdown of Bugs. See Appendix A, Table 19, for full variable definitions.

High Violations is defined from critical-violation density in the month before the placebo event. Large Round refers to the amount raised at the earlier Seed or Series A treatment round. The quality cutoff uses baseline firm–cohort observations of firms that appear as placebo-treated; the round-size cutoff uses one observation per placebo-treated firm.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Table 32: Series B+ Placebo Difference-in-Differences Estimates for Technical Debt

| | **Technical Debt per Line ($)** | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Treated × Post | −0.13 | −0.15 | −0.32 |
| | (0.41) | (0.21) | (0.40) |
| Post × High Violations (=1) | | −0.37 | |
| | | (0.62) | |
| Treated × Post × High Violations (=1) | | 0.09 | |
| | | (0.76) | |
| Post × Large Round | | | −0.56 |
| | | | (0.36) |
| Treated × Post × Large Round | | | 0.85 |
| | | | (0.62) |
| Cohort × Firm FE | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes |
| **# Obs** | 331,893 | 330,508 | 331,893 |
| $R^2$ | 0.84 | 0.84 | 0.84 |

This table reports Series B+ placebo estimates from the specification $M_{i,c,t} = \beta VC_{i,c,t} + \chi_{c,i} + \delta_{c,t} + \gamma X_{i,c,t} + \varepsilon_{i,c,t}$, where $VC_{i,c,t}$ is the Series B+ Treated × Post indicator. *Technical Debt per Line of Code* measures calibrated remediation costs in dollars per line of code. The Technical Debt Ratio equals $40/11$ times this outcome on the same observations, so the table reports the dollar unit. See Appendix A, Table 19, for full variable definitions.

High Violations is defined from critical-violation density in the month before the placebo event. Large Round refers to the amount raised at the earlier Seed or Series A treatment round. The quality cutoff uses baseline firm–cohort observations of firms that appear as placebo-treated; the round-size cutoff uses one observation per placebo-treated firm.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

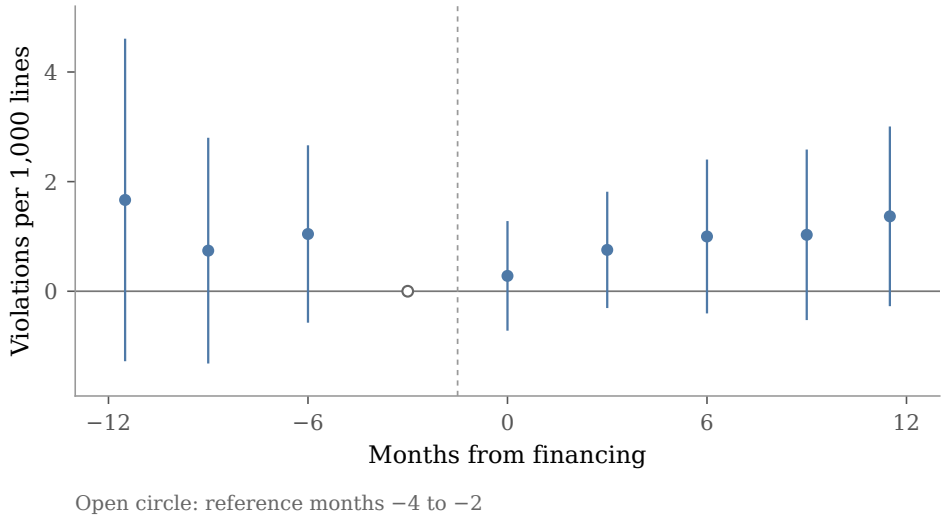



Figure 35: Later Financing: Maintainability Violations per 1,000 Lines of Code. Controls remain without a completed Seed or Series A round through the event window. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

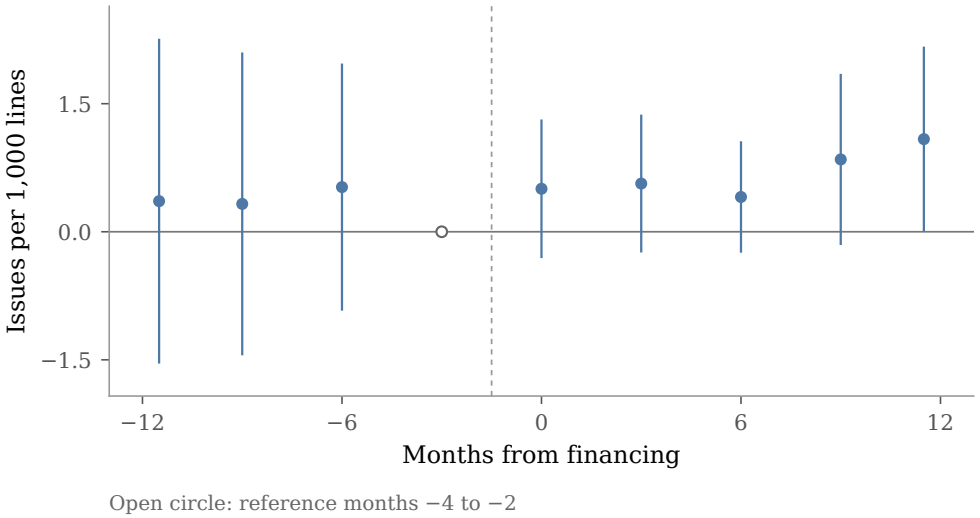



Figure 36: Later Financing: Maintainability Issues (High-Severity) per 1,000 Lines of Code. This measure uses the alternative classification described in Appendix M.4. Controls remain without a completed Seed or Series A round through the event window. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

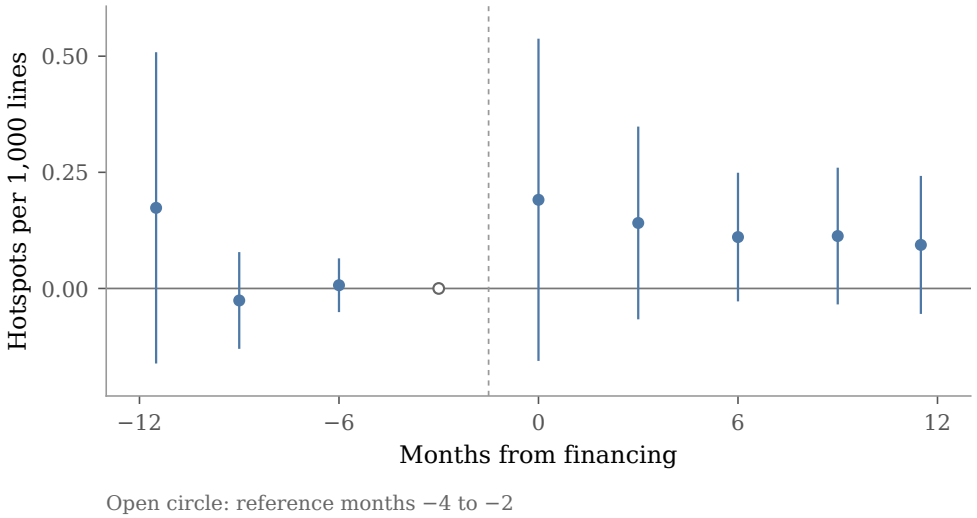
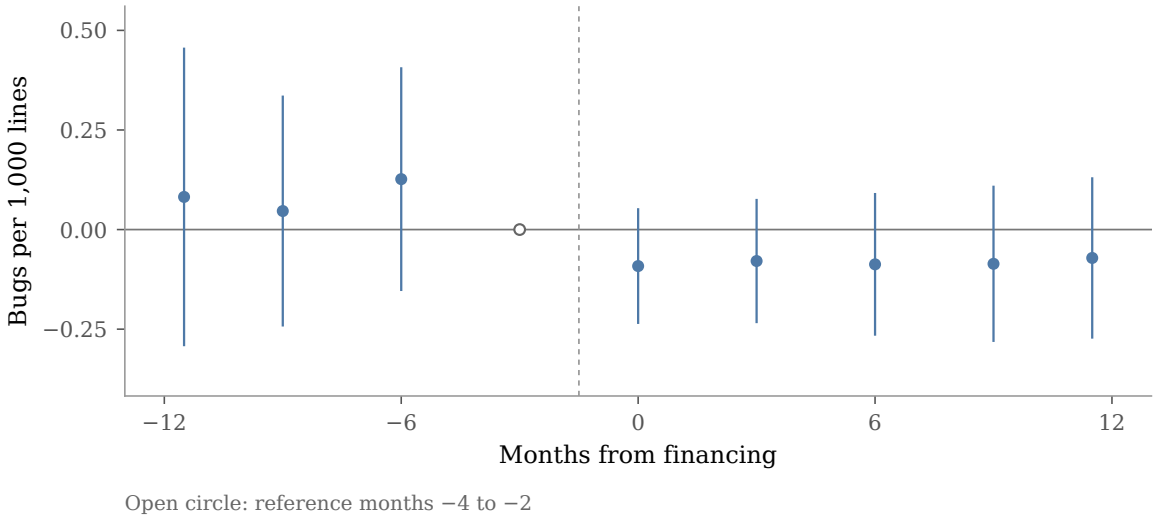
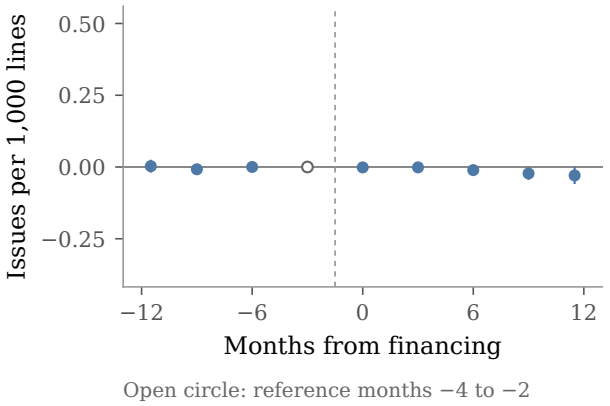


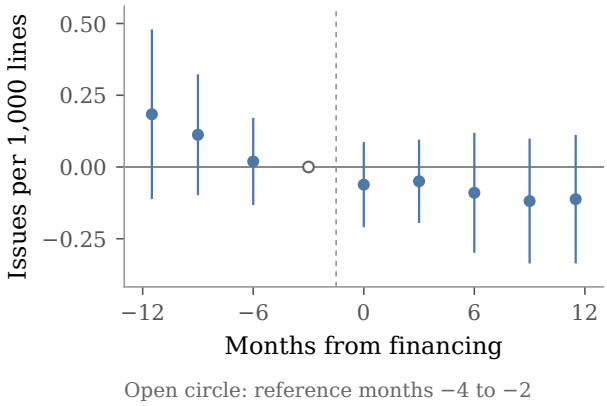

Figure 37: Later Financing: Security Hotspots per 1,000 Lines of Code. Controls remain without a completed Seed or Series A round through the event window. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.




(a) Bugs per 1,000 lines of code




(b) Reliability Issues (High Severity) per 1,000 lines of code




(c) Reliability Issues (Low Severity) per 1,000 lines of code

Figure 38: Later Financing: Bugs and Reliability Issues. Bugs use the traditional classification; the reliability measures use the alternative classification described in Appendix M.4. Controls remain without a completed Seed or Series A round through the event window. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

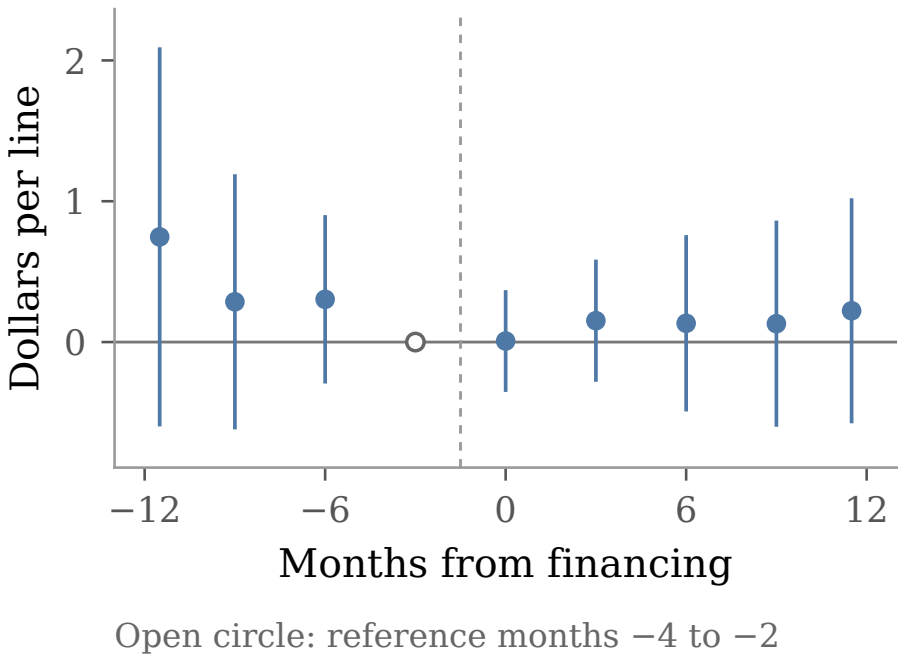



Figure 39: Later Financing: Technical Debt per Line of Code ($). Controls remain without a completed Seed or Series A round through the event window. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

# K Lifecycle and Commercialization: Additional Evidence

This appendix examines whether the changes around first institutional financing coincide with a shift toward commercialization. It complements the commercial-stage comparisons in Section 4.3 with LinkedIn employment records and commit histories. We use the panel of firms with one listed commercialized project and the estimator in Equation (7). This panel contains 212 treated firms; after restricting to observed commit measures, excluding shared repositories and removing fixed-effect singletons, the commercialization regressions contain 177,676 stacked observations from 260 firms, including 155 treated firms, in 83 cohorts. These analyses cover a subset of the main sample; each table reports its own coverage.

## K.1 Headcount Composition

We use employment records to test whether commercialization shifts resources from engineering toward selling. We estimate Equation (7) on each role group's share of active headcount. Table 33 reports a 2.4 percentage-point increase in the business-development share and a 6.6-point increase in the engineering share, both $p < 0.01$. Business-development roles also account for 4.2 percentage points more of monthly hires. The point estimates show commercial expansion alongside a larger

increase in engineering's share of employment. The event study shows that the business-development share was already increasing relative to controls before financing (Figure 40).

Table 33: Business-Development and Engineering Shares of Headcount

| | **Business Development Share of Headcount** | **Engineering Share of Headcount** | **Business Development Share of Monthly Hires** |
|---|---|---|---|
| | (1) | (2) | (3) |
| Treated × Post | 2.35*** | 6.65*** | 4.23*** |
| | (0.68) | (1.49) | (1.19) |
| Cohort × Firm FE | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes |
| **# Obs** | 157,553 | 157,553 | 68,416 |
| **# Firms** | 287 | 287 | 279 |

This table estimates the baseline stacked difference-in-differences specification of Equation (7) on LinkedIn employment spells for firms with one listed commercialized project and LinkedIn coverage. The outcomes in Columns (1) and (2) are the business-development and software-engineering categories' shares of a firm's active headcount in a month; the two categories are mutually exclusive. Column (3) uses the flow of role starts rather than the stock of active employees. Joint tests do not reject zero pre-treatment coefficients for these three share outcomes. They reject for the corresponding log-count outcomes, which are not reported. These outcomes derive from employment records and contain no SonarQube input.

Coefficients and standard errors are reported in percentage points.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.

Table 34 decomposes the commercial expansion into sales-related and marketing roles and identifies human-resources staff within the residual category. Sales-related roles account for 1.5 percentage points of the increase in headcount share ($p = 0.02$), and marketing for 0.9 points ($p < 0.01$). The human-resources estimate is not statistically significant ($p = 0.73$). The share of all other classified roles falls by 7.6 points ($p < 0.01$), and the reported hiring-flow estimates have the same signs. These changes indicate joint investment in engineering and commercial capacity, with no transfer of employment share from engineering toward selling.

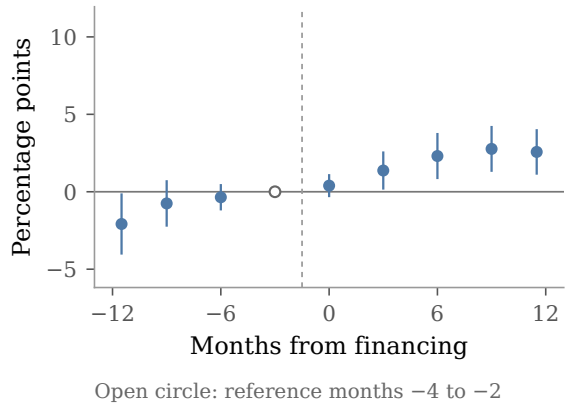
Table 34: Decomposition of the Business-Development Share of Headcount and Hires

| | **Sales-Related** | **Marketing** | **Human Resources** | **All Other Roles** |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| *Panel A: Share of active headcount* | | | | |
| Treated × Post | 1.50** | 0.87*** | −0.21 | −7.62*** |
| | (0.64) | (0.31) | (0.60) | (1.55) |
| **# Obs** | 157,553 | 157,553 | 157,553 | 157,553 |
| **# Firms** | 287 | 287 | 287 | 287 |
| *Panel B: Share of monthly hires* | | | | |
| Treated × Post | 2.34** | 1.89*** | — | −10.59*** |
| | (1.14) | (0.60) | | (2.52) |
| **# Obs** | 68,416 | 68,416 | — | 68,416 |
| **# Firms** | 279 | 279 | — | 279 |
| Cohort × Firm FE | Yes | Yes | Yes | Yes |
| Month × Cohort FE | Yes | Yes | Yes | Yes |

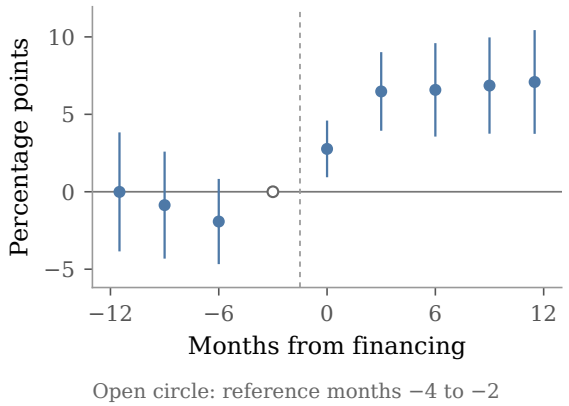
This table decomposes the business-development share from Table 33, using the same stacked difference-in-differences specification, sample, and employment records. Columns (1) and (2) split the business-development group into sales-related roles (sales, account management, sales development, business-development titles, partnerships, customer success, go-to-market, and revenue operations) and marketing roles (marketing, product marketing, growth, and demand generation); the two coefficients sum to the aggregate business-development estimate up to a small overlap from employees who hold roles in both components. Column (3) is a human-resources group (human resources, recruiting, talent acquisition, and people operations) drawn from the previously unclassified residual, and Column (4) is the remaining residual. Engineering-management roles are excluded from every group, following the role classification in Table 33, so the reported shares and the engineering share do not sum to one. Panel A uses each group's share of a firm's active headcount; Panel B uses the share of monthly role starts. Joint tests do not reject zero pre-treatment coefficients for the reported outcomes. They reject for the human-resources share of monthly hires ($p = 0.03$), which is not reported. These outcomes derive from employment records and contain no SonarQube input.

Coefficients and standard errors are reported in percentage points.

*Note:* Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Standard errors are clustered by cohort and firm.




(a) Business-development share of headcount

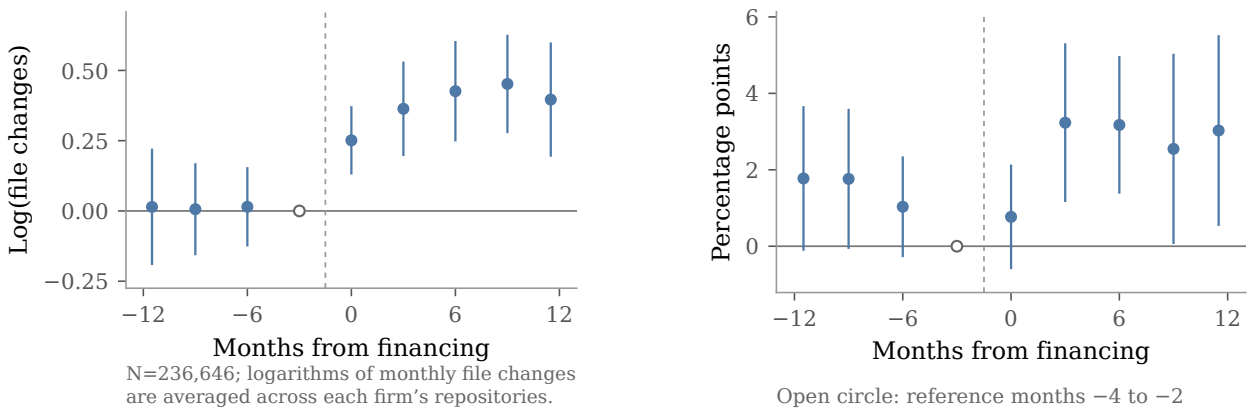



(b) Engineering share of headcount

Figure 40: Headcount Composition. Firms with one listed commercialized project; shares in percentage points. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

## K.2 Commercialization in the Codebase

We also examine commercialization within the software asset using commit messages, changed file paths, and dependency manifests—the files that list the external software packages used by a project. Table 18 shows no statistically significant increase in messages mentioning billing, payments, metering, subscriptions, or upgrades, or in the share of changed files associated with commercial features, payment libraries, enterprise functions, or growth. In the year after financing, 1.0% of repositories add a payment library, while 7.3% add a frontend framework used to build the user-facing interface. Development activity nevertheless expands. The growth rate in monthly file changes is approximately 34 percentage points greater than in controls. Engineers modify files in more top-level folders, and the share of changes to frontend code increases. Figure 41 traces these changes over the event window. The observable expansion is therefore broader than a shift toward the measured monetization features.




(a) Log file changes per month

(b) Frontend share of changed files

Figure 41: Development Activity and the Frontend Share of Changed Files. Firms with one listed commercialized project; shares in percentage points. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

Together, the employment and codebase evidence does not support a narrow lifecycle account in which firms shift resources from engineering toward selling at their first round. Engineering gains employment share, development expands, and the public codebase shows no detectable increase in the measured monetization features. Combined with the quality improvements among firms already generating revenue in Section 4.3, these findings indicate that the engineering response extends beyond the initial transition to commercialization.

# L Engineering Practice around VC Entry

This appendix examines the changes in development practices behind Section 3.8. We report commit-message components, headquarters-local weekend comparisons, and bug introduction among engineers active before financing. Several message measures use trailers: standardized lines, such as "Signed-off-by:" or "Co-authored-by:", that record review, sign-off, acknowledgement, approval or coauthorship. We link firms to their listed commercialized open-source projects using a fixed company–project mapping, held constant across event months. These analyses include firms with multiple listed projects and aggregate the available contribution histories to the firm level. Outcome coverage depends on the histories and measurements available for each firm. Table 35 reports observations, firms, treated firms, cohorts, and overlap with the main sample separately for each analysis.

Table 35: Sample Coverage of Contribution-Based Comparisons

| Comparison | Observations | Firms | Treated | Cohorts |
|---|---|---|---|---|
| Primary message-marker comparisons | 300,380 | 371 | 213 | 89 |
| Primary HQ-local weekend share | 171,116 | 273 | 199 | 85 |
| Primary bug introduction by pre-financing engineers: rate and components | 202,085 | 299 | 165 | 72 |
| Primary fixed-roster pre/post comparisons | 17,238 | 298 | 161 | 70 |
| Common sample for core outcomes | 125,114 | 239 | 135 | 63 |
| Common sample for broader outcomes | 76,633 | 178 | 125 | 60 |
| Single-project message-marker components | 216,934 | 325 | 191 | 79 |
| Single-project firm-wide attribution rate | 182,434 | 284 | 162 | 69 |
| Single-project post-entry newcomer gap | 71,426 | 271 | 146 | 66 |
| Single-project changed-file comparison | 236,646 | 335 | 190 | 78 |
| Single-project HQ-local weekend repair comparison | 2,339,208 | 201 | — | — |

All firms represented in this table belong to the main 484-firm sample. The primary analyses include firms with one or more listed commercialized projects and pool available contribution histories across their covered repositories. Their fitted samples vary with outcome measurement and contribution availability. The core and broader intersections use identical firm–cohort–month observations across their included outcomes (Table 36). The primary message, weekend and pre-financing-engineer comparisons include 72, 47 and 13 multi-project firms, respectively. Attribution measurement is available for 315 firms, including 13 multi-project firms, before the pre-financing-engineer and cohort restrictions.

The fixed-roster row counts paired firm–cohort units; the weekend repair row counts contributions and has no financing-cohort treatment. Other rows count firm–cohort–months. The primary analyses exclude singleton observations and cohorts without both treated and control firms. The remaining message-marker, firm-wide attribution, newcomer, changed-file and weekend-repair comparisons use firms with one listed project; they are distinct from the broader primary samples.

The sustained-documentation analysis in Appendix M.5 covers 394 main-sample firms with available commit-message measures. Marker-specific risk sets contain 334 firms and 19,801 months for numbered references, 384 and 37,381 for trailers, 378 and 39,468 for standardized prefixes, and 382 and 33,819 for testing/continuous integration. These duration models have firm clustering and no stacked treated-firm count.

Table 36 compares the principal process, bug-introduction, and code-quality outcomes on identical firm–cohort–month observations.

Table 36: Process and Engineering Quality on Identical Observations

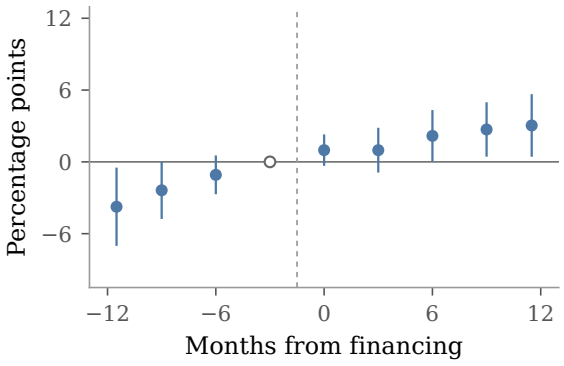
| Measure | Core estimate | $p$-value | Broader estimate | $p$-value |
|---|---|---|---|---|
| Contribution documentation (combined) | 3.92*** | < 0.01 | 4.13*** | < 0.01 |
| | (1.13) | | (1.16) | |
| Developer sign-off | 2.97** | 0.01 | 3.14*** | < 0.01 |
| | (1.13) | | (1.15) | |
| Recorded coauthorship | 1.22* | 0.06 | 1.29* | 0.06 |
| | (0.65) | | (0.68) | |
| Reviewed-by marker | −0.03 | 0.31 | −0.05 | 0.31 |
| | (0.03) | | (0.05) | |
| Pre-financing-engineer bug introduction | −1.94* | 0.05 | −1.38 | 0.18 |
| | (0.98) | | (1.03) | |
| Critical violations per 1,000 lines | −1.69 | 0.13 | −1.78 | 0.14 |
| | (1.12) | | (1.22) | |
| Numbered issue/task references | — | — | 5.73** | 0.02 |
| | | | (2.56) | |
| Pull-request references | — | — | 5.93** | 0.02 |
| | | | (2.58) | |
| Merge-or-squash markers | — | — | 5.98** | 0.01 |
| | | | (2.38) | |
| HQ-local weekend share | — | — | −1.92** | 0.01 |
| | | | (0.78) | |
| Explicit pull-request URLs | — | — | 0.07 | 0.41 |
| | | | (0.08) | |
| Observations | 125,114 | | 76,633 | |
| Firms / treated firms / cohorts | 239 / 135 / 63 | | 178 / 125 / 60 | |

Every outcome within a column uses the same firm–cohort–month observations, excluding singleton observations and cohorts without both treated and control firms. The core intersection combines trailer components, bug introduction among pre-financing engineers and recorded code quality. The broader intersection additionally requires numbered references, integration markers, explicit pull-request URLs and headquarters-local weekend shares. Control-role counts are 201 and 146; roles can overlap across cohorts. Shares and bug-introduction rates are percentage points; critical violations are counts per 1,000 lines. Both columns use firm-by-cohort and event-month-by-cohort fixed effects.

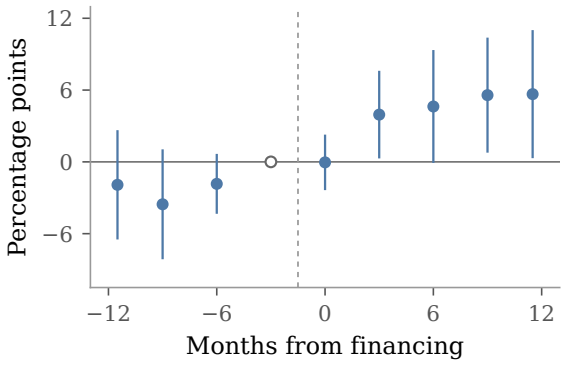
Coefficients and standard errors are percentage points unless otherwise stated. Standard errors in parentheses are clustered by original firm and cohort. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Displayed $p$-values use two decimals; stars use unrounded values.

## L.1 Engineering Process Measures

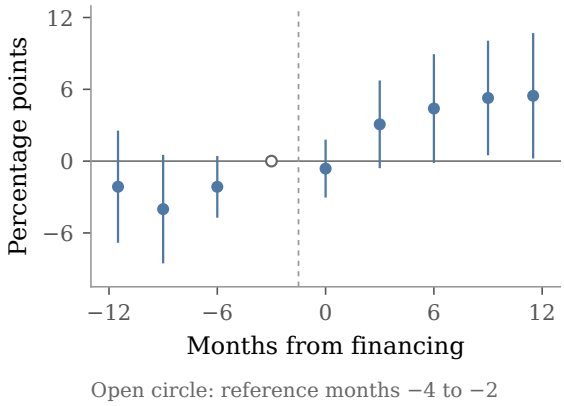
We use commit messages to examine how engineers document and coordinate changes. The primary comparisons cover firms with one or more listed projects and report review-related message markers, issue references, integration-related references and headquarters-local weekend activity (Tables 11 and 38). Separate tables report individual message components and workflow measures for firms with one listed project (Tables 37 and 39). Figure 42 presents event studies for the single-project sample. The estimates for review-related commit-message markers rise before financing toward the omitted reference period.




(a) Review, sign-off or co-authorship trailers




(b) Share of messages with pull-request references




(c) Issue-reference share

Figure 42: Engineering Process Event Studies: Single-Project Sample. The broad trailer measure includes review, sign-off, acknowledgement, approval and co-authorship. Issue references are a hash sign followed by digits, or a number following close(s), fix(es), resolve(s) or ref(s), ignoring case. Firms with one listed commercialized project; shares in percentage points. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

The later-round comparison produces a different pattern. Around Series B and later rounds, the issue-reference share falls by 4.6 percentage points ($p = 0.01$) and the merge-or-squash share by 4.7 percentage points ($p < 0.01$). The latter compares with a 6.0-point increase at first entry, measured the same way (Table 39). The integration marker therefore does not show recurring gains with each financing round, consistent with an initial change in how development is organized under institutional backing.

Table 37: Message-Marker Components on Identical Observations: Single-Project Sample

| Measure | Estimate (pp) | $p$-value |
|---|---|---|
| Combined trailer share | 3.32*** | < 0.01 |
| | (0.86) | |
| Combined share, trailer-line matches | 3.32*** | < 0.01 |
| | (0.86) | |
| Reviewed-by | 0.01 | 0.87 |
| | (0.07) | |
| Signed-off-by | 2.02** | 0.01 |
| | (0.84) | |
| Acked-by | 0.01 | 0.26 |
| | (0.01) | |
| Approved-by | < 0.01 | 0.32 |
| | (< 0.01) | |
| Co-authored-by | 1.40** | 0.01 |
| | (0.53) | |
| Different reviewer/approver email | 0.04 | 0.36 |
| | (0.05) | |
| Author email in review/approval trailer | −0.02 | 0.63 |
| | (0.05) | |
| Explicit pull-request URL | 0.03 | 0.76 |
| | (0.09) | |
| Numbered issue or task reference | 6.79*** | < 0.01 |
| | (2.03) | |
| External task-tracking identifier | 0.82 | 0.14 |
| | (0.55) | |

Every outcome is the fraction of a firm's monthly commits with the corresponding marker, among firms with one listed project: 216,934 observations, 325 firms, 191 treated firms and 79 cohorts. Trailer-line matches require a marker at the start of a line. The different-email category requires a Reviewed-by, Acked-by or Approved-by line containing an email other than the commit author's normalized email. Explicit pull-request URLs are matched separately from generic numbered references and external ticket keys. Markers can overlap within a commit. The table reports all message-marker categories.
Coefficients and standard errors are reported in percentage points.
*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

Table 38: Commit-Message Workflow Comparisons

| Measure | Estimate (pp) | $p$-value |
|---|---|---|
| Numbered issue-reference marker | 5.51***<br>(1.89) | < 0.01 |
| Pull-request-reference marker | 6.23***<br>(1.89) | < 0.01 |
| Merge-or-squash marker | 6.47***<br>(1.76) | < 0.01 |

Monthly shares pool qualifying contributions across the mapped projects covered for each firm. Numbered references match a hash sign followed by digits anywhere in a message. The pull-request-reference marker matches a parenthesized numbered reference such as `(#123)`, the words "pull request" separated by whitespace, or `/pull/` followed by digits, ignoring case. A parenthesized number alone qualifies without verifying whether it identifies a pull request or an issue. Explicit pull-request URLs are a separate measure in Table 37. The merge-or-squash marker equals one when a commit is a recorded merge or when its first message line ends in a parenthesized number such as `(#123)`, the default title of a pull request consolidated into a single commit. Each outcome uses 300,380 identical firm–cohort–month observations, 371 firms, 213 treated firms, 308 control-role firms and 89 cohorts. Firm-by-cohort and event-month-by-cohort fixed effects are absorbed, and singleton observations and cohorts without both treated and control firms are excluded. Table 39 separately reports the single-project comparisons.
Coefficients and standard errors are percentage points unless otherwise stated. Standard errors in parentheses are clustered by original firm and cohort. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Displayed $p$-values use two decimals; stars use unrounded values.

Table 39: Commit-Message Workflow Comparisons: Single-Project Sample

| Measure | Estimate (pp) | $p$-value |
|---|---|---|
| Numbered issue-reference marker | 5.75**<br>(2.25) | 0.01 |
| Pull-request-reference marker | 5.93**<br>(2.27) | 0.01 |
| Any tracker-reference marker | 5.82**<br>(2.22) | 0.01 |
| Merge-or-squash marker | 6.00***<br>(2.14) | <0.01 |

Each dependent variable is the fraction of a firm's monthly commits carrying the stated marker. Numbered references and merge-or-squash markers are defined as in Table 38. Pull-request references record message patterns referring to proposals for integrating code. The tracker-reference marker matches issue-tracker keys such as `ABC-123`, a number following a closing word such as "fixes" or "resolved," a link to a hosted issue tracker, or a pull-request reference. The single-project comparison excludes shared repositories and absorbs firm-by-cohort and event-month-by-cohort fixed effects, with 177,676 fitted observations, 260 firms and 155 treated firms in 83 cohorts. The outcome-specific comparisons in Table 11 exclude cohorts without both arms.
Coefficients and standard errors are reported in percentage points.
*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

Commit records provide a separate measure of merges, which join parallel versions of the code. The monthly merge-commit share changes by $-0.1$ percentage points ($p = 0.90$; not statistically significant; 95% interval $[-1.5, +1.4]$), using 260,755 observations, 358 firms, 200 treated firms and 79 cohorts. This outcome counts recorded merges alone, whereas the merge-or-squash marker in Table 38 also counts pull requests consolidated into a single commit and uses a different fitted sample.

## L.2 Weekend Contributions and Subsequent Repairs

We compare weekend and weekday contributions using Saturdays and Sundays at each firm's reported headquarters. We use the time each contribution was committed, converted to the time zone of the firm's headquarters and accounting for daylight-saving changes. The outcome records linkage to a bug fix within 730 elapsed days. The regression compares contributions by the same developer in the same repository, with calendar-month fixed effects and controls for message-length deciles. Table 40 reports a 0.5-percentage-point higher repair probability for weekend contributions ($p < 0.01$).

Table 40: Headquarters-Local Weekend Contributions and Subsequent Repairs

| Dep Var: | **Bug repair within 730 days** |
|---|---|
| Headquarters-local weekend | 0.54*** |
| | (0.17) |
| Author × Repository FE | Yes |
| Calendar-month FE | Yes |
| Message-length deciles | Yes |
| **# Obs** | 2,339,208 |
| **# Firms / Repositories** | 201 |
| **# Authors** | 39,027 |

The dependent variable equals one when a non-merge, non-bot contribution is linked to a bug fix within 730 elapsed days. The contribution sample ends in June 2024, and we track associated bug fixes through June 2026. Weekend equals Saturday or Sunday at the firm's reported headquarters; weekday is the comparison category. Conversion uses the time the contribution was committed and the headquarters time zone, accounting for daylight-saving changes. The model includes author-by-repository and calendar-month fixed effects and message-length-decile controls. The fitted sample contains 2,339,208 contributions, 201 firms and repositories, and 39,027 authors. The coefficient and standard error are reported in percentage points.
*Note:* Standard errors in parentheses, clustered by firm. * p<0.1, ** p<0.05, *** p<0.01.

## L.3 Bug Introduction Among Pre-Financing Engineers

We trace bug-fix commits to the earlier contributions that last modified the repaired lines, using `git blame`, the version-control operation that identifies the last recorded editor of each line. The resulting numerator counts attribution links, once per introducing commit, rather than repaired lines. Fix commits are identified by bug-fix message markers, excluding merge and squash commits. Attribution ignores differences in spacing and follows moved or copied code; fixes touching more than 30 files or deleting more than 1,000 lines are excluded. We classify engineers as pre-financing if their first contribution precedes the firm event, or the cohort event for controls. Newcomers first contribute at or after that event.

We exclude merge commits and accounts identified as bots from their names and email addresses.

Across a firm's projects, we link developers using recorded email addresses or normalized usernames extracted from GitHub's anonymized email addresses when repository records support the match; otherwise, we keep their identities separate across repositories. We classify engineers using their first recorded contribution across the firm's included projects, rather than a LinkedIn employment match.

Table 41 reports a 1.8-percentage-point decline in bug introduction among engineers active before financing, or 12.4% of the treated pre-treatment mean ($p = 0.03$). The within-engineer component accounts for 1.1 points ($p = 0.03$), while changes in engineers' participation and shares of work account for 0.6 points, a component that is not statistically significant ($p = 0.19$). All three estimates use the same 202,085 observations, 299 firms, 165 treated firms and 72 cohorts. The firm-wide rate among single-project firms declines by 1.2 points, which is not statistically significant ($p = 0.16$).

In the comparison restricted to firms with one listed project, we subtract the rate for pre-financing engineers from newcomers' rate and compare this difference between treated and control firms in post-event months with both groups observed. The difference between treated and control firms is not statistically significant ($p = 0.65$). The classification uses each firm's own cohort event and the first contribution month. With $r^N_{ice}$ and $r^I_{ice}$ denoting the newcomer and pre-financing-engineer rates, we estimate $r^N_{ice} - r^I_{ice} = \gamma \mathbf{1}\{E_i = c\} + \lambda_{ce} + u_{ice}$ for $e \geq 0$, retaining a row only when both contribution denominators are positive. Thus $\gamma$ is the treated-versus-control difference in the observed post-entry group gap.

Table 41: Bug Introduction Among Pre-Financing Engineers and Post-Entry Group Gaps

| Measure | Estimate (pp) | $p$-value | # Obs | # Firms | # Treated |
|---|---|---|---|---|---|
| Pre-financing engineers, total | −1.77** (0.81) | 0.03 | 202,085 | 299 | 165 |
| Within-engineer component | −1.12** (0.51) | 0.03 | 202,085 | 299 | 165 |
| Work-composition component | −0.65 (0.49) | 0.19 | 202,085 | 299 | 165 |
| Single-project post-entry newcomer gap | 0.27 (0.59) | 0.65 | 71,426 | 271 | 146 |
| Single-project firm-wide rate | −1.15 (0.81) | 0.16 | 182,434 | 284 | 162 |

The bug-introduction rate is the fraction of an engineer group's commits identified by tracing later bug fixes to earlier contributions, at calendar-month lags of 0–24. Pre-financing engineers first appear across included projects before the firm event cutoff; controls are classified relative to their cohort event. The total and its two additive components use identical monthly observations, firm-by-cohort and month-by-cohort fixed effects, and 72 cohorts. The decomposition compares each month's observed engineer rates and shares with the group's recorded pre-event history. For engineers active at both endpoints, it averages the endpoint shares for the rate component and the endpoint rates for the share component, splitting the rate–share interaction equally. Activity observed at only one endpoint is allocated to composition. The components sum to the primary total; inference retains their joint covariance. The treated pre-treatment mean of the total rate is 14.2%. Event windows end by June 2024. The last two rows use firms with one listed project. The post-entry gap uses only months with both groups observed and regresses newcomers' rate minus the pre-financing-engineer rate on treated status with cohort-by-event-month fixed effects, in 66 cohorts. Newcomer status uses the firm's own cohort event and the engineer's first contribution month. The firm-wide rate pools engineers and uses 69 cohorts. Mapped-project contributions and introducing commits are pooled and deduplicated by commit identity within firms. Zero contribution denominators yield missing rates. Cohorts lacking either arm are excluded in every comparison.
Coefficients and standard errors are reported in percentage points.
*Note:* Standard errors in parentheses, clustered by cohort and firm. * p<0.1, ** p<0.05, *** p<0.01.

**Shared contribution histories.** To check whether cross-firm reuse of contribution histories drives the result for pre-financing engineers, we use the shorter bug-fix follow-up and exclude firms linked to an introducing commit also mapped to another firm. The total and within-engineer estimates remain −2.0 percentage points ($p = 0.04$) and −1.3 points ($p = 0.04$), respectively, among 233 firms, compared with −1.8 and −1.1 points in the full panel using that follow-up.

**Fixed-roster comparisons.** Table 42 compares the same pre-financing engineers before and after financing. The share of contributions introducing bugs declines by 1.7 percentage points when each engineer receives equal weight (12.8% of the pre-treatment mean; $p = 0.03$) and by 2.4 points when engineers are weighted by their contribution shares before financing (15.9% of the pre-treatment mean; $p < 0.01$). These comparisons complement the monthly decomposition by holding both the team and the engineers' weights fixed.

Table 42: Attribution Changes with the Pre-Financing Engineer Roster Fixed

| Weighting | Estimate (pp) | 95% interval (pp) | Paired units |
|---|---|---|---|
| Equal engineer weights | −1.73** | [−3.31, −0.15] | 17,238 |
| | (0.79) | | |
| Fixed pre-financing contribution weights | −2.44*** | [−4.04, −0.83] | 17,238 |
| | (0.80) | | |

An engineer enters the fixed roster if active before the cohort event and if total contributions are positive in each of two periods: months −12 through −1 and months 0–12. We compute each engineer's attribution rate in each phase counting an introducing commit once if any recorded fix qualifies at calendar-month lags 0–24, then subtract the pre rate from the post rate. Aggregation either weights these same engineers equally in both phases or uses their pre-phase commit shares in both phases. Rates with zero denominators are missing. The comparisons retain 298 firms, 161 treated firms and 70 cohorts, and event windows end by June 2024. This rule differs from the monthly decomposition, which selects the first link in the data's row order before applying the lag restriction, rather than selecting the earliest dated fix. The paired units and weighting choices also differ from the monthly decomposition in Table 41. Coefficients, standard errors and intervals are percentage points; standard errors are clustered by original firm and cohort. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

**Bug-fix follow-up.** The firm-month attribution outcomes link contributions to subsequent bug fixes at calendar-month lags of 0–24. Cohort-by-month fixed effects compare treated and control firms in the same calendar months. The contribution sample ends in June 2024. We observe subsequent bug fixes through June 2026 for 304 repositories; for Odoo, we use the original bug-fix observations. All comparisons use the same contribution denominators, engineer identities, rosters, event windows and estimation specifications. The primary decline is 1.8 percentage points with both the shorter and the longer follow-up. Including additional fixes within the earlier observation period yields a decline of 1.8 points ($p = 0.03$ in each comparison). The within-engineer component remains 1.1 points in all three comparisons ($p = 0.03$ in each).

Using the longer bug-fix follow-up, the same contribution counts and 183,219 observations, firm-wide rates measured over 12 calendar months change by −0.8 percentage points ($p = 0.32$), over 24 calendar months by −1.2 points ($p = 0.15$) and over 730 days by −1.1 points ($p = 0.17$). None of these three estimates is statistically significant. These comparisons concern all contributors, rather than the pre-financing-engineer outcome in Table 41. This attribution comparison covers non-merge contributions. On months with both non-merge and merge denominators observed, the non-merge contribution rate changes by −0.3 points ($p = 0.66$; 137,295 observations); this estimate is not statistically significant.

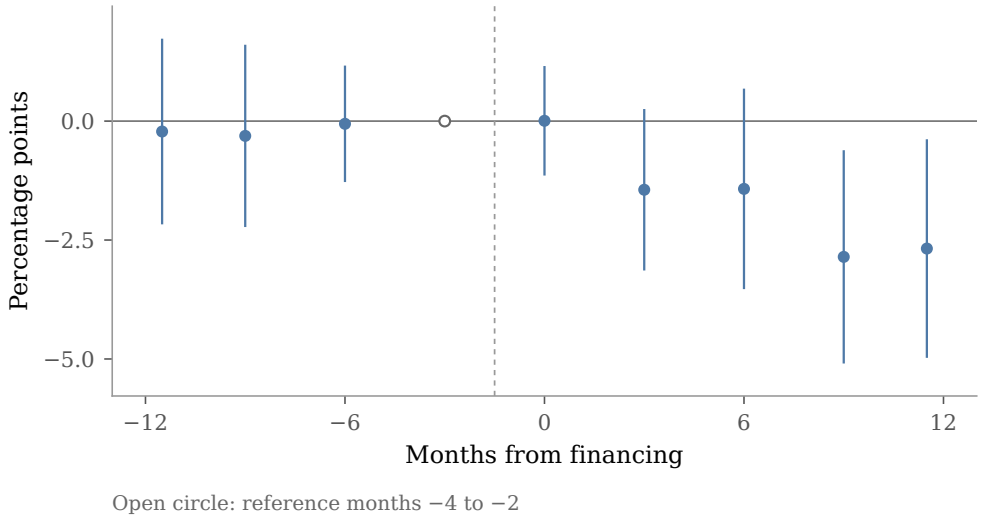



Figure 43: Attribution to Pre-Financing Engineers. The event study uses the shorter bug-fix follow-up to track the bug-introduction rate among these engineers. Firms with one listed commercialized project; shares in percentage points. Quarterly estimates are relative to months −4 to −2; the zero bin spans months −1 to +1. The dashed line marks the end of the reference months. Bars show 95% confidence intervals; standard errors are clustered by firm and cohort.

Together, the findings on documentation, weekend activity and existing engineers support the professionalization interpretation. Existing engineers become less likely to introduce bugs, reflecting improvements within engineers and changes in who does how much work.

# M Additional Outcomes and Design Sensitivity

This appendix reports sensitivity to the event window, treatment timing, and control reuse, followed by additional outcomes and evidence on sustained process documentation, repository allocation, and pre-financing-engineer retention. The outcome and repository-allocation analyses use the main sample. The sustained-documentation analysis starts with 394 main-sample firms covered by the commit corpus and applies marker-specific risk-set filters.

## M.1 Design and Inference Sensitivity

Table 43 collects the principal variations of the baseline design. The tests address three distinct concerns: sensitivity to observations immediately around the deal, the period over which firms are compared, and the weight placed on repeatedly used controls. Estimates remain negative under donut exclusions, alternative windows, and inverse-frequency weighting. The persistence across these choices indicates that the quality improvement is not confined to the financing month or to one construction of the comparison sample.

Table 43: Additional Design Sensitivity

| Specification | Critical Violations | Technical Debt | Violations |
|---|---|---|---|
| Baseline (±12 months) | −1.85** | −1.57** | −4.74** |
| | (0.86) | (0.77) | (2.02) |
| Donut radius 1 | −1.97** | −1.73** | −5.20** |
| | (0.94) | (0.85) | (2.22) |
| Donut radius 2 | −2.06** | −1.84** | −5.62** |
| | (0.99) | (0.91) | (2.43) |
| ±6 months | −1.71** | −0.92* | −2.47** |
| | (0.78) | (0.48) | (1.03) |
| ±9 months | −1.85** | −1.33** | −3.88** |
| | (0.80) | (0.61) | (1.50) |
| ±18 months | −2.25** | −2.23** | −6.14** |
| | (0.99) | (1.01) | (2.75) |
| ±24 months | −2.30* | −1.71** | −4.33** |
| | (1.16) | (0.81) | (1.77) |
| Inverse control-frequency weights | −1.98** | −1.83** | −5.29** |
| | (0.89) | (0.92) | (2.52) |

Critical Violations and Violations are counts per 1,000 lines of code; Technical Debt is calibrated dollars per line. Donut specifications omit observations within the stated number of months on both sides of treatment. Inverse-frequency weights down-weight controls in proportion to the number of cohort stacks in which they appear. Standard errors are clustered by cohort and firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Across 99 leave-one-cohort-out estimates, the coefficient ranges are $[-2.00, -1.25]$ for Critical Violations, $[-1.67, -1.06]$ for Technical Debt, and $[-5.00, -3.15]$ for Violations; every estimate is negative. These ranges summarize coefficient sensitivity. Table 58 separately reports pointwise and familywise baseline tests.

## M.2 Prior Angel Financing and Angel Entry

Table 44 reports the subsequent Seed/Series A comparison for firms with a prior Angel (individual) round. Four firms have an Angel (individual) round as their first qualifying capital round; three have observations before and after the institutional event. Table 45 instead centers the comparison on angel financing itself, with five treated firms and five cohorts. The tables also report estimates excluding each treated firm in turn.

Table 44: Prior-Angel Comparison and Individual-Firm Exclusions

| Sample | Quantity | Critical | Debt | Log code volume | N | Cohorts |
|---|---|---|---|---|---|---|
| Full sample | Interaction | 1.71 | 2.46** | 0.00 | 487,577 | 99 |
| | | (1.82) | (1.11) | (0.24) | | |
| Full sample | Prior-angel group | −0.16 | 0.87 | 0.23 | 487,577 | 99 |
| | | (1.60) | (0.76) | (0.23) | | |
| Exclude GitGuardian | Interaction | 1.71 | 2.46** | 0.00 | 487,568 | 99 |
| | | (1.82) | (1.11) | (0.24) | | |
| Exclude GitGuardian | Prior-angel group | −0.16 | 0.87 | 0.23 | 487,568 | 99 |
| | | (1.60) | (0.76) | (0.23) | | |
| Exclude Testsigma | Interaction | 1.65 | 2.47** | 0.03 | 487,565 | 99 |
| | | (1.92) | (1.14) | (0.25) | | |
| Exclude Testsigma | Prior-angel group | −0.22 | 0.88 | 0.26 | 487,565 | 99 |
| | | (1.71) | (0.80) | (0.25) | | |
| Exclude Traefik Labs | Interaction | 3.79*** | 3.39*** | −0.29*** | 482,004 | 98 |
| | | (0.98) | (0.92) | (0.10) | | |
| Exclude Traefik Labs | Prior-angel group | 1.92*** | 1.80*** | −0.06 | 482,004 | 98 |
| | | (0.43) | (0.43) | (0.08) | | |
| Exclude Tyk | Interaction | −0.34 | 1.59* | 0.27* | 486,866 | 99 |
| | | (1.09) | (0.94) | (0.16) | | |
| Exclude Tyk | Prior-angel group | −2.20*** | 0.00 | 0.50*** | 486,866 | 99 |
| | | (0.60) | (0.42) | (0.15) | | |

Critical violations are counts per 1,000 non-comment lines; debt is calibrated dollars per line; code volume is log(lines). Interaction is the additional Treated × Post coefficient for firms with prior angel financing; group effect sums it with the reference-group coefficient, using their covariance. Prior angel financing requires the firm's first qualifying capital round to be Angel (individual), preceding the first completed Seed or Series A. GitGuardian contributes nine post-event observations and none before; Testsigma contributes one before and 11 after; Traefik Labs and Tyk contribute 12 before and 13 after. The full-sample prior-angel group's 95% intervals are critical violations $[-3.33, 3.01]$, debt per line $[-0.64, 2.37]$, log code volume $[-0.23, 0.70]$. Every specification includes firm-by-cohort and month-by-cohort fixed effects. The stars and intervals use the conventional clustered reference distribution; individual-firm exclusions show sensitivity to the three firms supplying pre/post variation.

Standard errors in parentheses, clustered by cohort and firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 45: Quality at Angel Entry and Individual-Firm Exclusions

| Sample | Critical | Total | Debt | Maintainability | N | Cohorts |
|---|---|---|---|---|---|---|
| Full sample | 0.13 | 2.26* | 0.44 | 2.29* | 31,678 | 5 |
| | [−1.13, 1.40] | [−0.19, 4.71] | [−0.16, 1.05] | [0.00, 4.58] | | |
| Exclude Blender | 0.19 | 2.41 | 0.47 | 2.49* | 27,367 | 4 |
| | [−1.62, 1.99] | [−1.12, 5.94] | [−0.43, 1.36] | [−0.76, 5.73] | | |
| Exclude Moodle | 0.08 | 2.63* | 0.60* | 2.61* | 24,527 | 4 |
| | [−1.82, 1.99] | [−0.74, 6.00] | [−0.17, 1.36] | [−0.60, 5.81] | | |
| Exclude Traefik Labs | 0.18 | 2.69* | 0.28* | 2.67** | 26,170 | 4 |
| | [−1.40, 1.76] | [−0.04, 5.41] | [−0.08, 0.63] | [0.09, 5.25] | | |
| Exclude Tyk | 0.44 | 1.52 | 0.48 | 1.59* | 25,555 | 4 |
| | [−0.73, 1.61] | [−0.65, 3.70] | [−0.41, 1.36] | [−0.37, 3.56] | | |
| Exclude erxes | −0.22 | 2.00 | 0.42 | 2.04 | 22,862 | 4 |
| | [−1.46, 1.01] | [−1.63, 5.63] | [−0.49, 1.33] | [−1.35, 5.44] | | |

Critical, total and maintainability outcomes are issue counts per 1,000 non-comment lines; debt is calibrated dollars per line. Each coefficient is Treated × Post around the angel date, using firm-by-cohort and month-by-cohort fixed effects and censoring treated firms at subsequent completed Seed or Series A. The full sample contains five treated firms, 461 control firms and five cohorts. Controls satisfy the no-Seed/Series A requirement; 174 control firms, representing 326 cohort roles, have an earlier financing record before the window. Financing records identify 12 controls with a prior angel record and 170 with a closed deal during at least one fitted window. Excluding one treated firm leaves four treated firms and four cohorts.

Brackets report 95% confidence intervals from standard errors clustered by cohort and firm and the Student-$t$ distribution (four degrees of freedom in the full sample, three in exclusions). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

## M.3 Additional SonarQube Outcomes

We examine 42 additional SonarQube outcomes grouped into nine families, using a screen based on expected signs, pre-trend tests and reporting criteria. Benjamini–Hochberg adjustment uses all candidate outcomes within each of the nine families before the reporting criteria are applied. The analysis applies the baseline specification, a Series B+ placebo, a five-repository cap, and a single-repository restriction. Table 46 reports selected results and multiplicity-adjusted significance levels, allowing us to assess whether the main findings extend to other dimensions of the codebase.

Table 46: Additional SonarQube Outcomes

| Outcome | First VC Entry | Series B+ Placebo |
|---|---|---|
| Log duplicated lines | 0.29*** | 0.04 |
| | (0.07) | (0.06) |
| Duplicated blocks per 1,000 LoC | 0.94** | 0.19 |
| | (0.45) | (0.31) |
| Maintainability violations (traditional classification) per 1,000 LoC | −4.80** | 0.10 |
| | (1.99) | (0.89) |
| Complexity per 1,000 LoC | −17.37* | 9.34 |
| | (9.94) | (5.73) |

Each cell is a Treated × Post coefficient. Log duplicated lines is ln(duplicated lines) for positive counts. Its first-entry and Series B+ samples contain 452,226 and 300,623 stacked observations, respectively. Duplicated blocks counts repeated code blocks; complexity counts linearly independent paths through the code. Both are divided by non-comment lines of code and multiplied by 1,000. The maintainability count under the traditional classification is SonarQube's "code smells" measure; Table 47 compares it with the alternative classification on identical observations. Log duplicated lines has a Benjamini–Hochberg $q$-value below 0.01; duplicated blocks has $q = 0.11$. Complexity has $p = 0.08$ and $q = 0.18$. The screen also finds null density estimates throughout the security family and nonmonotone estimates across severity levels. Standard errors are clustered by cohort and firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The duplication results describe a different margin of engineering quality. The number of duplicated lines rises as firms expand, while maintainability violations per line decline; the duplicated-block density estimate has a multiplicity-adjusted $q$-value of 0.11. Reusing code can accelerate development while creating a future maintenance burden, so these measures need not move together. The reduction in rule violations and calibrated remediation cost per line therefore coexists with an increase in the number of duplicated lines.

## M.4 Traditional and Alternative Classifications

SonarQube's traditional classification assigns issues to bugs, vulnerabilities or "code smells," the vendor's term for maintainability violations. The alternative classification is SonarQube's Multi-Quality Rule (MQR) classification. It classifies a rule's impacts on reliability, security and maintainability. A rule can affect more than one quality, so the alternative classification's domain counts overlap and should not be summed as unique violations. Both classifications are available in the same scans. Critical violations under the traditional classification and high-severity violations under the alternative classification provide closely aligned measures of serious violations, while maintainability remediation can differ between the classifications.

SonarQube's traditional measures report critical violations and each issue type separately, but

not critical violations by type. The alternative classification reports high-severity issues by the quality they affect, and its high-severity count closely matches critical violations: before financing, treated firms average 12.41 high-severity issues and 12.40 critical violations per 1,000 lines of code, and Table 47 shows that the two yield the same estimates to the reported precision. Maintainability issues account for 12.29 of these 12.41 high-severity issues. The traditional counts alone imply that maintainability violations make up at least 83% of critical violations, because treated firms average only 2.06 bugs and security vulnerabilities combined across all severity levels.

Table 47 compares the outcomes on identical baseline and matched observations. The high-severity estimates round to the same values, and the maintainability-count estimates differ by less than 0.1 violation per 1,000 lines. Baseline calibrated remediation declines by \$1.57 per line under the traditional classification and \$1.46 under the alternative classification, with a standard error of \$0.77 in both cases. Under the preferred matching specification, the corresponding declines are \$1.29 and \$1.25. The paired difference has $p = 0.26$ in the baseline and $p = 0.67$ after matching (neither difference is statistically significant). Across these classifications, the estimated quality gains are closely aligned in magnitude, direction and uncertainty.

Table 47: Traditional and Alternative Classifications on Identical Samples

| Outcome | Traditional classification | Alternative classification | Difference $p$ |
|---|---|---|---|
| *Panel A: Baseline* | | | |
| Critical / high severity | −1.85** | −1.85** | 0.15 |
| | (0.86) | (0.86) | |
| Maintainability violations | −4.80** | −4.76** | 0.48 |
| | (1.99) | (1.99) | |
| Remediation, dollars per line | −1.57** | −1.46* | 0.26 |
| | (0.77) | (0.77) | |
| *Panel B: Preferred matching* | | | |
| Critical / high severity | −2.32* | −2.32* | 0.44 |
| | (1.22) | (1.22) | |
| Maintainability violations | −4.46** | −4.41** | 0.55 |
| | (2.11) | (2.11) | |
| Remediation, dollars per line | −1.29 | −1.25 | 0.67 |
| | (0.98) | (0.98) | |

Critical violations under the traditional classification correspond to high-severity violations under the alternative classification. For maintainability, the traditional classification counts issues classified as code smells; the alternative classification counts maintainability impacts. Counts are per 1,000 non-comment lines. Remediation uses the same fixed calibration under both classifications and is dollars per line. Each pair uses identical observations, regression design and calibration; Difference $p$ uses the alternative estimate minus the traditional estimate and their paired covariance. The baseline has 487,577 observations, 483 firms, 273 treated firms and 99 cohorts; matching has 71,837 observations, 249 firms, 110 treated firms and 68 cohorts. Standard errors in parentheses are clustered by original firm and cohort. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

## M.5 Sustained Process Documentation

We examine when firms begin consistently referring to testing and related development practices in their commit messages. A firm qualifies when these references appear in at least 25% of its commits for three consecutive calendar months, with at least five commits in each month. We date the event to the first month of the first qualifying run. The message terms cover testing, automated integration, code checks and formatting, including references such as "test," "coverage," "pipeline," "lint" and "format."

We follow each firm from its first month with at least five commits until the event or its last month with at least five commits. We exclude firms whose qualifying run begins within the first three months of this observation period. Months with fewer commits or no observed commits remain in the intervening period and are recorded as months without an event; they account for 2,664 of the 33,819 firm-month observations. Financing is dated to the firm's own first financing month. The model estimates the monthly probability of beginning a qualifying run, controlling for time since observation began and calendar year, with standard errors clustered by firm.

Among 382 firms, this monthly probability rises by 0.2 percentage points after financing, from a 0.5% baseline—a relative increase of approximately 47% ($p = 0.04$). Comparable estimates for issue references, review-related message lines and standardized message prefixes are not statistically significant. These comparisons cover 334, 384 and 378 firms, respectively, with 19,801, 37,381 and 39,468 monthly observations.

The testing-related result is strongest at lower thresholds and is no longer statistically significant when references must appear in at least 50% of commits ($p = 0.76$). Adjustment for the four process measures tested also removes statistical significance (Holm-adjusted $p = 0.18$). The analysis therefore documents a change in sustained process-related references, complementing the contribution and tracking measures.

## M.6 Repository Allocation and Continued Contribution

The repository-allocation test asks whether the quality improvement reflects concentrating development in a core repository while pruning peripheral projects. We designate the largest repository before financing as the flagship and track its subsequent share of activity. The multi-repository panel contains 55 treated firms; commit-based outcomes are available for 42 of them. The flagship's commit share ($p = 0.32$) and within-firm commit concentration ($p = 0.63$) show no statistically significant change. Active repositories increase by 0.2 ($p = 0.06$), while the change in new-repository creation is not statistically significant ($p = 0.41$). These results provide no evidence of a systematic shift toward the flagship or withdrawal from peripheral repositories.

Finally, we test whether the improvement among pre-financing engineers reflects the selective retention of engineers whose prior commits were less frequently linked to later fixes. Fixing the

engineer cohort at $t = -1$ and controlling for pre-period activity, we find no statistically significant change in 12-month retention ($p = 0.29$). A direct comparison of retained engineers with leavers shows no statistically significant treated-firm difference in pre-period attribution rates ($p = 0.91$). The fixed-engineer comparisons in Table 42 hold the engineer roster and pre-financing contribution weights fixed and retain a decline in attribution rates.

# N Observation Support, Aggregation and Subgroup Comparisons

## N.1 Observation Support and Aggregation

The initial universe contains 649 firms. The company–repository crosswalk links 609 firms to 703 distinct repositories; 569 firms supply covered codebase months, and 484 enter the stacked panel. The 274 treated and 412 control roles overlap because firms can serve as controls before their own financing. Fixed-effect singleton deletion leaves 273 treated firms in the fitted baseline.

Table 48: Sample Construction and Regression Coverage

| Sample stage | Firms | Difference from preceding row |
|---|---|---|
| OSSRank source universe | 649 | — |
| Company–repository crosswalk | 609 | 40 |
| At least one month with positive lines of code | 569 | 40 |
| Estimation sample before singleton deletion | 484 | 85 |
| Fitted baseline after singleton deletion | 483 | 1 |

The crosswalk links 703 distinct repositories. Differences are counts between the listed groups of firms. The source stack has 274 treated and 412 control firms in 99 cohorts. The fitted baseline has 273 treated and 412 control firms in 99 cohorts, after deleting 18 fixed-effect singleton rows. Treated and control roles can overlap across cohorts.

On the 25-month grid of firm–cohort units represented in the estimation panel, recorded-scan availability rises by 6.6 percentage points after financing relative to controls ($p < 0.01$, $N = 521{,}375$). A recorded firm–month scan means that at least one repository was scanned; the grid is conditional on the firm–cohort units already represented in the stacked panel. Table 50 reports the observation and repository counts by event month. Table 49 compares the baseline DiD coefficient with complete observation windows, paired changes, nonnegative treated-firm aggregation, and fixed repository sets. Each restriction defines its own supported population.

Table 49: Quality Estimates by Observation Support and Aggregation

| Specification | **Critical** | **Total** | **Debt** | **Maintainability** | **N** | **Treated firms** | **Cohorts** |
|---|---|---|---|---|---|---|---|
| Baseline | −1.85** | −4.74** | −1.57** | −4.80** | 487,577 | 273 | 99 |
| | (0.86) | (2.02) | (0.77) | (1.99) | | | |
| Observed before and after | −1.85** | −4.74** | −1.57** | −4.80** | 483,358 | 253 | 98 |
| | (0.86) | (2.02) | (0.77) | (1.99) | | | |
| Complete 25-month windows | −2.19** | −5.61** | −1.87** | −5.60** | 375,175 | 182 | 90 |
| | (1.05) | (2.45) | (0.89) | (2.42) | | | |
| Balanced, treated-firm weights | −2.19** | −5.58** | −1.87** | −5.57** | 15,007 | 182 | 90 |
| | (1.05) | (2.43) | (0.88) | (2.40) | | | |
| Paired changes from month −1 | −2.37** | −3.46** | −1.26 | −3.44** | 248,792 | 244 | 95 |
| | (1.16) | (1.56) | (0.76) | (1.51) | | | |
| Fixed baseline repositories | −1.89** | −5.07** | −1.70** | −5.09** | 476,758 | 247 | 98 |
| | (0.87) | (2.04) | (0.78) | (2.02) | | | |
| All baseline repositories observed | −2.01** | −5.21** | −1.77** | −5.24** | 458,153 | 247 | 98 |
| | (0.90) | (2.10) | (0.80) | (2.08) | | | |
| Original, common repository support | −1.99** | −5.00** | −1.68** | −5.06** | 458,153 | 247 | 98 |
| | (0.90) | (2.10) | (0.80) | (2.07) | | | |

Critical, total and maintainability outcomes are issue counts per 1,000 non-comment lines; debt is calibrated dollars per line. Observed before and after requires at least one recorded month on each side. Complete windows require every month −12 through +12. Balanced aggregation subtracts each firm's mean in months −12 to −1 from its mean in months 0–12, with cohort weights proportional to treated-firm counts. Its N counts firm–cohort units. Paired changes compare each post month with the same firm's month −1 observation, give post months equal weight and fix cohort weights at baseline treated-firm counts; its N counts paired post observations. Repository sets are fixed to repositories with positive scans at month −1. The all-observed and original common-support rows require each baseline repository to remain observed. Cohorts without both arms are dropped after each restriction.

*Note:* Standard errors in parentheses, clustered by original firm and cohort. * p<0.1, ** p<0.05, *** p<0.01.

Table 50: Recorded Codebase Coverage by Event Month

| Event month | Treated rows | Control rows | Control firms | Repos, treated | Repos, control |
|---|---|---|---|---|---|
| −12 | 199 | 17,057 | 367 | 1.4 | 1.7 |
| −11 | 201 | 17,281 | 369 | 1.4 | 1.7 |
| −10 | 208 | 17,528 | 376 | 1.4 | 1.7 |
| −9 | 217 | 17,806 | 377 | 1.4 | 1.7 |
| −8 | 221 | 18,060 | 380 | 1.4 | 1.7 |
| −7 | 223 | 18,225 | 382 | 1.4 | 1.7 |
| −6 | 226 | 18,434 | 384 | 1.4 | 1.7 |
| −5 | 232 | 18,595 | 390 | 1.4 | 1.7 |
| −4 | 237 | 18,755 | 392 | 1.4 | 1.7 |
| −3 | 235 | 19,024 | 399 | 1.4 | 1.7 |
| −2 | 241 | 19,228 | 402 | 1.4 | 1.7 |
| −1 | 247 | 19,686 | 410 | 1.4 | 1.7 |
| 0 | 245 | 19,414 | 410 | 1.4 | 1.6 |
| 1 | 245 | 19,347 | 409 | 1.4 | 1.6 |
| 2 | 245 | 19,364 | 409 | 1.4 | 1.6 |
| 3 | 245 | 19,336 | 409 | 1.4 | 1.6 |
| 4 | 241 | 19,349 | 408 | 1.4 | 1.6 |
| 5 | 243 | 19,332 | 408 | 1.4 | 1.6 |
| 6 | 246 | 19,319 | 408 | 1.4 | 1.6 |
| 7 | 245 | 19,320 | 407 | 1.4 | 1.6 |
| 8 | 244 | 19,302 | 409 | 1.4 | 1.6 |
| 9 | 245 | 19,307 | 409 | 1.4 | 1.6 |
| 10 | 241 | 19,314 | 409 | 1.4 | 1.6 |
| 11 | 245 | 19,267 | 408 | 1.4 | 1.6 |
| 12 | 243 | 19,248 | 408 | 1.4 | 1.6 |

Counts cover comparisons with recorded scans and positive code-line counts for which a repository set can be defined at month −1. A treated firm enters its own cohort once, so treated rows also count treated firms. Controls can appear in several cohorts in the same event month; control rows count those cohort appearances, while control firms counts unique original firms. Repos is the mean number of observed baseline repositories per firm–cohort row. Observation of any repository qualifies a firm–month as recorded; this table does not impose observation of every baseline repository. The all-repositories-observed restriction is reported separately in Table 49.

## N.2 Repository and Language Composition

A firm's aggregate violation density reflects both quality within its repositories and their relative sizes. The density can fall if violations decline within repositories or if repositories with fewer violations account for more of the observed codebase. To examine the first channel, we hold each repository's share of non-comment lines at its value in the month before financing. Quality can then change within the pre-financing repositories while their weights in the firm aggregate remain fixed.

Panel A of Table 51 compares the original and fixed-share measures on the same firm–cohort–month observations. We retain a month only when every repository observed in the month before financing remains observed. Critical violations decline by 2.0 per 1,000 lines under the original measure ($p = 0.02$) and by 1.8 with fixed repository shares ($p = 0.04$). Total violations also decline under both measures ($p = 0.01$ for each), as does technical debt ($p = 0.03$ for each). The estimated

response of the difference between the original and fixed-share measures is −0.18 for critical violations and is not statistically significant ($p = 0.29$). The quality gains therefore persist with the pre-financing repository mix.

We next ask whether quality improves among firms whose main programming language remains unchanged. We define the main language as the language accounting for the most observed lines and retain firm–cohorts for which it is the same in every observed month of the event window. Panel B of Table 51 reports this comparison on its own sample. Critical violations decline by 1.4 per 1,000 lines ($p = 0.03$) and total violations by 2.3 ($p = 0.02$). The technical-debt estimate is negative but not statistically significant ($p = 0.16$). This restriction holds the identity of the main language fixed while allowing the shares of other languages to change.

Table 51: Quality Estimates under Repository and Language Restrictions

| Comparison | **Critical** | **Total** | **Debt** | **N** | **Treated firms** | **Cohorts** |
|---|---|---|---|---|---|---|
| *Panel A: Fixed repository shares, identical observations* | | | | | | |
| Original measure | −1.99** | −5.00** | −1.68** | 458,153 | 247 | 98 |
| | (0.90) | (2.10) | (0.80) | | | |
| Fixed pre-financing shares | −1.82** | −5.04** | −1.70** | 458,153 | 247 | 98 |
| | (0.88) | (2.09) | (0.80) | | | |
| Original minus fixed | −0.18 | 0.04 | 0.02 | 458,153 | 247 | 98 |
| | (0.17) | (0.22) | (0.10) | | | |
| *Panel B: Unchanged main language, separate sample* | | | | | | |
| Unchanged main language | −1.44** | −2.33** | −0.57 | 353,940 | 217 | 92 |
| | (0.68) | (1.02) | (0.41) | | | |

Critical and total violations are counts per 1,000 non-comment lines; debt is calibrated dollars per non-comment line. Panel A uses identical firm–cohort–month observations for all three rows, retaining a month only when every repository with positive scanned lines at month −1 remains observed. The original measure uses current line shares. The fixed-share measure weights each pre-financing repository's current density by its share of lines at month −1; repositories absent at month −1 receive zero weight. The difference row estimates the treatment response of the original measure minus the fixed-share measure on those observations. Panel B is estimated separately on firm–cohorts whose recorded main (dominant) language, the language with the most observed lines, is unchanged across all observed event-window months. This restriction does not require an observation in every month. Both panels include cohort-by-firm and cohort-by-event-month fixed effects and exclude cohorts without both arms; the listed treated counts describe fitted support.
*Note:* Standard errors in parentheses, clustered by original firm and cohort. * p<0.1, ** p<0.05, *** p<0.01.

Finally, Table 52 measures changes in the language mix of the observed codebase. Each outcome is a language family's share of total non-comment lines. The JavaScript/TypeScript share falls by 0.9 percentage points after financing relative to controls ($p = 0.04$). The table reports all ten language families, providing a direct description of how codebase composition changes around financing.

Table 52: Changes in Programming-Language Shares of the Observed Codebase

| Language family | **Percentage points** | Language family | **Percentage points** |
|---|---|---|---|
| JavaScript/TypeScript | −0.89** | Rust | −0.01 |
| | (0.44) | | (0.20) |
| Python | 0.13 | Ruby | 0.02 |
| | (0.51) | | (0.12) |
| Java/JVM | 0.22 | C# | 0.00 |
| | (0.18) | | (0.00) |
| C/C++ | 0.35 | Web markup | 0.02 |
| | (0.22) | | (0.36) |
| Go | 0.51 | Other | −0.34 |
| | (0.40) | | (0.28) |

Each outcome is a language family's non-comment line count divided by total non-comment lines in the covered firm–month codebase. Coefficients and standard errors are percentage points. The ten mutually exclusive families sum to the observed lines; all are reported. Each column is estimated on the same firm–cohort–month rows with the baseline fixed effects, so the ten treatment coefficients sum to zero before rounding. All language-share estimates use the same 487,577 observations, 273 fitted treated firms and 99 cohorts. C# line share is zero throughout this sample, so its zero coefficient and standard error are arithmetic rather than an estimated significance test. These shares describe analyzer-recorded language totals; they do not inventory the source files present in the mapped repositories.
*Note:* Standard errors in parentheses, clustered by original firm and cohort. * p<0.1, ** p<0.05, *** p<0.01.

## N.3 Baseline Quality and Concentration

The high-versus-low critical-violation difference remains negative when the threshold is defined from one baseline observation at each treated firm's own financing or from average pre-financing quality. Table 53 reports these group effects and the full sequence excluding the firms with the highest month −1 critical-violation densities. Removing the seven highest-density treated firms changes the critical-violation DiD coefficient from −1.85 ($p = 0.03$) to −0.64 ($p = 0.11$; not statistically significant), a 65.2% reduction in its magnitude. The average decline is therefore sensitive to firms with high initial critical-violation density.

Table 53: Initial Quality and the Concentration of Improvements

| | Critical violations | | | Total violations | | |
|---|---|---|---|---|---|---|
| Quality classification | **Low** | **High** | **Difference** | **Low** | **High** | **Difference** |
| **Panel A: Response by baseline critical-violation density** | | | | | | |
| Baseline cutoff | −0.28 | −4.14** | −3.87** | −4.60 | −5.41** | −0.81 |
| | (0.39) | (1.85) | (1.89) | (2.98) | (2.40) | (3.74) |
| Own-financing baseline cutoff | −0.28 | −3.55** | −3.27* | −5.06 | −4.74** | 0.32 |
| | (0.42) | (1.64) | (1.69) | (3.36) | (2.13) | (3.89) |
| Six-month pre-financing average | −0.12 | −3.73** | −3.62** | −4.80 | −4.91** | −0.12 |
| | (0.18) | (1.66) | (1.67) | (3.27) | (2.16) | (3.83) |
| Average (at least three observed pre-financing months) | −0.09 | −3.63** | −3.54** | −4.86 | −4.79** | 0.07 |
| | (0.17) | (1.69) | (1.70) | (3.31) | (2.20) | (3.89) |

| **Panel B: Exclusions ranked by month −1 critical density** | | | | |
|---|---|---|---|---|
| | **Critical** | **95% interval** | **High minus low** | **N** |
| Exclude highest 1 | −1.75** | [−3.47, −0.04] | −3.64* | 487,563 |
| | (0.86) | | (1.91) | |
| Exclude highest 3 | −1.61* | [−3.30, 0.08] | −3.35* | 486,498 |
| | (0.85) | | (1.91) | |
| Exclude highest 5 | −1.30* | [−2.81, 0.22] | −2.65 | 481,099 |
| | (0.76) | | (1.74) | |
| Exclude highest 7 | −0.64 | [−1.46, 0.17] | −1.11 | 480,114 |
| | (0.41) | | (0.87) | |

All outcomes are issue counts per 1,000 non-comment lines. Panel A includes the Post interaction with high baseline critical-violation density and reports critical and total violation densities. The baseline cutoff uses the median across baseline firm–cohort observations of firms treated somewhere in the sample. The own-financing cutoff uses one month −1 observation per treated firm. The average uses months −6 through −1; the restricted version requires observations in at least three of those months. Group-effect standard errors account for the covariance of the base and treatment interaction. Panel B ranks treated firms at their own financing and removes each listed firm's treated and control appearances. It reports the critical-violation DiD coefficient and the high-minus-low critical interaction using the baseline cutoff. Each resulting sample retains both arms in every cohort.

*Note:* Standard errors in parentheses, clustered by original firm and cohort. * p<0.1, ** p<0.05, *** p<0.01.

Table 54: Baseline-Quality Groups: Support and Effect Intervals

| Classification | Low, treated | High, treated | N | Low, 95% interval | High, 95% interval |
|---|---|---|---|---|---|
| Baseline cutoff | 141 | 106 | 477,107 | [−1.05, 0.50] | [−7.82, −0.47] |
| Own-financing baseline cutoff | 124 | 123 | 477,107 | [−1.12, 0.55] | [−6.79, −0.30] |
| Six-month pre-financing average | 126 | 126 | 483,145 | [−0.46, 0.23] | [−7.02, −0.44] |
| Average (at least three observed pre-financing months) | 121 | 120 | 475,847 | [−0.43, 0.26] | [−6.99, −0.27] |

Intervals concern the critical-violation effects per 1,000 non-comment lines reported in Panel A of Table 53. High-group intervals use the covariance of the base coefficient and treatment interaction. Counts describe unique fitted treated firms in each group; all four comparisons retain 98 cohorts. Cohort-by-firm and cohort-by-event-month fixed effects, and clustering by original firm and cohort, follow the corresponding interaction regression.

## N.4 Matching Eligibility and Timing

Table 55 separates the requirement for observable growth features from the selection of comparable controls. Its second row retains precisely the treated firms entering the matched regression while leaving their eligible candidate controls unmatched. The common-population specification comparison in Appendix E separately fixes the treated population and cohort weights across all matching configurations.

Table 55: Matching Eligibility and Control Selection

| Comparison | **Critical** | **Total** | **Debt** | **Maintainability** | **N** | **Treated** | **Controls** | **Cohorts** |
|---|---|---|---|---|---|---|---|---|
| Matching-eligible, unmatched | −2.52* | −4.26** | −1.14 | −4.05** | 177,278 | 112 | 206 | 70 |
| | (1.43) | (1.86) | (0.82) | (1.75) | | | | |
| Final treated firms, unmatched | −2.57* | −4.36** | −1.14 | −4.15** | 172,134 | 110 | 206 | 68 |
| | (1.45) | (1.88) | (0.83) | (1.78) | | | | |
| Preferred matched comparison | −2.32* | −4.80** | −1.29 | −4.46** | 71,837 | 110 | 196 | 68 |
| | (1.22) | (2.37) | (0.98) | (2.11) | | | | |

Critical, total and maintainability outcomes are counts per 1,000 non-comment lines; debt is calibrated dollars per line. Eligibility requires an observation in month −1 and finite headcount-growth measures for treated and control firms. The final-treated unmatched row uses the 110 treated firms retained after the caliper and cohort support rule with all eligible controls. The preferred comparison selects controls within a Mahalanobis-distance caliper of one from the cohort's average treated headcount-growth profile. Counts are fitted unique firms by arm, which can overlap across cohorts.

*Note:* Standard errors in parentheses, clustered by original firm and cohort. * p<0.1, ** p<0.05, *** p<0.01.

The main quarterly event study places month −1 in the same bin as financing month 0; Table 56 estimates monthly coefficients that separate the two.

Table 56: Monthly Timing Relative to Month −4

| Event month | **Critical** | **Total** | **Debt** | **Maintainability** |
|---|---|---|---|---|
| Month −3 | 0.07 | −0.17 | −0.01 | −0.23 |
| | (0.19) | (0.34) | (0.12) | (0.29) |
| Month −2 | 0.14 | −0.27 | −0.07 | −0.34 |
| | (0.22) | (0.45) | (0.16) | (0.41) |
| Month −1 | 0.33 | 0.13 | −0.16 | 0.03 |
| | (0.66) | (0.89) | (0.29) | (0.85) |
| Month 0 | −0.24 | −0.65 | −0.35 | −0.83 |
| | (0.88) | (1.13) | (0.39) | (1.08) |
| Month 12 | −2.60** | −4.75** | −1.87* | −4.67** |
| | (1.28) | (1.94) | (1.01) | (1.85) |
| All pre-financing months, joint $p$ | 0.93 | 0.81 | 0.83 | 0.74 |
| Months −3, −2, −1, joint $p$ | 0.91 | 0.88 | 0.94 | 0.79 |

Critical, total and maintainability outcomes are counts per 1,000 non-comment lines; debt is calibrated dollars per line. Monthly treatment interactions omit month −4, preserving a separate coefficient for month −1. The regression uses the baseline 487,577 observations across 99 cohorts. Joint tests of zero pre-financing coefficients use the full two-way-clustered covariance matrix and a chi-square reference distribution, covering all eleven nonreference pre-financing months and, separately, months −3, −2, −1.
*Note:* Standard errors in parentheses, clustered by original firm and cohort. * p<0.1, ** p<0.05, *** p<0.01.

## N.5 Financing Amount and Minor Maintainability Violations

Table 57 makes the round-size gradient explicit. Low-severity maintainability violations decline by 1.0 per 1,000 lines in the smaller-round group ($p = 0.05$), while the larger-round estimate is close to zero and not statistically significant ($p = 0.83$). The groups differ by 1.1 violations per 1,000 lines ($p = 0.03$). Restricting to recorded financing amounts retains a positive difference of 1.1 ($p = 0.02$). In the Series B+ tables in Appendix J, Large Round continues to refer to the earlier first-round amount; those interactions do not classify the size of the later financing itself.

Table 57: First-Round Size and Low-Severity Maintainability Violations

| Sample | Smaller round | Larger round | Difference | Treated small / large | N |
|---|---|---|---|---|---|
| Original missing-size coding | −1.03* | 0.08 | 1.11** | 157 / 116 | 487,577 |
| | [−2.09, 0.04] | [−0.70, 0.86] | [0.08, 2.13] | | |
| Recorded-size observations | −0.92** | 0.20 | 1.13** | 138 / 116 | 132,023 |
| | [−1.79, −0.06] | [−0.43, 0.84] | [0.13, 2.13] | | |

Outcomes are low-severity maintainability issue counts under the alternative classification per 1,000 non-comment lines. Large Round uses the treated-firm median of the first qualifying financing amount. The original coding puts missing amounts in the smaller-round group; the recorded-size comparison removes rows with missing amount. The larger-round effect sums the base and interaction with their covariance. The two rows retain 99 and 95 cohorts, respectively. Brackets are 95% confidence intervals; standard errors are clustered by original firm and cohort. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

## N.6 Outcome Families and Inference

Stars in the regression tables report two-sided pointwise tests using standard errors clustered by original firm and cohort.

A conjunctive claim requires both critical-violation density and technical debt to decline. Its null is that at least one of these effects is nonnegative. An intersection–union test rejects only when each component test rejects in the negative direction. Using the retained two-sided tests, both negative baseline estimates meet the 5% threshold required for this conjunction, without assuming that the outcomes are independent. This conclusion concerns the joint requirement that both outcomes improve; it does not provide adjusted significance for selecting individual outcomes from a family.

Table 58 instead reports Holm adjustments for separate discoveries within the stated outcome families, controlling the probability of any false individual rejection. Technical debt per line and the Technical Debt Ratio express the same outcome in different units on common support and enter a family once. The two headline outcomes both have Holm-adjusted $p = 0.06$; adding total violations or maintainability leaves the adjusted values between 0.06 and 0.07. These sensitivity calculations leave the coefficients and pointwise tests unchanged.

Table 58: Pointwise and Holm-Adjusted Baseline Tests

| Outcome | Pointwise $p$ | Two outcomes | Three outcomes | Four outcomes |
|---|---|---|---|---|
| Critical violations | 0.03 | 0.06 | 0.06 | 0.07 |
| Technical debt per line | 0.04 | 0.06 | 0.06 | 0.07 |
| Total violations | 0.02 | — | 0.06 | 0.07 |
| Maintainability violations | 0.01 | — | — | 0.07 |

The two-outcome family contains critical violations and technical debt per line. The three-outcome family adds total violations; the four-outcome family also adds the maintainability count. The Technical Debt Ratio is a fixed rescaling of debt per line on identical support and is counted once. Holm adjustments control familywise rejection using the retained two-sided pointwise tests without assuming independent outcomes. These are sensitivity calculations for the stated families, rather than a claim that a family was selected before analysis. All tests use the same 487,577 fitted rows, 483 firms and 99 cohorts and the firm-and-cohort clustered covariance.

For positive covered lines, the defined Technical Debt Ratio is

$$\mathrm{TDR}_{it} = \frac{100\,\mathrm{Debt}_{it}}{w\ell\,\mathrm{LoC}_{it}} = \frac{40}{11}\,\mathrm{DebtPerLoC}_{it}, \qquad \ell = 0.5.$$

We construct the ratio from the same covered lines as dollar debt per line. It is available for all 61,768 positive-line firm–months in the descriptive panel. The baseline and later-round tables report the dollar outcome on its own support; multiplying its coefficient and standard error by $40/11$ gives the ratio in percentage points on those same rows.

## N.7 Baseline Selection Regression

We examine whether pre-financing engineering quality and codebase size predict which firms receive financing within a cohort. The dependent variable equals one for that cohort's treated firms and zero for its controls. The regressions use 19,933 firm–cohort observations at month $-1$, representing 462 original firms and 98 cohorts. Both specifications absorb cohort fixed effects and cluster standard errors by firm. The quality-only specification includes critical-violation density, total-violation density and debt per line; the second adds log(LoC). The defined debt ratio is collinear with debt per line and adds no independent regressor. Joint tests use $F(k, 461)$, with $k$ equal to the number of retained tested coefficients and 461 equal to the number of firm clusters minus one. The tests give $F(3, 461) = 1.38$ ($p = 0.24$; not statistically significant) and $F(4, 461) = 3.86$ ($p < 0.01$). The corresponding within-cohort explained variance is 0.02% and 0.1%.

## N.8 Denominators for Relative Effect Sizes

We express each quality coefficient as a percentage of that outcome's own treated pre-financing mean on the fitted sample: $100\widehat{\beta}/\bar{Y}_{T,\text{pre}}$. The mean averages the available treated observations in months $-12$ through $-1$, giving each observed row equal weight. It therefore differs from the month $-1$ balance-table mean and from a mean giving each firm equal total weight. Table 59 reports the actual denominators separately for the baseline and preferred matching comparisons, including both maintainability classifications. The percentage is calculated from unrounded estimates and means before display rounding.

Table 59: Pre-Financing Means Used for Relative Effect Sizes

| Outcome | Pre-financing mean | Change (%) | Treated pre-financing observations | Regression observations |
|---|---|---|---|---|
| *Baseline* | | | | |
| Critical violations | 12.40 | -14.9 | 2,719 | 487,577 |
| Total violations | 32.89 | -14.4 | 2,719 | 487,577 |
| Maintainability (traditional classification) | 30.83 | -15.6 | 2,719 | 487,577 |
| Maintainability (alternative classification) | 30.70 | -15.5 | 2,719 | 487,577 |
| Debt, dollars per line | 13.09 | -12.0 | 2,719 | 487,577 |
| *Preferred matching* | | | | |
| Critical violations | 16.09 | -14.4 | 1,315 | 71,837 |
| Total violations | 36.62 | -13.1 | 1,315 | 71,837 |
| Maintainability (traditional classification) | 34.54 | -12.9 | 1,315 | 71,837 |
| Maintainability (alternative classification) | 34.09 | -12.9 | 1,315 | 71,837 |
| Debt, dollars per line | 14.81 | -8.7 | 1,315 | 71,837 |

Critical, total and maintainability means are violations per 1,000 non-comment lines; the debt mean is calibrated dollars per line. Each reported treated pre-financing count is the number of observations used to calculate that mean. Baseline comparisons have 273 fitted treated firms in 99 cohorts; preferred matching has 110 treated firms in 68 cohorts. The debt ratio is a fixed rescaling of debt per line on this support and has the same relative change, so it is not repeated. All means use unweighted arithmetic averaging of the specified fitted pre-period rows.

## N.9 Calendar-Month Dependence

The baseline fitted observations cover 178 calendar months, 99 financing cohorts and 483 original firms. Table 60 keeps the coefficients and observations fixed and adds calendar-month clustering to the original firm and cohort dimensions. Critical violations retain $p = 0.03$ and technical debt retains $p = 0.04$; their confidence intervals are slightly narrower than under baseline clustering.

Table 60: Calendar-Month Clustering on Identical Baseline Observations

| | Critical violations | | Technical debt | |
|---|---|---|---|---|
| Clustering | Estimate (SE) | 95% interval | Estimate (SE) | 95% interval |
| Firm and cohort | −1.85 (0.86) | [−3.56, −0.14] | −1.57 (0.77) | [−3.08, −0.05] |
| Firm and calendar month | −1.85 (0.78) | [−3.39, −0.31] | −1.57 (0.73) | [−3.01, −0.12] |
| Firm, cohort and calendar month | −1.85 (0.85) | [−3.53, −0.17] | −1.57 (0.76) | [−3.07, −0.06] |

All rows use the same 487,577 observations and baseline fixed effects. Critical violations are counts per 1,000 non-comment lines; technical debt is calibrated dollars per line. Calendar months use the observation's original year and month. Covariance combines original score cross-products for every marginal clustering dimension and intersection by inclusion–exclusion, preserving repeated observations of each original firm. The common minimum-cluster CRV1 correction and Student-$t$ reference follow the baseline convention: degrees of freedom are 98 for rows including cohort and 177 for firm/calendar clustering.